

CompoDock

Technical Manual

Software version: 2.4.x

Edition: 12/06.20

Part no.: M650701

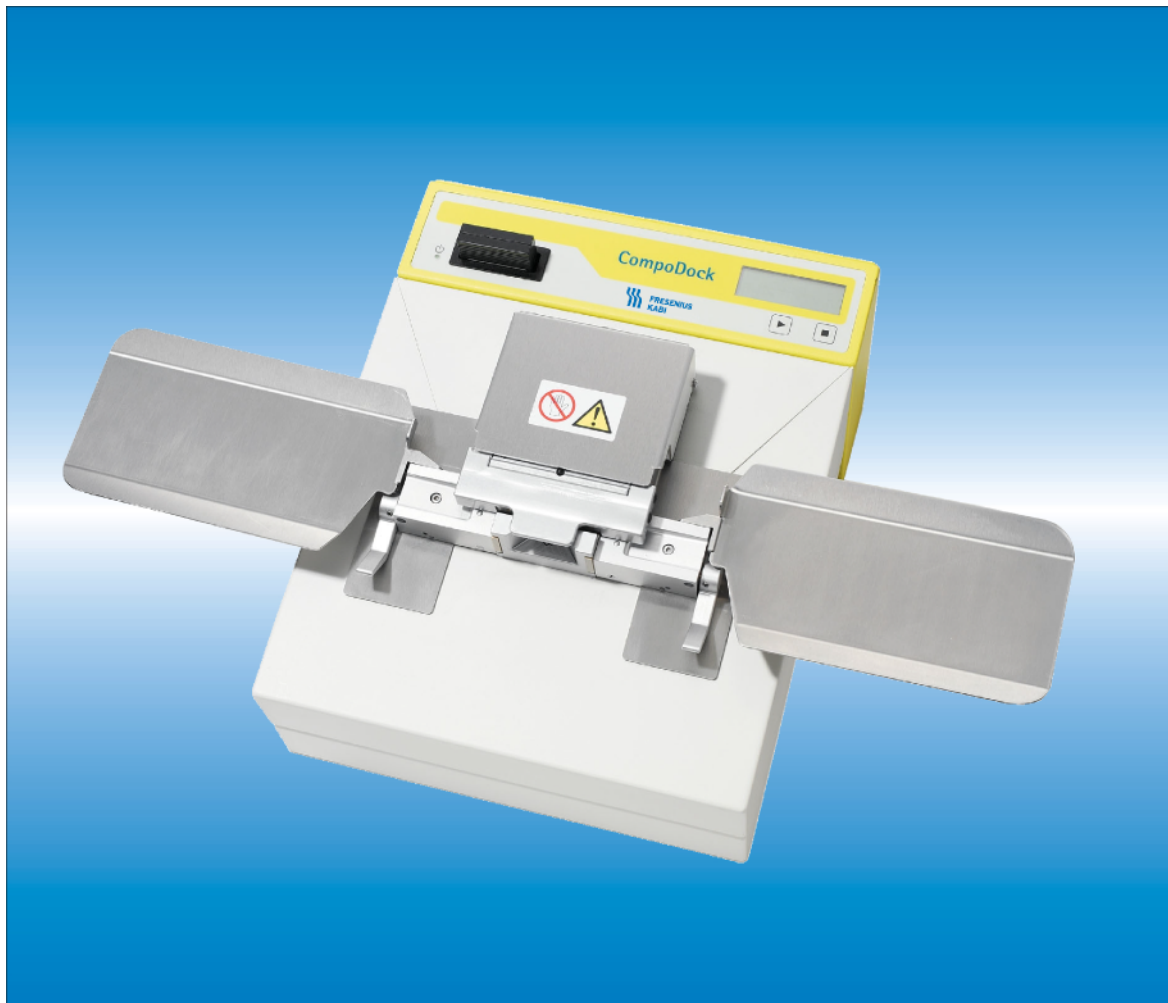


Table of Contents

1 Index

2 Important information

2.1	Important information on the Technical Manual	2-1
2.1.1	How to use the Technical Manual	2-1
2.1.2	Explanation of the Note and Caution symbols used	2-1
2.2	Precautions for working on the system	2-2
2.3	Addresses	2-3

3 Block diagrams

4 Service menu

4.1	Introduction.....	4-1
4.2	Preparation.....	4-1
4.3	Starting the service menu.....	4-2
4.3.1	Procedure with E-code < 069	4-2
4.3.2	Procedure with E-code ≥ 069	4-2
4.4	Main menu	4-4
4.4.1	Parameters	4-4
4.4.2	Flow diagram	4-6
4.4.3	Set Default Values	4-7
4.4.4	Seal times	4-7
4.4.5	Seal stretch tubes	4-7
4.4.6	Heating Parameter	4-8
4.4.7	Xmotor Heat pos.....	4-8
4.4.8	Tube heating time	4-8
4.4.9	Heater dock temp.	4-8
4.4.10	Xmotor pos.dock.....	4-9
4.4.11	Dock cooldown time	4-9
4.4.12	HF motor.....	4-9
4.4.13	Y motor	4-10
4.4.14	Xmotor Outside pos.....	4-10
4.4.15	Language.....	4-11
4.4.16	Blower current	4-11
4.4.17	Voltage supply	4-12
4.4.18	Runcounter	4-12
4.4.19	Seal test.....	4-12
4.4.20	Life test	4-13
4.4.21	Autostart	4-13
4.4.22	Dock End Beep Time.....	4-13

5 Repair / adjustment / maintenance

5.1	Opening the device	5-1
5.2	Power supply	5-4
5.2.1	Power connection < E-code 069.....	5-4
5.2.2	Power connection ≥ E-code 069.....	5-4
5.2.3	Power connection ≥ E-code 140.....	5-5
5.3	DC voltage supply	5-6
5.3.1	Measurement and adjustment points P.C.B. LP 1058 (< E-code 069).....	5-6
5.3.2	Measurement and adjustment points P.C.B. LP 1058-1.....	5-7
5.3.3	Measurement and adjustment points P.C.B. LP 1058-1 (≥ E-code 069).....	5-8
5.3.4	Measurement and adjustment points P.C.B. LP 1058D (≥ E-code 140)	5-9
5.3.5	Unregulated DC voltage < E-code 069.....	5-10
5.3.6	Unregulated DC voltage < E-code 140.....	5-10
5.3.7	Regulated DC voltage.....	5-10
5.3.8	Reference voltage.....	5-11
5.3.9	Display contrast	5-11
5.3.10	Display backlighting (only with P.C.B. LP 1058, layout C).....	5-11
5.3.11	Display backlighting (only with P.C.B. LP 1058D, layout D).....	5-11
5.3.12	Blower current.....	5-11
5.4	Position switches	5-12
5.4.1	Layout of the position switches.....	5-12
5.4.2	Tube clamp positions.....	5-13
5.4.3	Electrode positions	5-14
5.4.4	Heating element positions	5-15
5.4.5	Light-emitting diodes P.C.B. LP 1058.....	5-16
5.4.6	Light-emitting diodes P.C.B. LP 1058D	5-17
5.5	Visual inspection / functional test	5-18
5.5.1	Checking the position of the tube clamps.....	5-18
5.5.1.1	Checking the alignment of the tube clamps in docking position.....	5-18
5.5.1.2	Checking the alignment of the tube clamps with the electrodes	5-18
5.5.1.3	Checking the alignment of the tube clamps with the central block.....	5-19
5.5.1.4	Checking the alignment of the tube clamp height position	5-19
5.5.2	Performing a sealing test.....	5-20
5.5.3	Performing a docking test.....	5-20
5.5.4	Reading out parameters / saving parameters with service tool	5-20
5.6	Tubing cover	5-21
5.6.1	Tubing cover switch < E-code 030	5-21
5.6.2	Tubing cover ≥ E-code 030.....	5-23
5.7	Central block	5-25
5.7.1	Removal of the central block	5-25
5.7.2	Installation of the central block	5-26
5.8	HF electrodes unit	5-27
5.8.1	Checking the HF mechanism.....	5-27
5.8.2	Removal of the electrodes.....	5-29
5.8.3	Installation of the electrodes	5-29
5.8.4	Adjustment of the HF mechanism.....	5-30
5.8.4.1	Adjusting the electrode distance	5-30
5.8.4.2	Adjusting the mechanical end position of the HF unit	5-31

5.9	Tube clamps	5-33
5.9.1	Test.....	5-34
5.9.2	Removal of the tube clamps	5-34
5.9.3	Installation of the tube clamps	5-34
5.9.4	Adjustment of the tube clamps (central position).....	5-35
	5.9.4.1 Basic adjustment of the tube clamps.....	5-35
	5.9.4.2 Adjustment of the central block	5-35
	5.9.4.3 Adjustment of the Y0 position switch.....	5-36
5.9.5	Basic adjustment of the HF unit.....	5-39
5.9.6	HF-0 position switch	5-41
5.9.7	HF-1 position switch	5-42
5.10	X system	5-43
5.10.1	X1 / Y0 position - separation of the tubings.....	5-44
5.10.2	Docking of the tubings - X3 / Y1 position.....	5-45
5.11	Y system	5-46
5.11.1	Y0 position switch.....	5-46
5.11.2	Y1 position switch.....	5-48
5.11.3	Y1 position switch.....	5-49
5.12	Heating element	5-50
5.12.1	Test of the heating element.....	5-50
5.12.2	Adjustment of the heating temperature	5-52
5.12.3	Removal of the heating element.....	5-52
5.12.4	Installation of the heating element.....	5-52
5.12.5	Adjustment of the H0 position switch.....	5-52
5.12.6	Adjustment of the H1 position switch.....	5-53
5.12.7	Adjustment of the heating position	5-54
5.13	HF voltage	5-56
5.13.1	Checking the HF voltage	5-57
5.13.2	Adjustment of the HF voltage	5-59
5.13.3	Removal of the seal head box	5-61
5.13.4	Installation of the seal head box	5-61
5.14	Sealing test	5-62
5.15	Docking test	5-63
5.15.1	Visual check	5-63
5.15.2	Tensile strength	5-63
5.15.3	Tightness	5-64
5.16	Electrical test	5-65
5.17	Initial start-up	5-66
5.17.1	Seal test.....	5-67
5.17.2	Dock test.....	5-68
5.18	Alarm processing	5-69
5.18.1	Fatal (system) error	5-69
5.19	Software update	5-71
5.19.1	Software versions 2.20.R11 and 2.20.R12.....	5-71
5.19.2	Software versions 2.21.R0, 2.21.R1 and 2.21.R2	5-72
5.20	Modification to CompoDock with interface (E-code 059 or higher)	5-74
5.21	Software upgrade (E-code 069 - 140)	5-77

5.22	Software upgrade (E-code 140 and higher)	5-80
5.22.1	Installation of the Service-Tool V2	5-80
5.22.2	GUI Elements	5-84
5.23	Backup (E-code 069 - 140)	5-87
5.24	Backup (E-code 140 and higher)	5-88

6 Initial Start-up Test report

7 Maintenance Test report

8 Spare Parts Catalog

1 Index

A

Accessories 2-2
Adjusting gauge 5-38
Adjusting pin 5-22

C

Changes 2-1
Circuit diagrams 2-1
Clean Position 5-21
Cold electrode 5-29

D

Docking position 5-18

E

Editorial information 2-1
ESD 2-2

F

Flow diagram 4-6

H

Heating temperature 5-50

I

Identification 2-1
Initial position 5-18

K

Key functions 4-3

M

Main menu 4-4

O

Optical sensor 5-43

P

Page identification 2-1
Perforated disk 5-43

Precautions 2-2

R

Reference position 5-43

S

Service menu 4-2

Spare Parts Catalog 6-1

Specifications 2-1

V

Voltage indicator 5-4

Voltage selector switch 5-4

2 Important information

2.1 Important information on the Technical Manual

2.1.1 How to use the Technical Manual

Identification	The document can be identified by the following information on the title page and on the labels, if any: – Edition of the technical document – Part number of the technical document
Page identification	Page number 1-3 is to be interpreted as chapter 1, page 3.
Editorial information	The editorial information 12/06.20, for example, refers to: 12th edition, June 2020. Valid for CompoDock software V2.4 x and higher.
Changes	Manual changes will be released as new editions or supplements. In general, this manual is subject to change without notice.
Purpose	This Technical Manual is intended for service technicians. It can be used both for first studies (to acquire a basic knowledge) and for reference purposes (for TSC, maintenance and repair). The Technical Manual, however, does not replace the training courses offered by the manufacturer.
Requirements	Knowledge of the current Operating Instructions of the respective system. Background experience in mechanics, electrical and medical engineering.
Specifications	For the specifications of the respective device, refer to the current Operating Instructions.
Circuit diagrams and component layouts	The identification on the PCB permits the operator/technician to verify if the circuit diagram/component layout matches the PCB actually installed in the system.

2.1.2 Explanation of the Note and Caution symbols used



Note

Informs the operator that in case of a failure to follow the steps as described, a specific function will be executed incorrectly or will not be executed at all, or will not produce the desired effect.



Caution

Advises the operator against certain procedures or actions that could cause damage to the equipment or may have adverse effects on individuals.

2.2 Precautions for working on the system

Authorized persons	Assembly, extensions, adjustments, modifications or repairs may only be carried out by the manufacturer or persons authorized by him.
Test equipment and accessories	The activities described in the Technical Manual require the availability of the necessary technical test equipment and accessories.
Precautions	When working on the open system, the following precautions must be respected: – Protect the components against ingress of fluids. – Do not touch live parts.
ESD precautions	When repairing the device and when replacing spare parts, observe the applicable ESD precautions (e.g. EN 100 015-1).

2.3 Addresses

Please address any inquiries to:

Manufacturer

Fresenius Kabi AG

61346 Bad Homburg

Germany

Phone: +49 6172 / 608 - 0

Local service:



National service

Fresenius HemoCare GmbH

Technischer Service

Grüner Weg 10

61169 Friedberg

Germany

International Service Hotline (08:30 -17:00)

Phone: +49 6172 / 608 8469

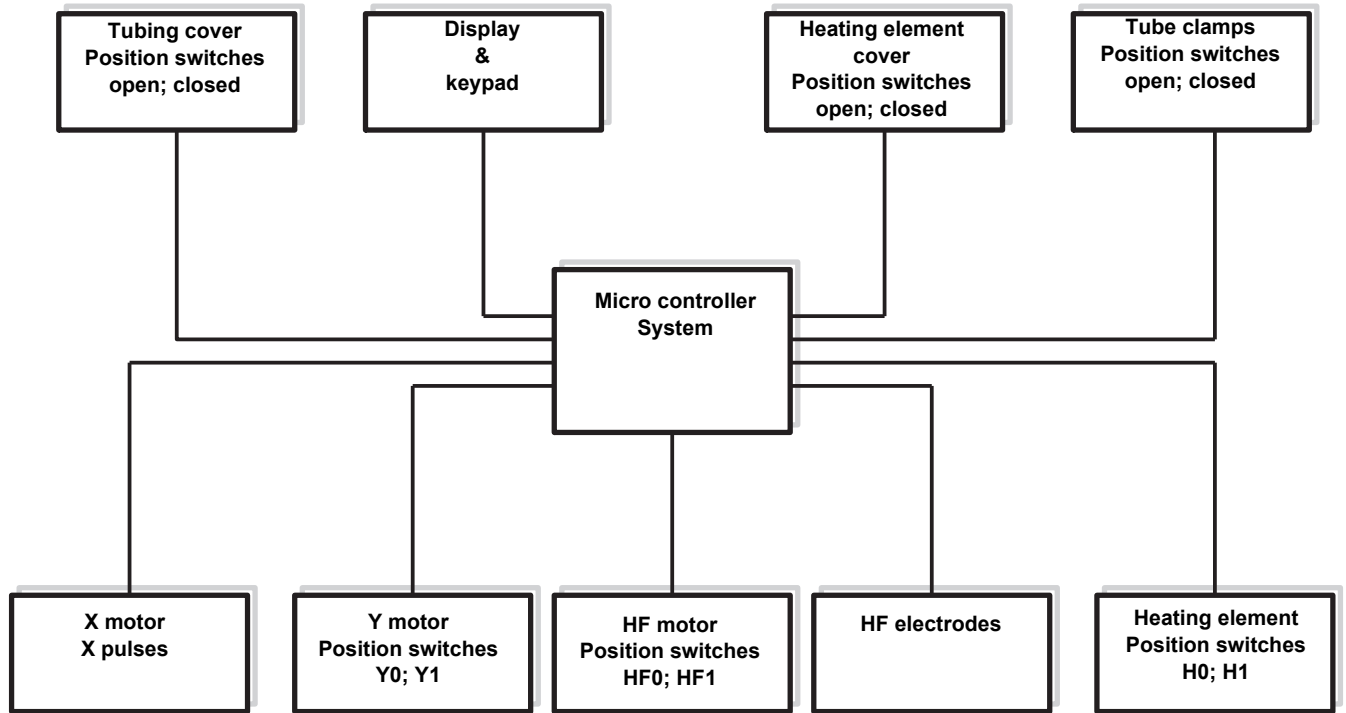
Fax: +49 6172 / 608 8539

E-mail:

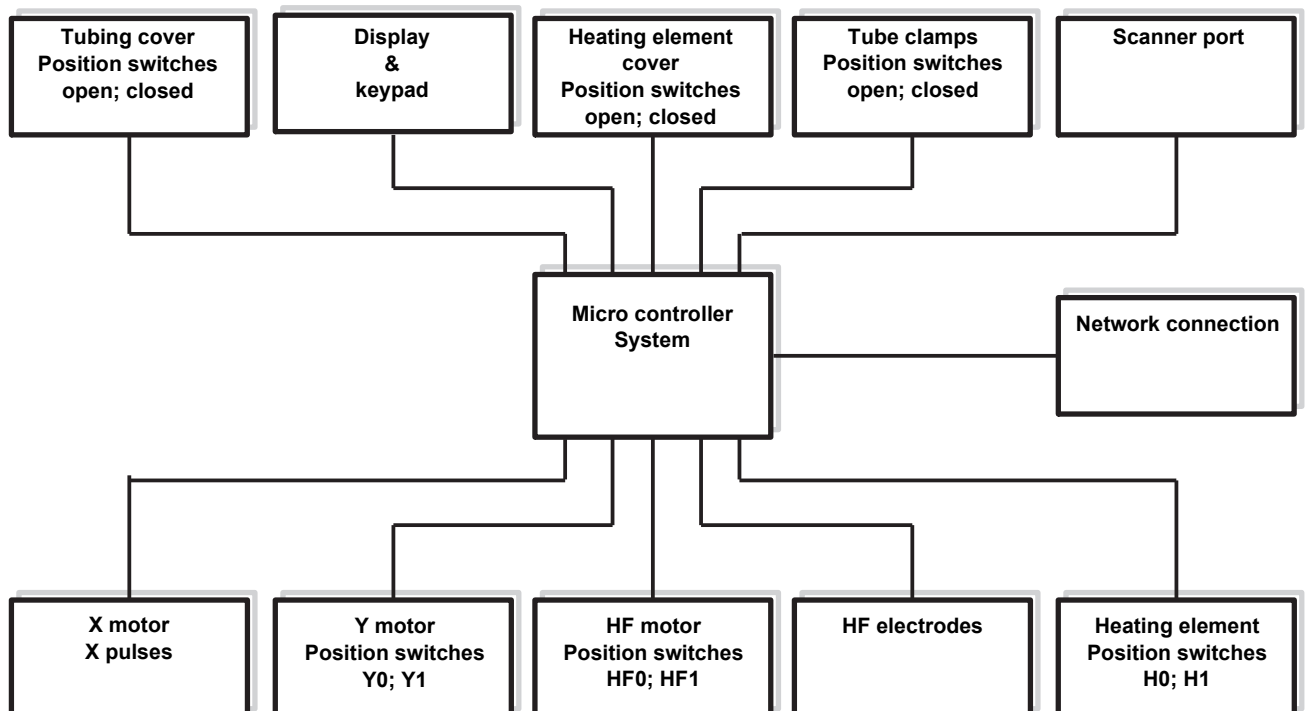
technical_support_medical_device@fresenius-kabi.com

3 Block diagrams

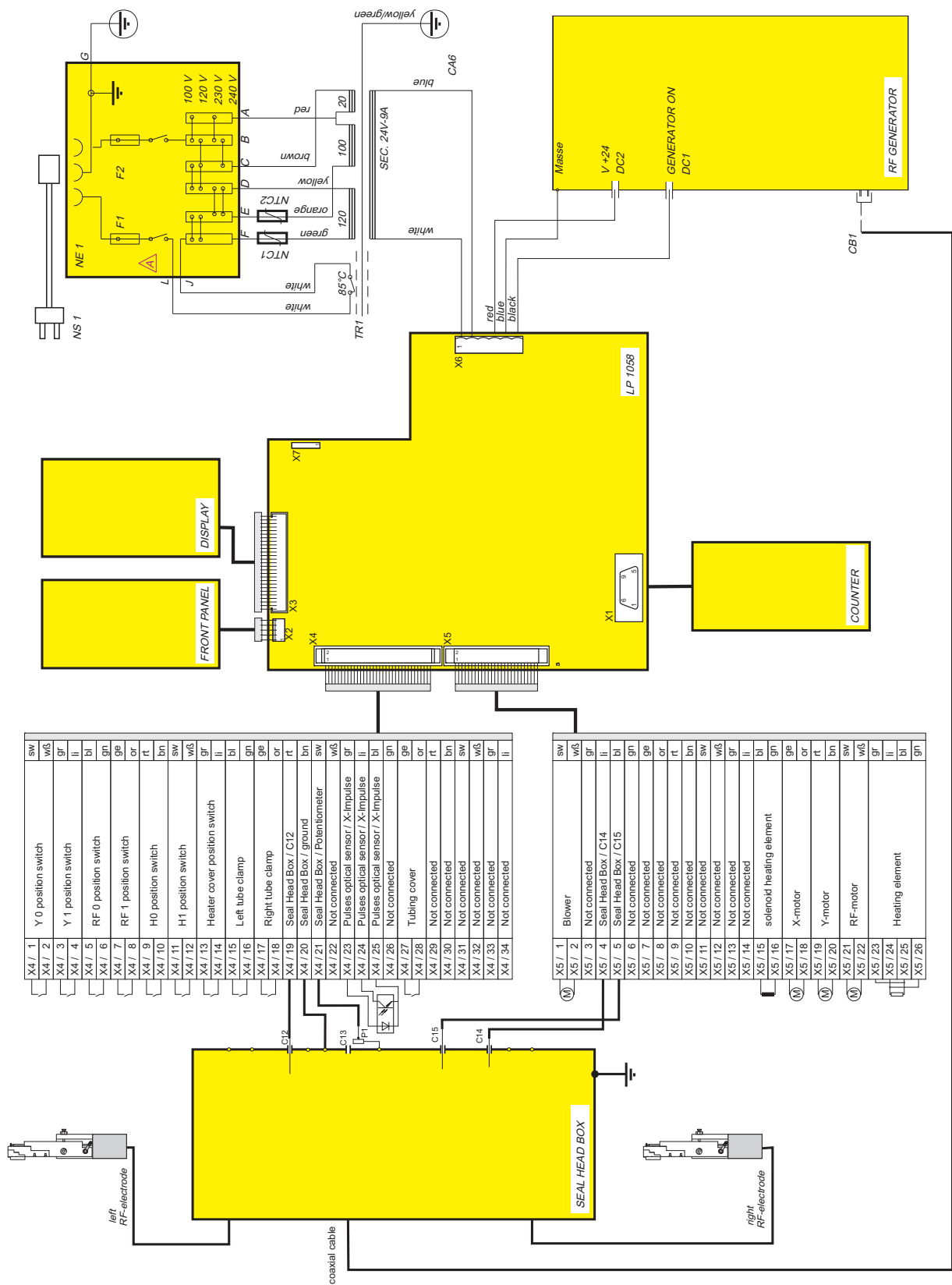
- Functional diagram < E-code 069



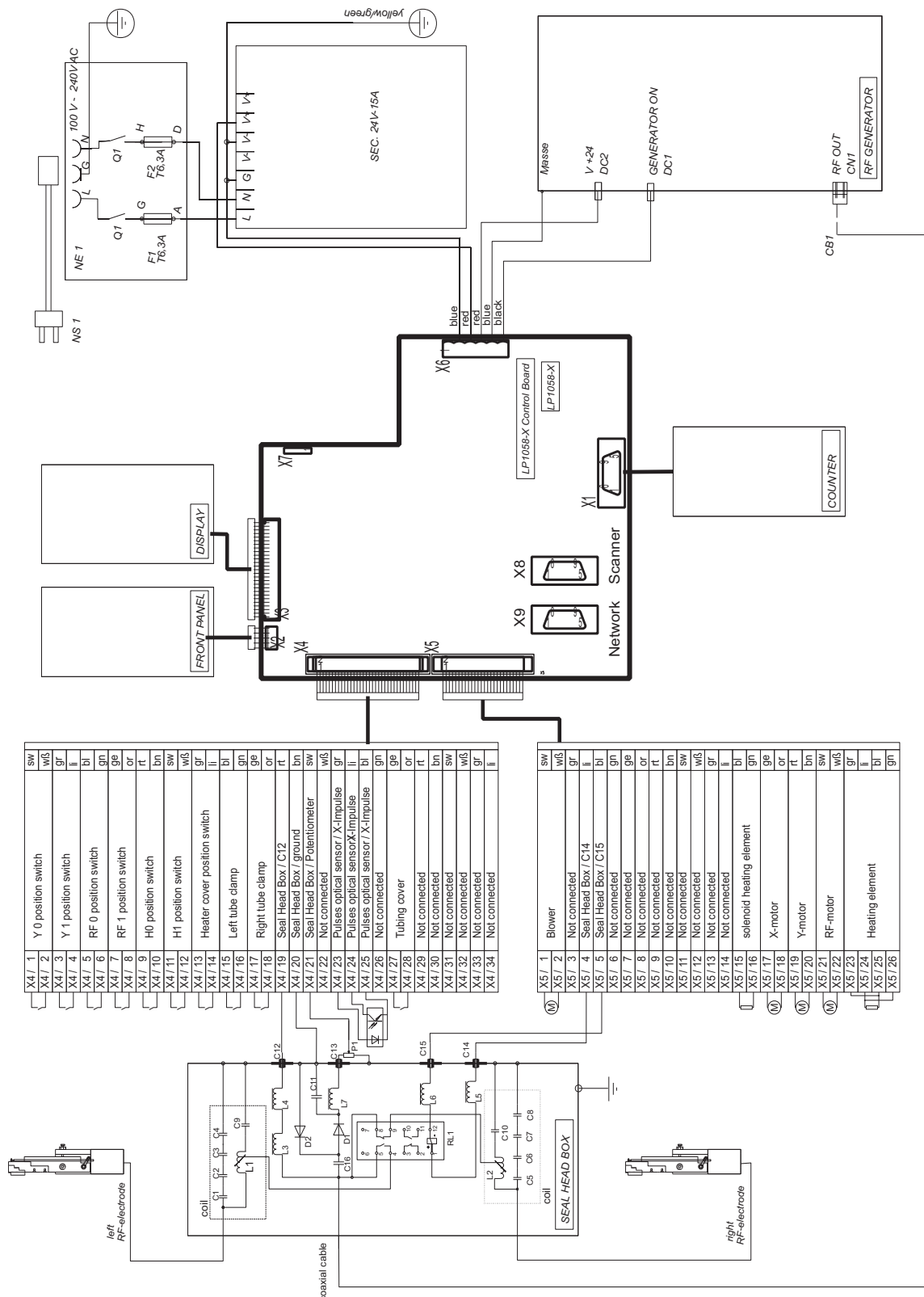
- Functional diagram ≥ E-code 069



● Device block diagram < E-code 069



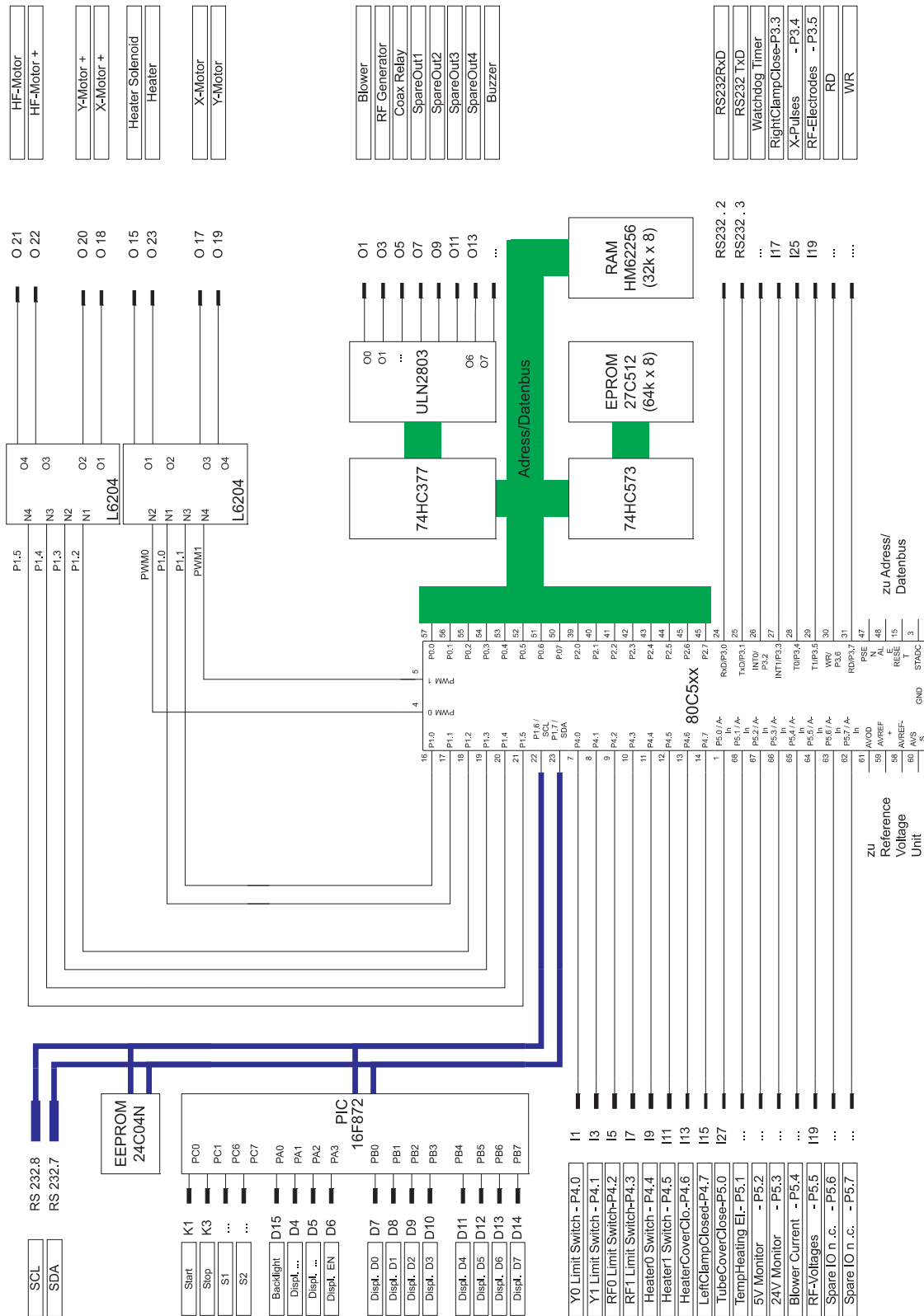
● Device block diagram ≥ E-code 069



LP1058-1 entspricht E-Code ≥ 069

LP1058D entspricht E-Code ≥ 140

● Block diagram, P.C.B. LP1058



Legend:
RS232.X: RS232Plug [X1]; Kxx: KeyPlug [X2]; Dxx: DisplayPlug [X3]; Ixx: InputPlug [X4]; Oxx: OutputPlug [X5]

●



RS232.X: RS232Plug [X1]; Kxx: KeyPlug [X2]; Dxx: DisplayPlug [X3]; Ixx: InputPlug [X4]; Oxx: OutputPlug [X5];

4 Service menu

4.1 Introduction

The CompoDock includes a service menu. Via this menu, it is possible to activate individual movements and to set operating parameters.



Note

The parameters entered must be verified by the operator, i.e. he must check the values entered to ensure they are correct. If the verification reveals a deviation between the desired parameters and the parameters displayed, the setting must be corrected before activating the function.

The actual values displayed must be compared with the desired values specified.

4.2 Preparation

- **Opening the device**

(see chapter 5.1 on page 5-1)

● **Option 2 (with the housing rear panel open):**

- Actuate the main power switch to turn the system on.

The **System check** is performed, the clamps move to initial position.

Message **Searching Dockmaster** appears for approx. 5 sec.

- Simultaneously press service keys S1 and S2 / P.C.B. LP 1058.

The first menu item of the main menu appears in the display.

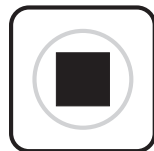
*** Main menu ***
Set Default Values

● **Key functions**



Start key

- Select a submenu
- Activate selected items



Stop key

- Exit service menus
- Exit submenus
- Save parameter values



Key S1 / S2 P.C.B. LP 1058

- Select main menu
- Modify adjustable parameter values
 Press **S2** to reduce a value
 Press **S1** to increase a value

4.4 Main menu

4.4.1 Parameters

	Default values*)	Alarm limits**)	
Menu		Minimum	Maximum
Voltage supply			
5 Volt supply	5 V	4.60 V	5.40 V
E-Code ≥ 140	5 V	4.8 V	5.1 V
Voltage supply			
24 Volt supply	24 V	23.65 V	24.31 V
Blower current			
Blower current	15 mA	8.6 mA	20.9 mA
E-Code ≥ 140	-----	28.6 mA	98.1 mA
Xmotor Outside pos			
time (= X motor in outer position)			
< E-code 030	95 counts*)	0.2 s	1.5 s
$\geq$ E-code 030 or $\geq$ software 2.2x	100 counts*)	0.2 s	1.5 s
Y motor			
time (= Y motor)	Front: 2.0 s Back: 2.0 s	0.8 s	2.5 s
HF Motor			
time (= HF motor)	close 0.8 s: open: 0.8 s	0.5 s	1.7 s
Dock cooldown time			
time (= sealing seam cooldown time)	4 s	-	-
Xmotor pos.dock			
time xxx counts	628*) counts	0.8 s	1.0 s
Heater dock temp.			
heater counts (= heater element temperature)	120*) counts 465 °C $\pm$ 5 °C (China version: 470 °C $\pm$ 5 °C)	-	-
Tube heating time			
time (= sealing seam heating time)	12.0 s	-	-

	Default values*)	Alarm limits**)	
Menu		Minimum	Maximum
Xmotor Heat pos °) time xxx counts (= X motor in heating position)	555*) counts	0.8 s	2.6 s
heating Parameter			
Offset Heating Pos	30 counts	-	-
Pre heating Time	4 s	-	-
X-MotorSpeed Heat	speed 015*)	-	-
Seal stretch Tubes			
counts	5 counts	-	-
Seal Times			
Pulse time	026	-	-
Hf-Motor Move In	999	-	-
Hf-Motor Move Time	000	-	-
Pausetime	008	-	-

*) Depending on the device and variable, the actual value may deviate slightly.

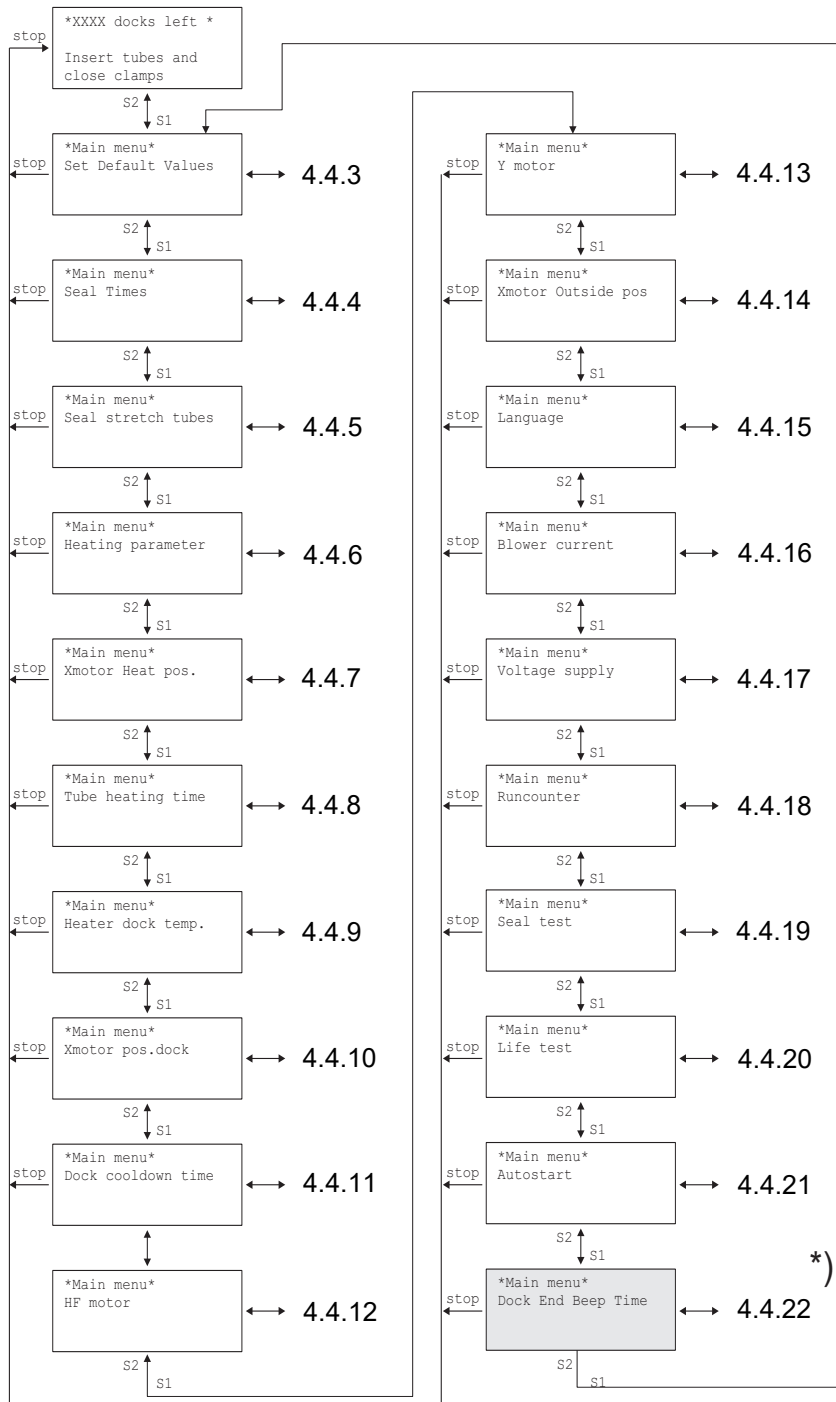
**) Do not enter any parameter values outside the adjustment range.

°) This is the only menu option in which the heating element is not active.

The other values must not be changed.

4.4.2 Flow diagram

● Main menu



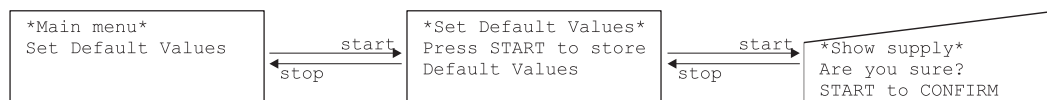
*) E-code 069 and higher

● Values shown in the flow diagrams

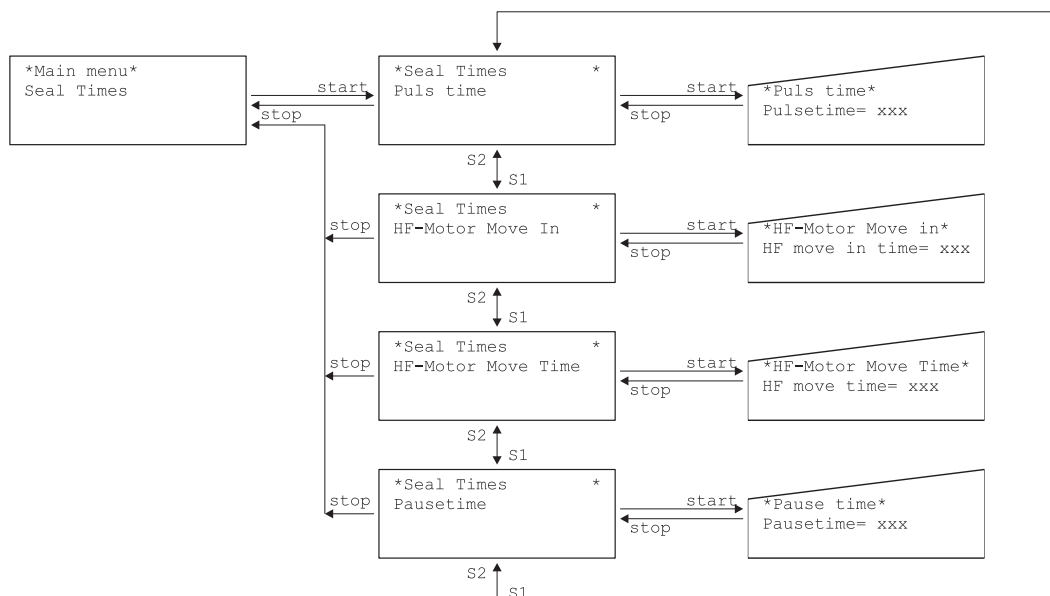
YY: values set, actually measured

XX: values adjustable via the service keys

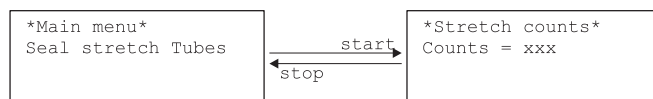
4.4.3 Set Default Values



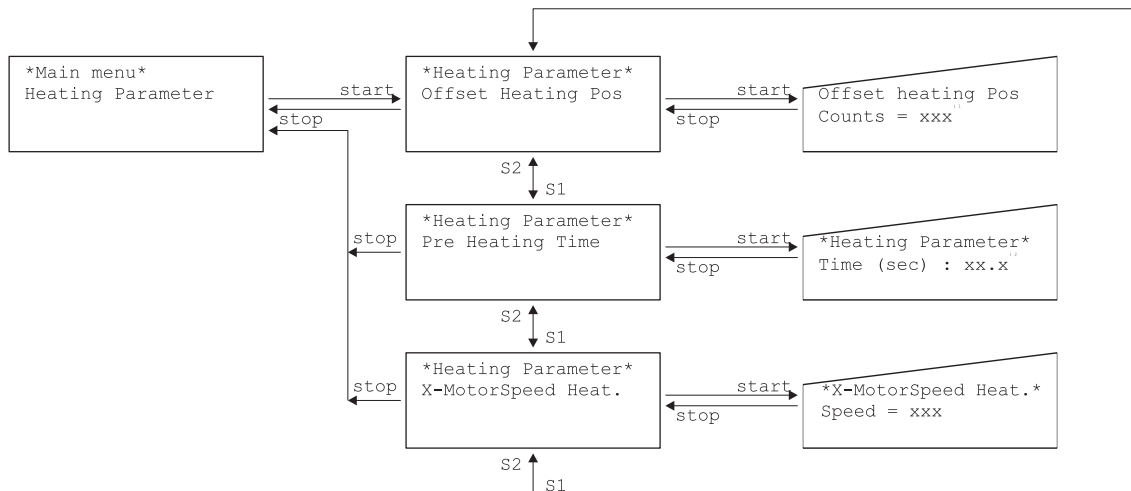
4.4.4 Seal times



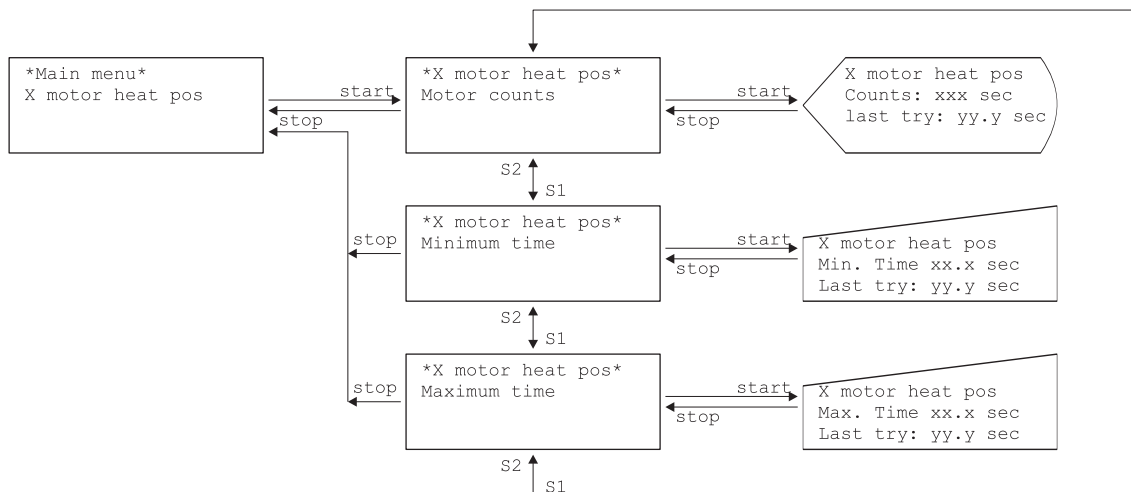
4.4.5 Seal stretch tubes



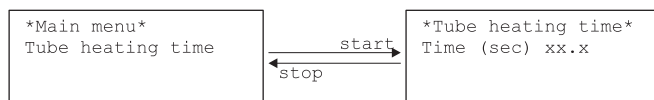
4.4.6 Heating Parameter



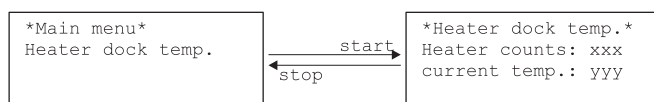
4.4.7 Xmotor Heat pos



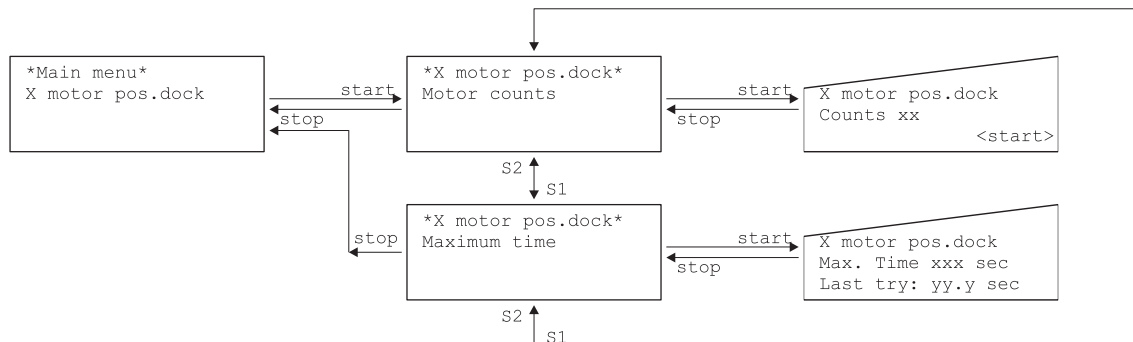
4.4.8 Tube heating time



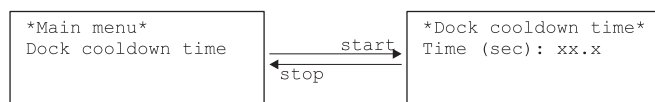
4.4.9 Heater dock temp.



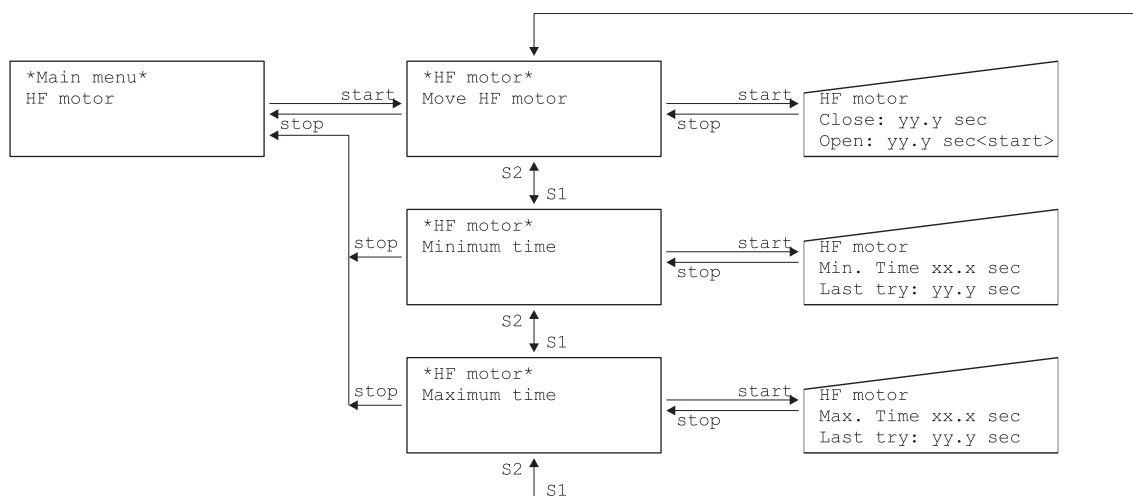
4.4.10 Xmotor pos.dock



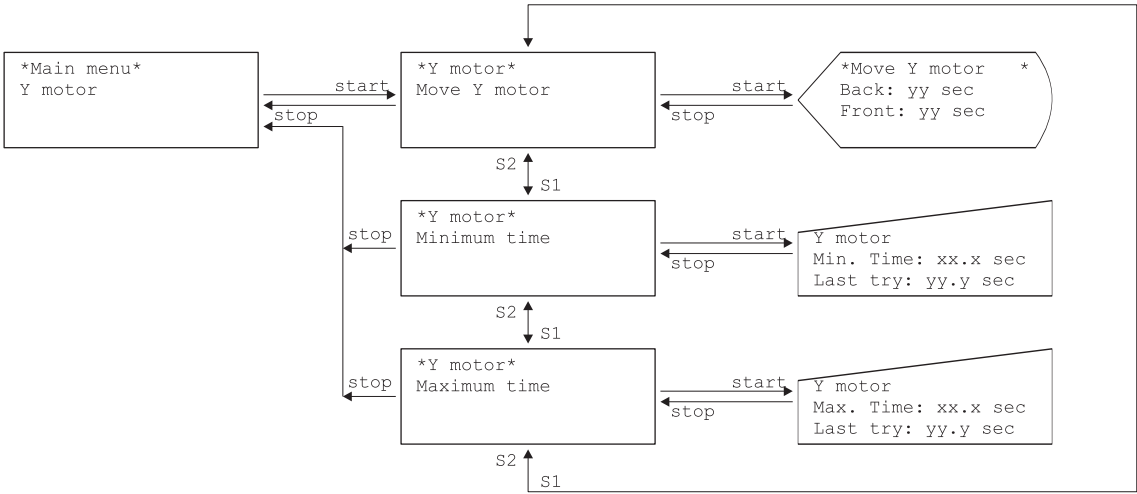
4.4.11 Dock cooldown time



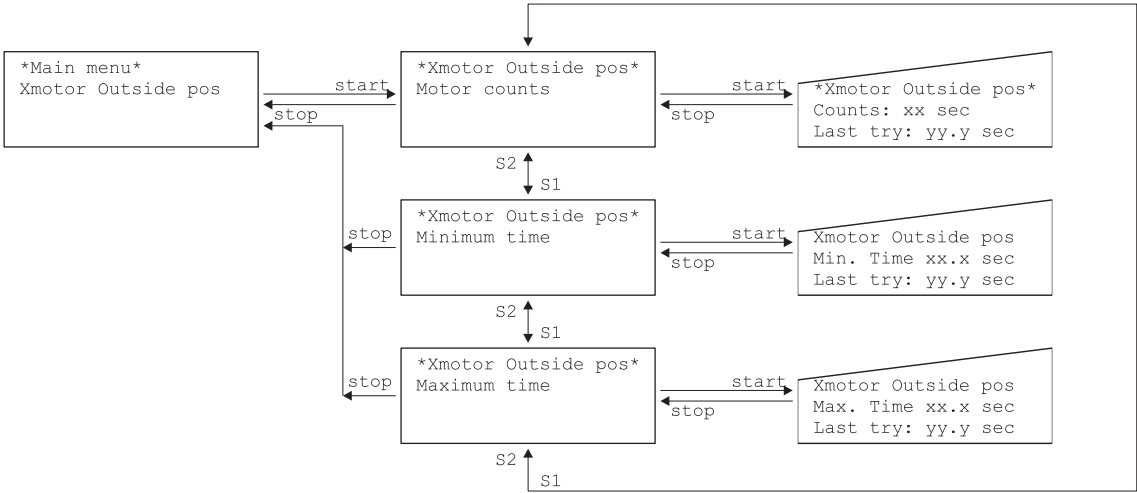
4.4.12 HF motor



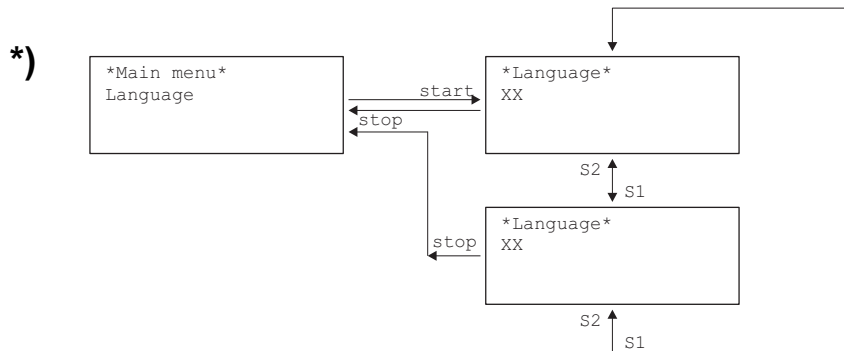
4.4.13 Y motor



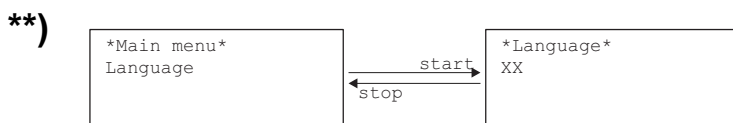
4.4.14 Xmotor Outside pos



4.4.15 Language



Displays the installed language.



The language can only be changed via the **CompoDock ServiceTool**.

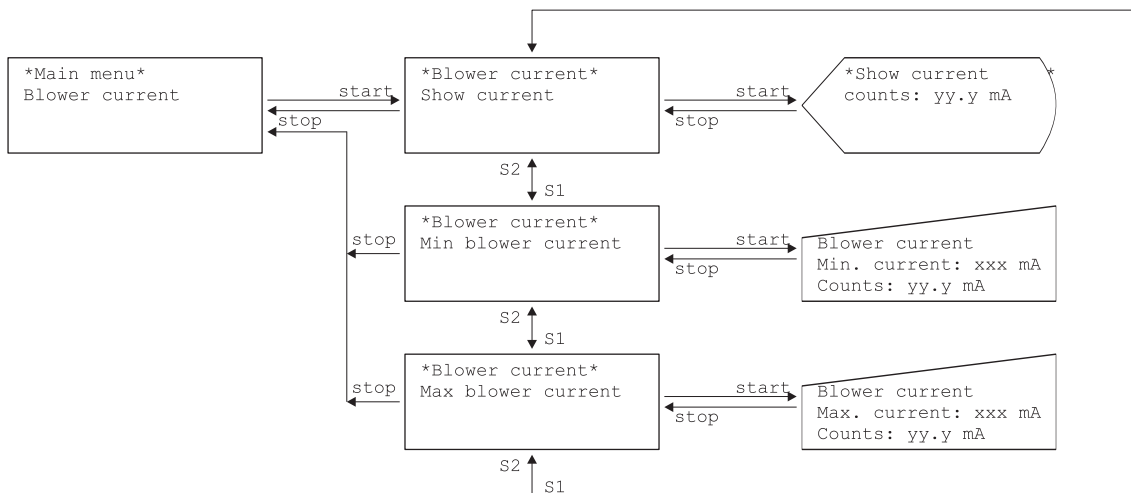
***) Starting from E-Code 140 all languages are available

*) < E-code 069

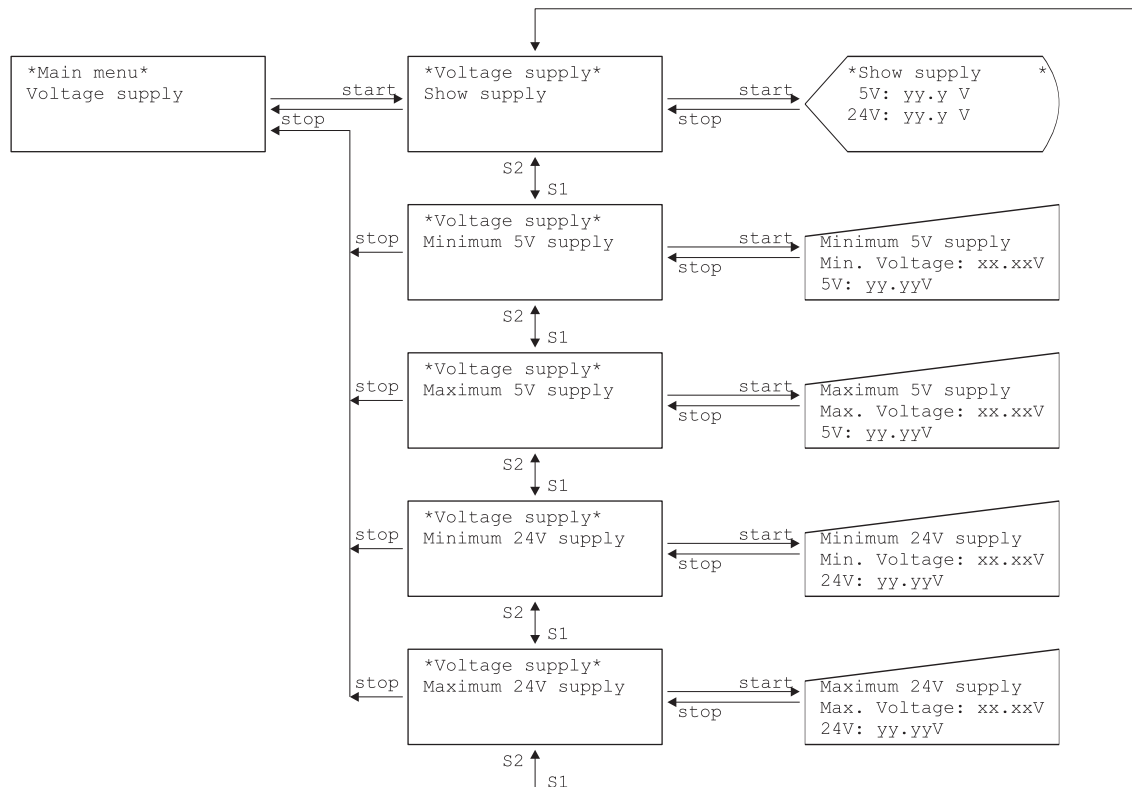
**) ≥ E-code 069

***) ≥ E-Code 140

4.4.16 Blower current

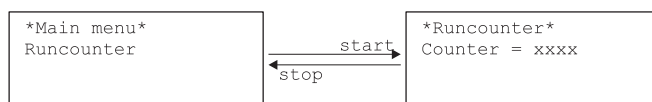


4.4.17 Voltage supply



4.4.18 Runcounter

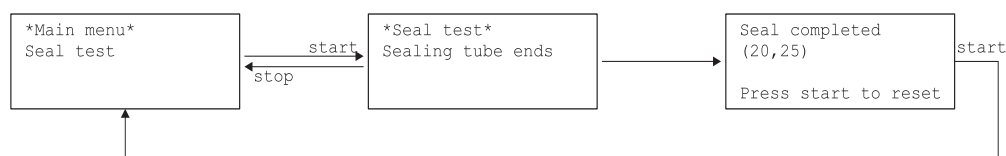
In the Runcounter menu, the number of docks effected since the last reset is displayed.



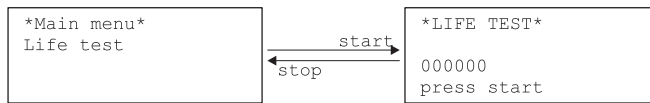
Resetting the counter

- In the Runcounter menu, press the **Start** key
The counts will be displayed.
- Simultaneously press the **S1** and **S2** keys, in order to reset the counter to zero.

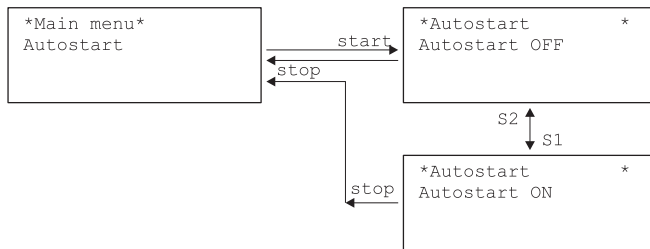
4.4.19 Seal test



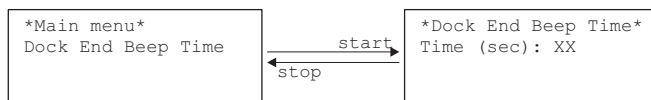
4.4.20 Life test



4.4.21 Autostart



4.4.22 Dock End Beep Time



5 Repair / adjustment / maintenance

5.1 Opening the device

● Opening the rear panel

The diagrams may vary slightly depending on the E-code

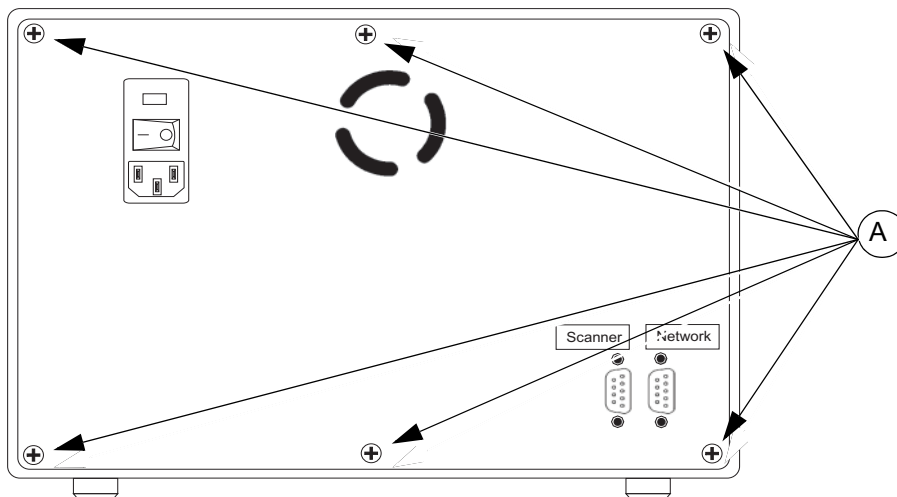


Caution

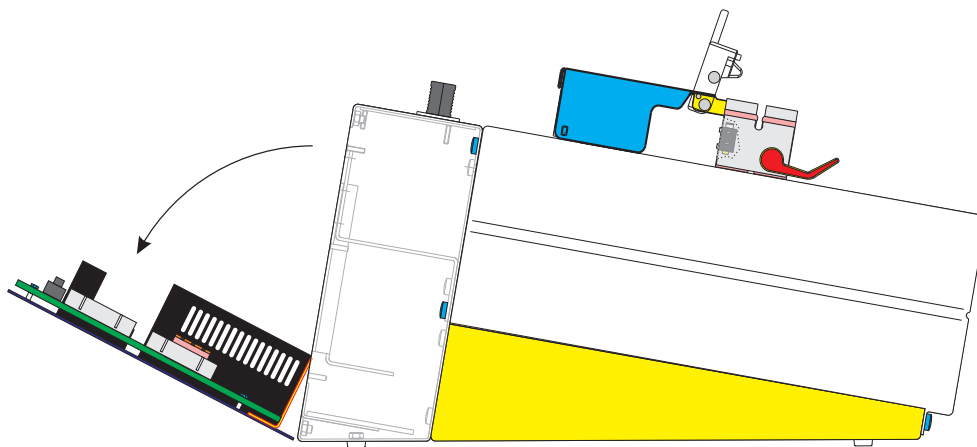
Electric shock!

Turn the device off, remove the power plug!

- Remove the six screws (A) M5 x 6 mm on the rear panel.



- Hinge the rear panel down.



● **Closing the base plate and the rear panel**



Note

The cables must be laid without any mechanical tension and without kinks! Make sure not to jam the cables.

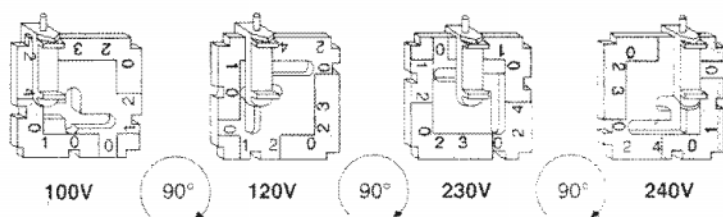
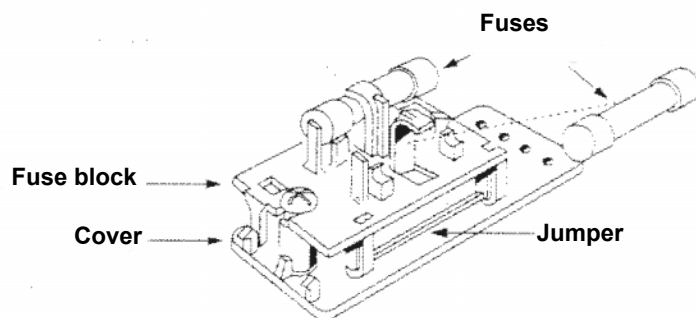
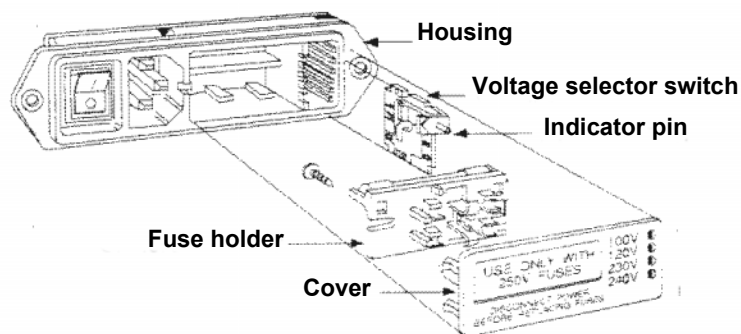
- Ensure proper cable routing when closing the base plate and the rear panel.

5.2 Power supply

5.2.1 Power connection < E-code 069

The default setting of the CompoDock line voltage is 230 V.

Adjust the white voltage indicator pin of the voltage selector switch to the voltage required.



5.2.2 Power connection ≥ E-code 069

The CompoDock is fitted with a wide-range power supply unit.

The voltage range is 100-240 V / 50-60 Hz.

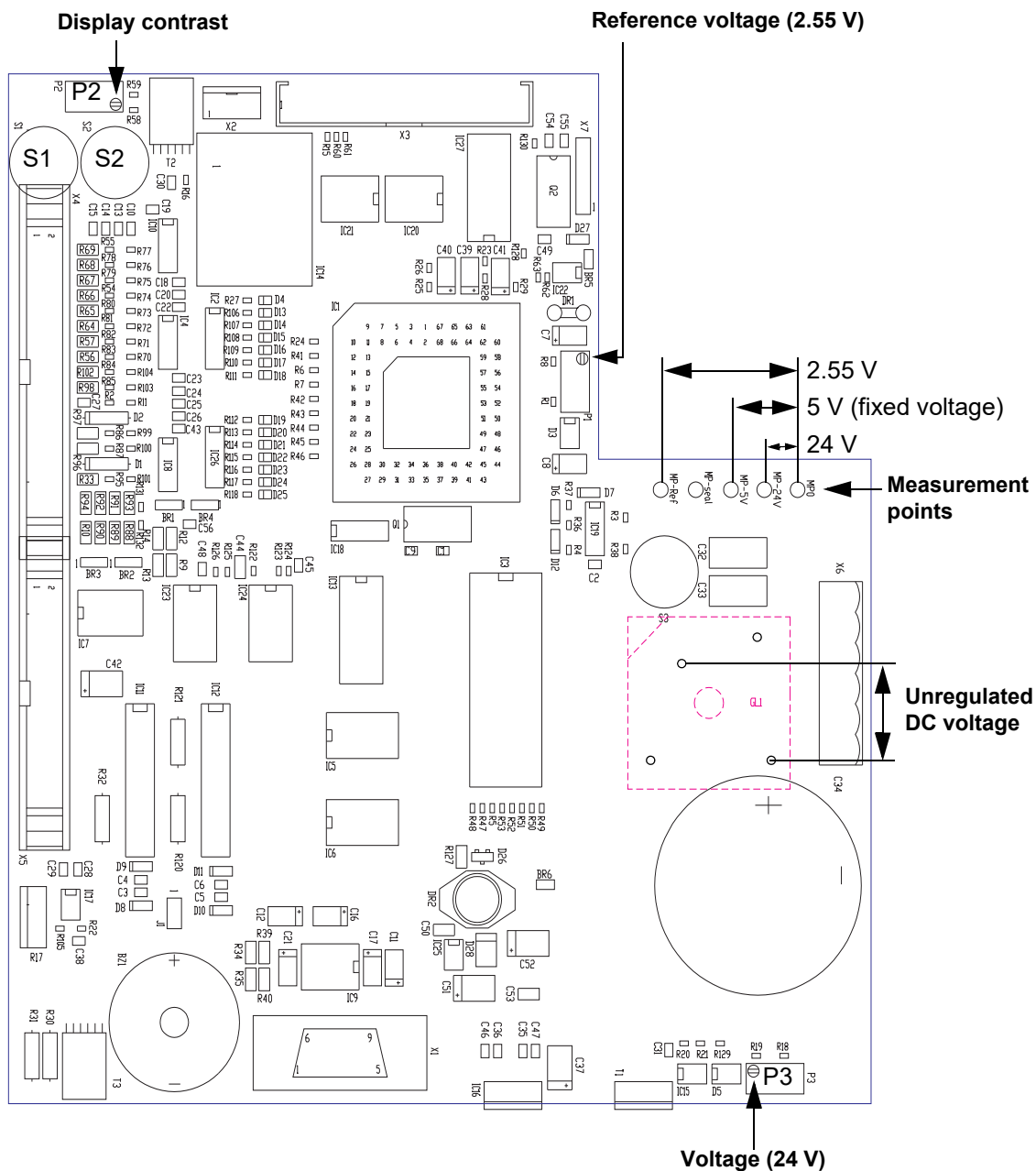
5.2.3 Power connection ≥ E-code 140

The CompoDock is fitted with a wide-range power supply unit.

The voltage range is 100-240 V / 50-60 Hz.

5.3 DC voltage supply

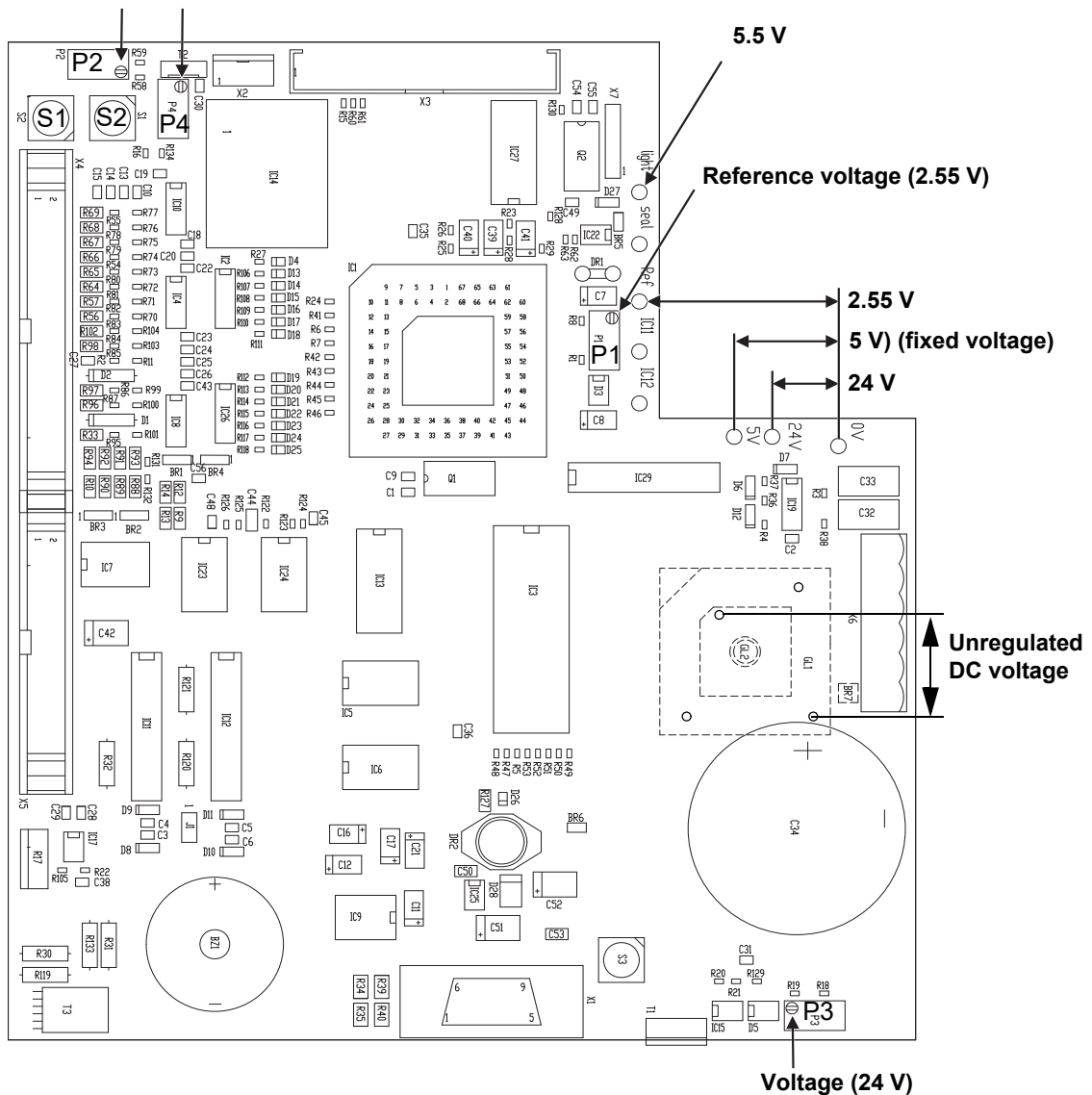
5.3.1 Measurement and adjustment points P.C.B. LP 1058 (< E-code 069)



5.3.2 Measurement and adjustment points P.C.B. LP 1058-1

Replacement for
P.C.B. LP 1058

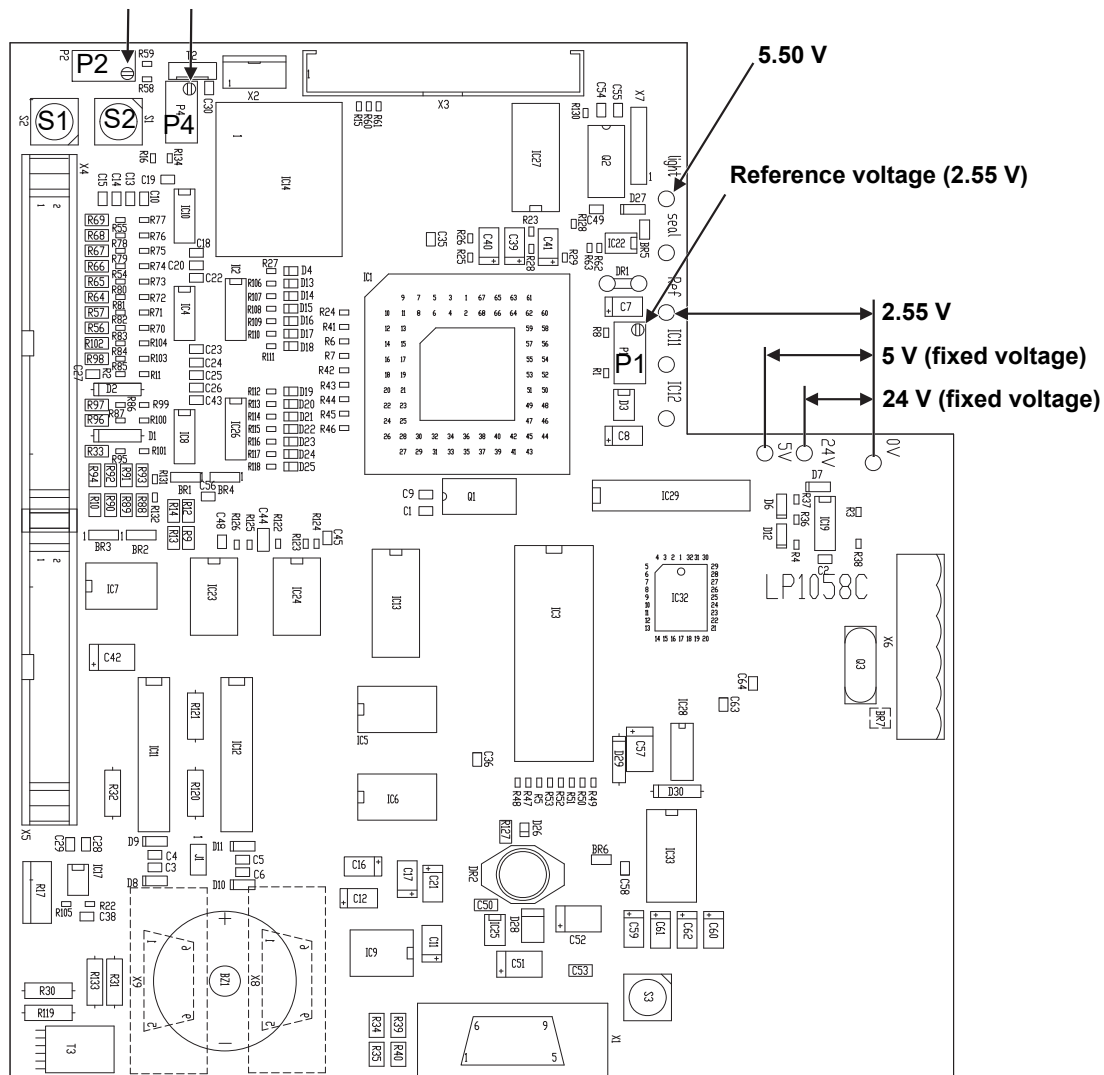
Display contrast Brightness of display backlighting (5.5V)



5.3.3 Measurement and adjustment points P.C.B. LP 1058-1 ($\geq$ E-code 069)

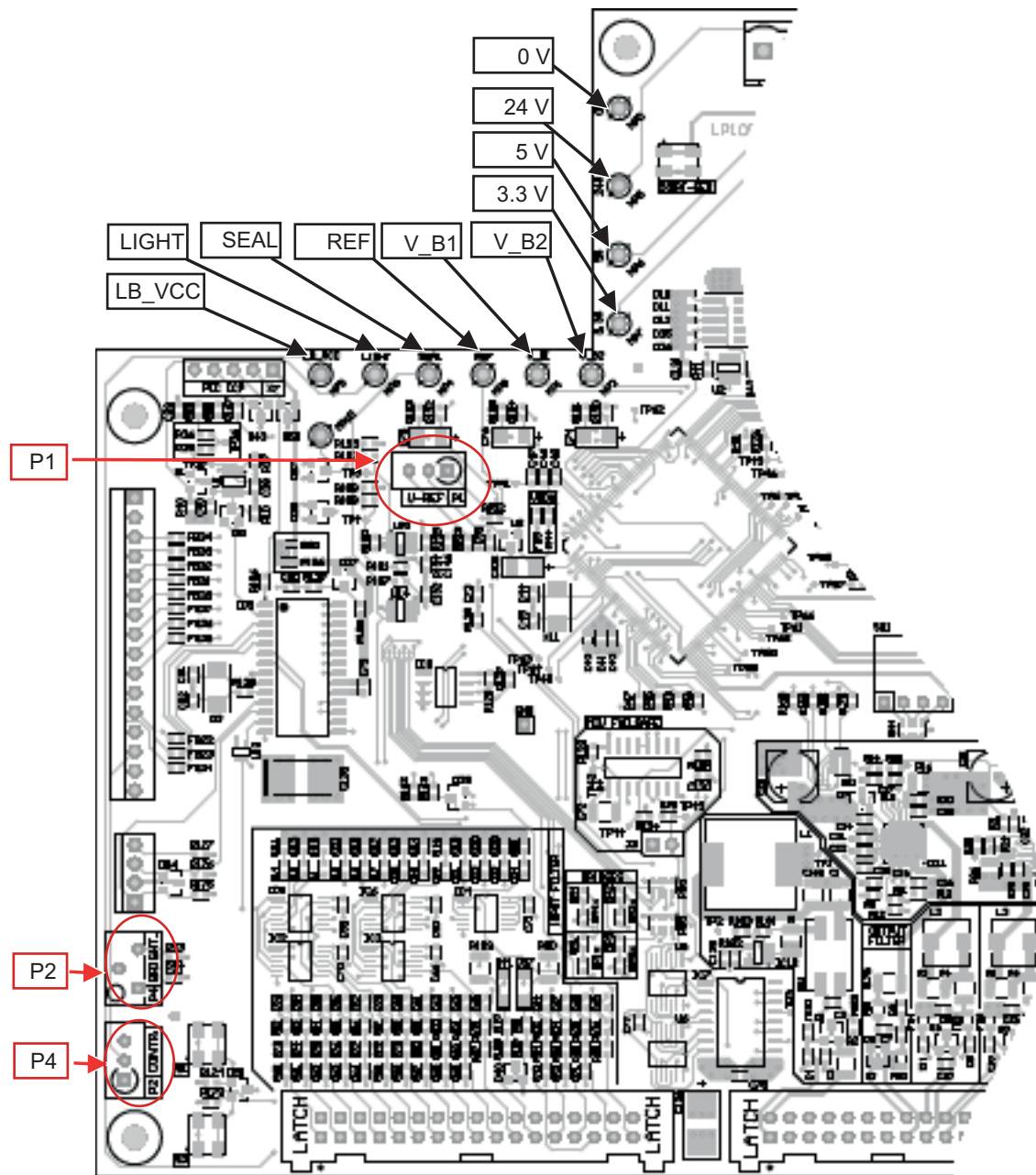
LP 1058-1 with interface

Display contrast Brightness of display backlighting (5.50 V)



5.3.4 Measurement and adjustment points P.C.B. LP 1058D ($\geq$ E-code 140)

LP 1058D with interface



5.3.5 Unregulated DC voltage < E-code 069

- **Unregulated DC voltage**

- Measurement points: positive pole of the bridge rectifier / negative pole of the bridge rectifier
- Desired value: 31 V – 35.5 V



Caution

With < E-code 069 devices, the negative pole of the bridge rectifier is not connected to the ground. The 0-voltage is regulated by the power supply FET (T3). Floating ground is used!

5.3.6 Unregulated DC voltage < E-code 140

- **Unregulated DC voltage**

- Check 3.3V; measurement point MP7
- Desired value: 3,18V - 3,42V

5.3.7 Regulated DC voltage

- **5 V voltage**

- Measurement points: MP 0 / MP 5 V
- Desired value: 5 V $\pm$ 0.1 V

- **24 V voltage**

- Measurement points: MP 0 / MP 24 V
- Desired value: 23.65 V - 24.31 V
- Use P3/P.C.B. LP1058 to set the 24V voltage to 24.0 V. **Not applicable to P.C.B. LP1058-1 with interface!**

- **5 V / 24 V voltage**

- Select the **Voltage supply** main menu.
- Desired value 5 V: 5 V $\pm$ 0.1 V
- Desired value 24 V: 23.65 V - 24.31 V

5.3.8 Reference voltage

- Measurement points: MP 0 / MP Ref
- Desired value: $2.55\text{ V} \pm 0.001\text{ V}$
- Use P1 / P.C. B. LP 1058 to set the reference voltage.

5.3.9 Display contrast

- Use P2 / P.C. B. LP 1058 to set the display contrast.

5.3.10 Display backlighting (only with P.C.B. LP 1058, layout C)

- Set the display background lighting using P4 / P.C.B. LP 1058.
- Measurement points: MP 0 / light
- Desired value: $5.50\text{ V} \pm 0.1\text{ V}$

5.3.11 Display backlighting (only with P.C.B. LP 1058D, layout D)

- Set the display background lighting using P4 / P.C.B. LP 1058D.
- Measurement points: MP 0 / light
- Desired value: $100\text{ mV} \pm 0.1\text{ V}$

5.3.12 Blower current



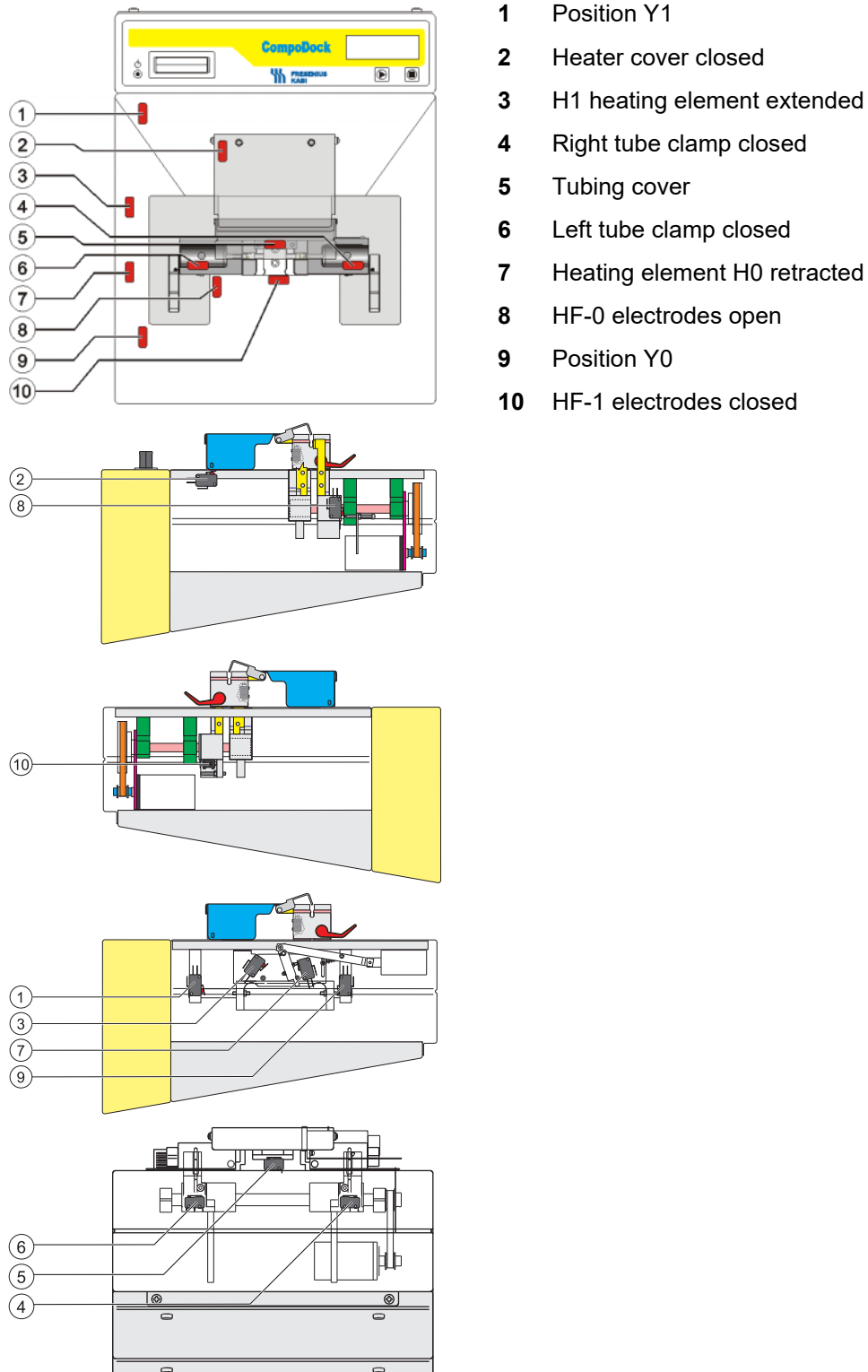
Note

Obstructions in the air stream will affect the blower current measurement. Measurements must be performed with the device assembled (rear panel fitted).

- Select the **Blower current** main menu.
- Desired value: 8.6 mA to 20.9 mA (E-Code 069)
- Desired value: 28.6 mA to 98.1 mA (E-Code 140)

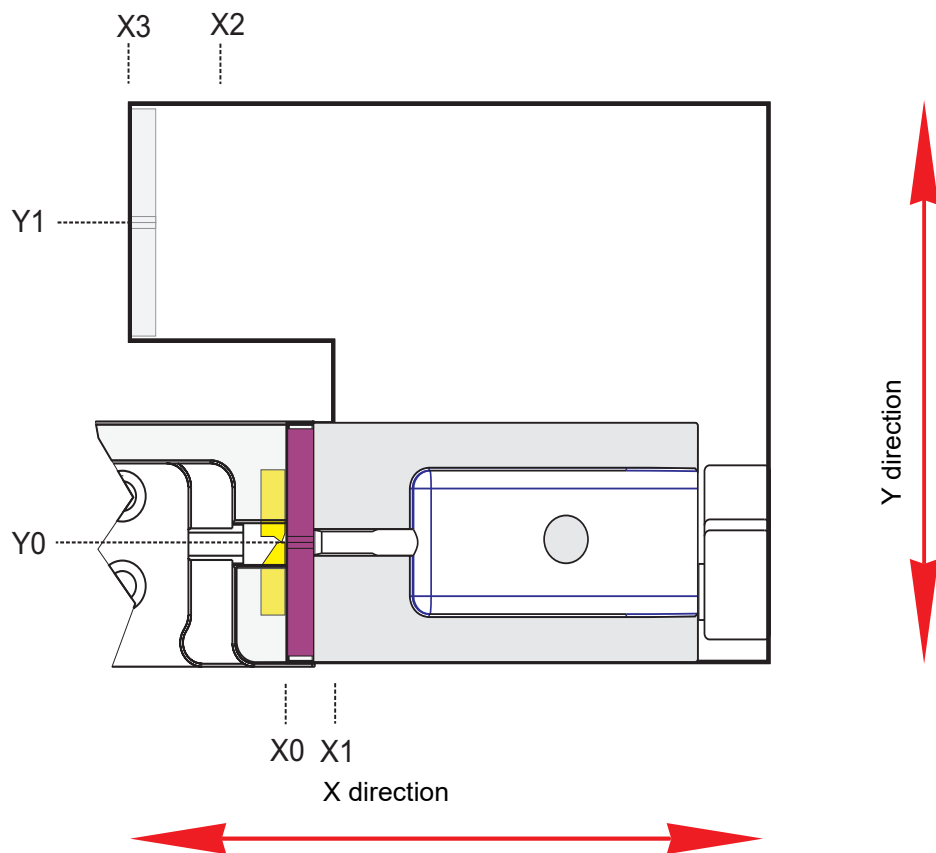
5.4 Position switches

5.4.1 Layout of the position switches



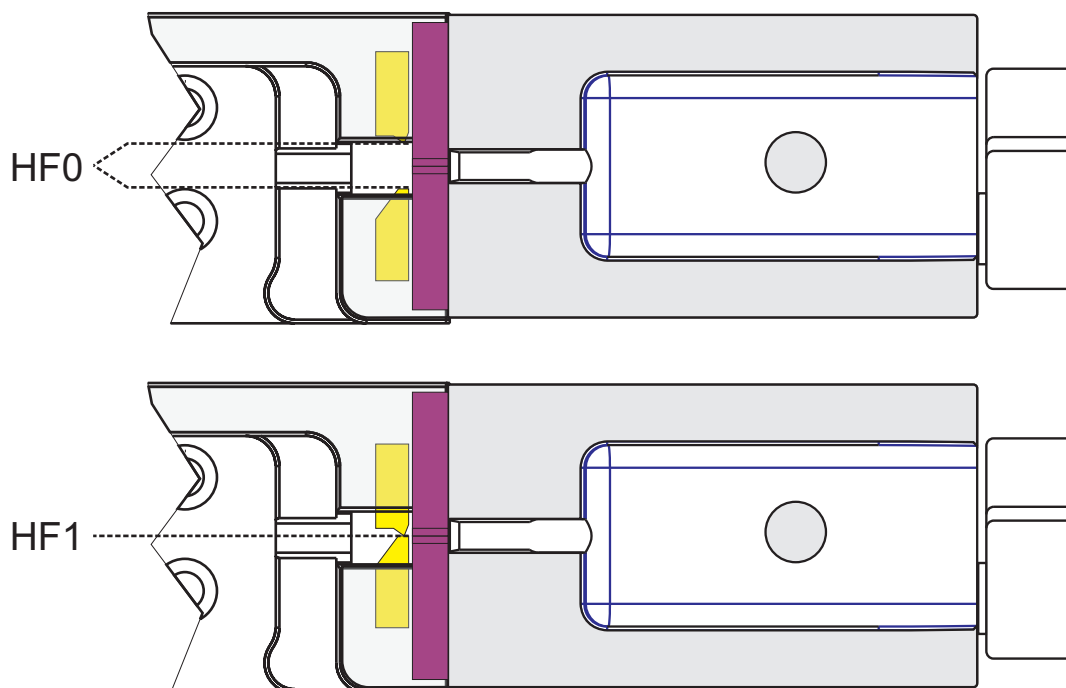
5.4.2 Tube clamp positions

Positions		Menu
X0 / Y0	Initial position of the tube clamps	Y motor
X1 / Y0	Outer left and right position of the tube clamps	Xmotor outside pos
X2 / Y1	Heating position of the tube clamps	Xmotor heat pos
X3 / Y1	Docking position of the tube clamps	Xmotor pos. dock



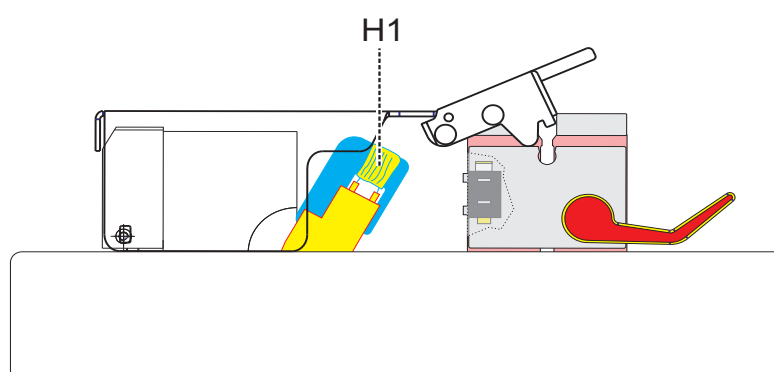
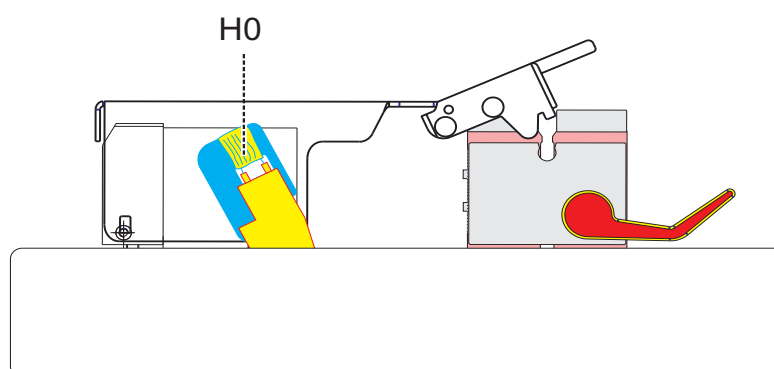
5.4.3 Electrode positions

Positions		Menu
HF0	Electrodes open	HF motor
HF1	Electrodes closed	HF motor

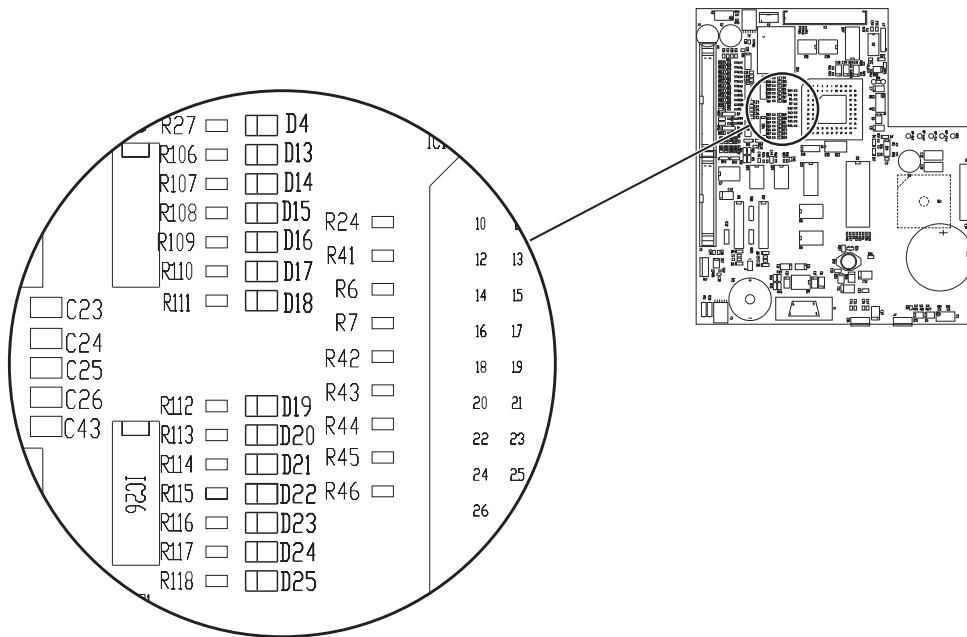


5.4.4 Heating element positions

Positions	
H0	Heating element retracted
H1	Heating element extended

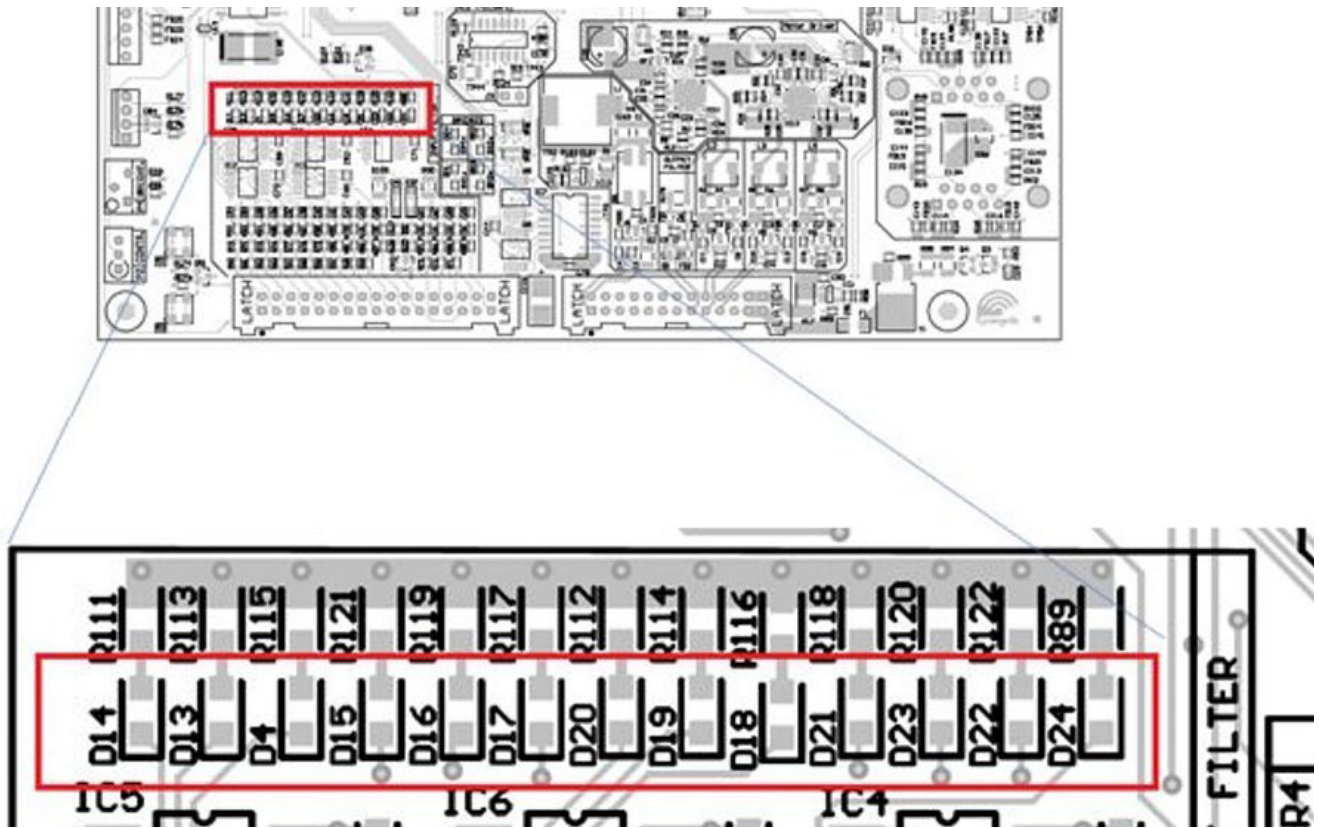


5.4.5 Light-emitting diodes P.C.B. LP 1058



Light-emitting diode	Description
D4	Position Y0
D13	Position Y1
D14	HF-0 electrodes open
D15	HF-1 electrodes closed The electrodes touch each other without bias voltage
D16	Heating element H0 retracted
D17	Heating element H1 retracted
D18	Heater cover
D19	Tube clamp, left
D20	Tube clamp, right
D21	HF short circuit The electrodes are biased
D22	HF voltage
D23	Pulses optical sensor X direction
D24	Tubing cover
D25	Heating element function

5.4.6 Light-emitting diodes P.C.B. LP 1058D



Light-emitting diode	Description
D4	Position Y0
D13	Position Y1
D14	HF-0 electrodes open
D15	HF-1 electrodes closed The electrodes touch each other without bias voltage
D16	Heating element H0 retracted
D17	Heating element H1 retracted
D18	Heater cover
D19	Tube clamp, left
D20	Tube clamp, right
D21	HF short circuit The electrodes are biased
D22	HF voltage

Light-emitting diode	Description
D23	Pulses optical sensor X direction
D24	Tubing cover
D25	Heating element function

5.5 Visual inspection / functional test



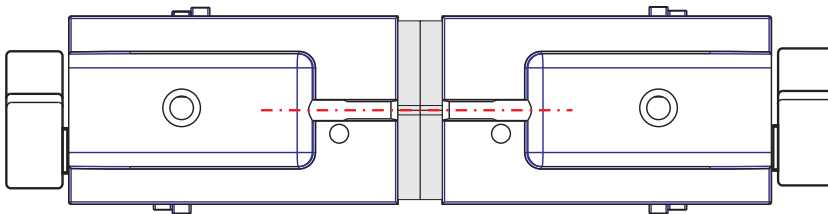
Note

Checks must be performed before or after maintenance works or setting of all items described in the chapter.

5.5.1 Checking the position of the tube clamps

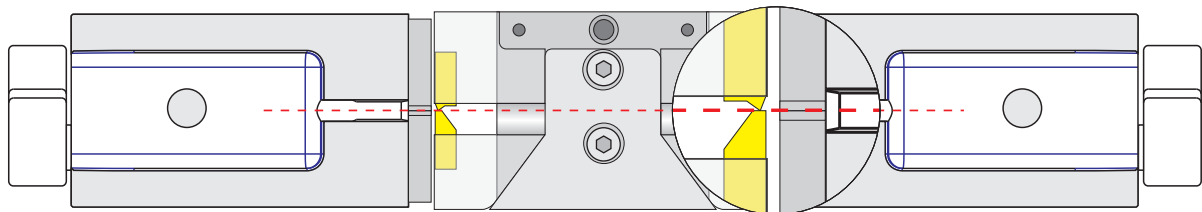
5.5.1.1 Checking the alignment of the tube clamps in docking position

- In initial position (X0/Y0), select the **Xmotor pos. dock** service menu and move the tube clamps to the docking position.
In the docking position the closed clamping jaws must be in one line.



5.5.1.2 Checking the alignment of the tube clamps with the electrodes

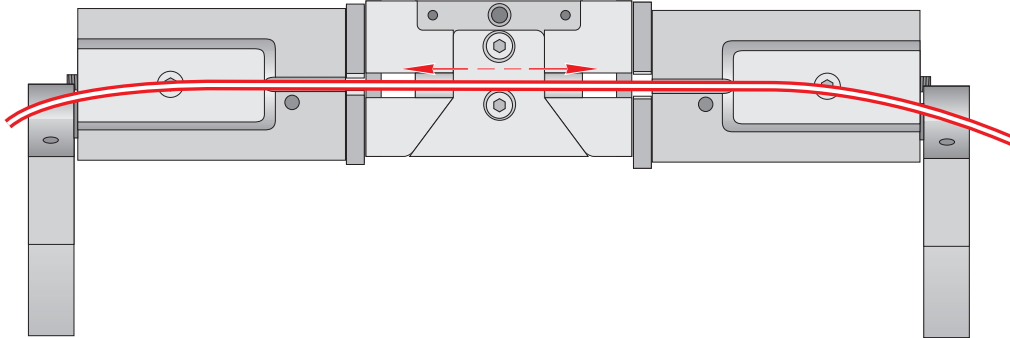
- In initial position (X0/Y0), select the **HF motor** service menu and close the electrodes.
The closed tube clamps and the closed electrodes are in one line.



5.5.1.3 Checking the alignment of the tube clamps with the central block

- In initial position (X0/Y0), select the **HF motor** service menu and open the electrodes.
- Open the tube clamps
- Insert a 4.2 mm tubing (part -no. M60 024 1) (China version: 4.55 mm tubing (part-no. M600251))

The tubing must be positioned in the guides in a straight line and must be moved easily in axial direction.

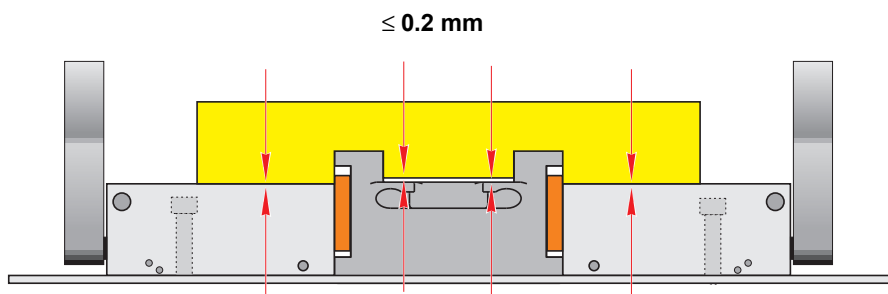


5.5.1.4 Checking the alignment of the tube clamp height position

In initial position (X0/Y0), the adjusting gauge is positioned evenly on the tube clamps and on the central block.

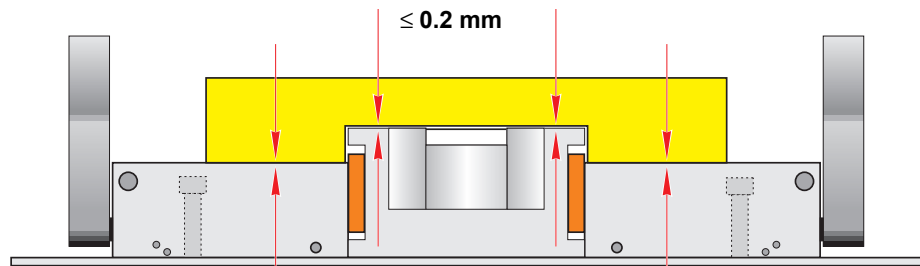
Applicable to tube clamps <E-code 030:

- Use adjusting gauge (part no. M651681)



Applicable to tube clamps > E-code 030:

Use adjusting gauge (part no. M659261)



5.5.2 Performing a sealing test

(see chapter 5.14 on page 5-62).

5.5.3 Performing a docking test

(see chapter 5.15 on page 5-63).

A docking test must be performed only on completion of all maintenance items.

5.5.4 Reading out parameters / saving parameters with service tool

Before performing maintenance measures or replacing the P.C.B., read out all parameter values from the service menu and record them in the maintenance report:

After replacing the P.C.B., readjust the following parameter values in the service menu. Then perform a visual inspection/functional check.

- Xmotor Outside pos
- Xmotor pos.dock
- Heater dock temp.
- Xmotor heat pos
- Seal-stretch-tubes Standard: 005 (China: 010)
- heating Parameter/X-MotorSpeed Heat.

If it is not possible to read out the parameter values from the service menu, the above-mentioned values must be determined or re-adjusted. The default setting will be used for all other parameter values. Then perform a visual inspection/functional check.

5.6 Tubing cover

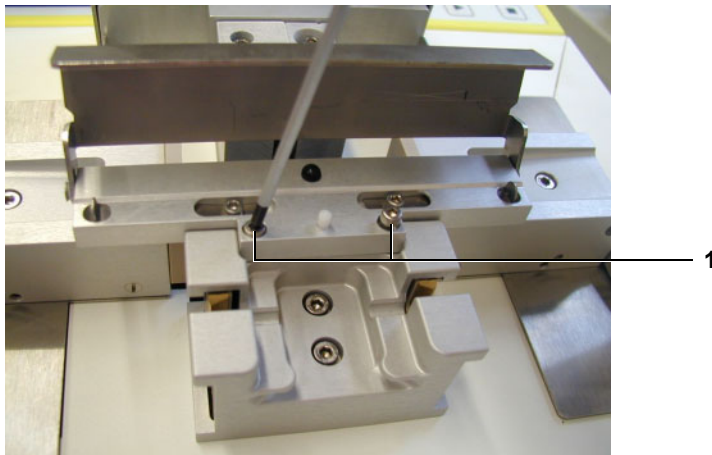
5.6.1 Tubing cover switch < E-code 030

● Test

- Insert one adjusting pin $\varnothing$ 4.0 mm (part no. 644 206 1) each into the insertion groove of the two tube clamps. Close the tubing cover. The position switch must switch (LED D24 / P.C.B. LP 1058 is illuminated).
- Insert one adjusting pin $\varnothing$ 4.3 mm (part no. M65 193 1) each into the insertion groove of the two tube clamps. Close the tubing cover. The position switch must not be switching.

● Removal

- Turn the device on.
- Simultaneously press the **Start** and the **Stop** key.
The device moves to the **Clean Position**.
- Open the heating element cover.
The clamps will move outwards.
- Remove the 2 Allen screws (1) of the tubing cover and remove the tubing cover.
- Remove the switching pin and the spring.

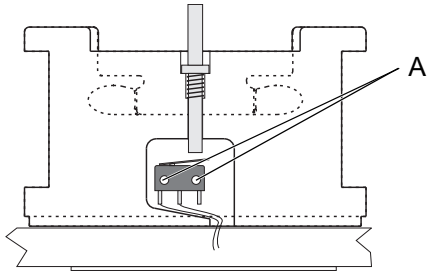


● Installation

- Simultaneously press the **Start** and the **Stop** key.
The device moves to the **Clean Position**.
- Open the heating element cover.
The clamps will move outwards.
- Insert the spring and the switching pin into the central block.
- Attach the tubing cover using the 2 Allen screws (1).

● **Adjustment**

- Insert the adjusting pin Ø 4 mm (part no. 644 206 1) into the groove of the two tube clamps. Close the tubing cover. Adjust the position switch Y0 so that it switches just in that moment (LED D4 / P.C.B. LP 1058 is illuminated).
- Insert the adjusting pin Ø 4.3 mm (part no. M65 193 1) into the groove of the two tube clamps. Close the tubing cover. The position switch must not switch (LED D24 / P.C.B. LP 1058 not illuminated).

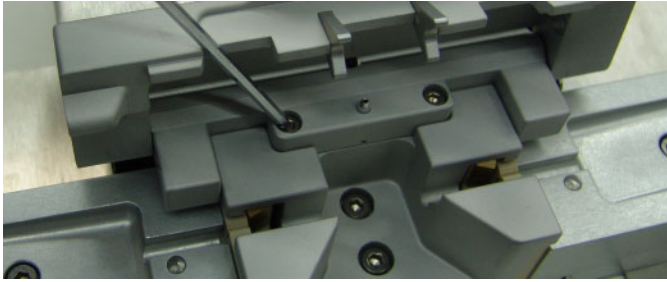


5.6.2 Tubing cover $\geq$ E-code 030

- Insert one adjusting pin $\varnothing$ 4.3 mm (part no. M651931) each (China version: $\varnothing$ 4.9 mm (part no. M667551)) into the insertion groove of the two tube clamps. Close the tubing cover. The position switch must switch (LED D24 / P.C.B. LP1058 is illuminated).

● Removal

- Open the tubing cover, remove the fixing screws (A) and then the tubing cover.



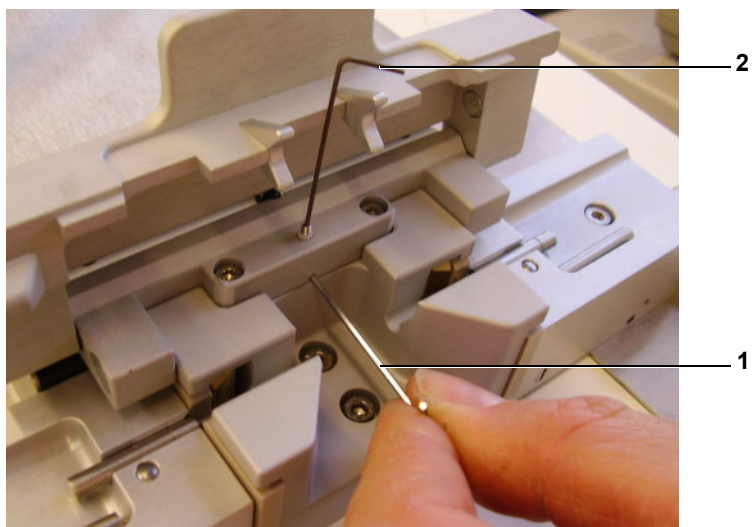
● Installation

- Simultaneously press the **Start** and the **Stop** key.
The device moves to the **Clean Position**.
- Open the heating element cover.
The clamps will move outwards.
- Place the switching pin with the threaded pin and the spring in the central block so that the hole of the switching pin is in the same position as the hole of the central block.
- Attach the tubing cover using the 2 Allen screws (1).

● Adjustment

- Pass a small size 1.3 Allen wrench (1) through the holes to prevent turning of the switching pin.
- Insert one adjusting pin $\varnothing$ 4.3 mm (part no. M651931) each (China version: $\varnothing$ 4.9 mm (part no. M667551)) into the insertion groove of the two tube clamps.
- Use a size 0.9 Allen wrench (2) to adjust the position switch so that it still switches with the tubing cover pressed (LED D24 / P.C.B. LP 1058 dark).

- Turn the threaded pin $190^\circ \pm 10^\circ$ (China version: $350^\circ \pm 10^\circ$) counterclockwise.



- Close the tubing cover.
The position switch must switch.

5.7 Central block

5.7.1 Removal of the central block

- **Requirements**

- Tubing cover removed

- **Removing the central block**



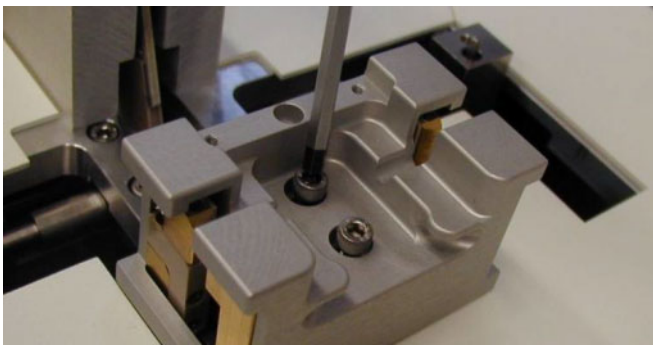
Note

Cable routing!

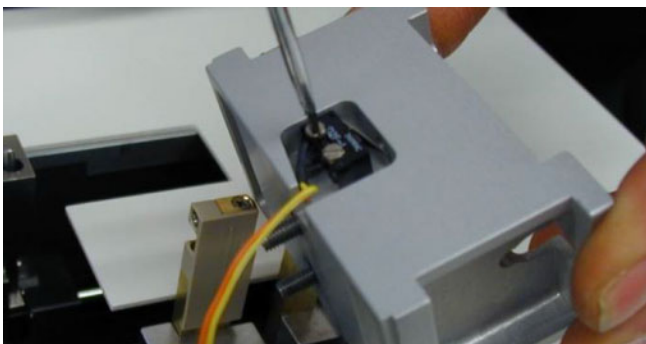
- Parting of the tubing cover switch cable may occur if it is not handled with care.

Carefully remove the central block.

- Remove the 2 Allen screws M4x16 of the central block.



- Carefully remove the central block by pulling it upward.
- Unscrew the tubing cover switch from the central block.



- Remove the central block

5.7.2 Installation of the central block

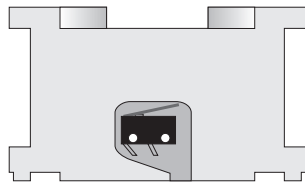
If only the central block was removed without any modification of all other settings, an adjustment of the HF mechanism may be omitted.

● Requirements

Before installing the central block, adjust the settings of the HF mechanism (see chapter 5.8.4 on page 5-30).

● Installation

- Screw the tubing cover switch onto the central block
Ensure proper cable routing
- Install the central block
- Insert and gently tighten the two Allen screws M4x16 of the central block.



● Adjustment

- Slightly loosen the two Allen screws, size 3, of the central block.
- Select the **Xmotor pos dock** service menu and move the tube clamps to the initial position X0 / Y0.
- Move the tube clamps by one turn at the maximum in X1 direction (to the outside)
- Manually move the tube clamps back in X0 direction, until they just touch the central block
- The central block can be moved easily in Y direction within the tolerance of its fixing holes
- Open the electrode pairs to their maximum
- Open the tube clamps
- Insert a 4.2 mm tubing (part no. M600241) (China version: 4.55 mm tubing (part-no. M600251))
- Adjust the central block until the tubing can easily be moved in the guide.
- Tighten the two Allen screws, size 3, of the central block



Note

Check the following after working on the central block:

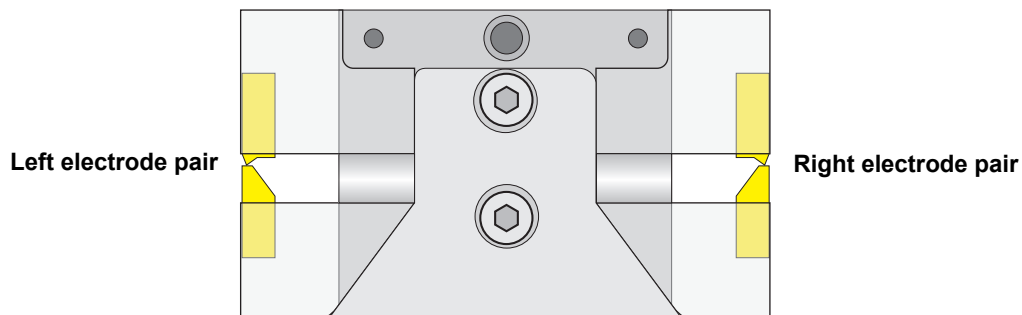
- Alignment of the tube clamps with the central block
(see chapter 5.5.1.3 on page 5-19)
 - Alignment of the tube clamp height position
(see chapter 5.5.1.4 on page 5-19)
-

5.8 HF electrodes unit

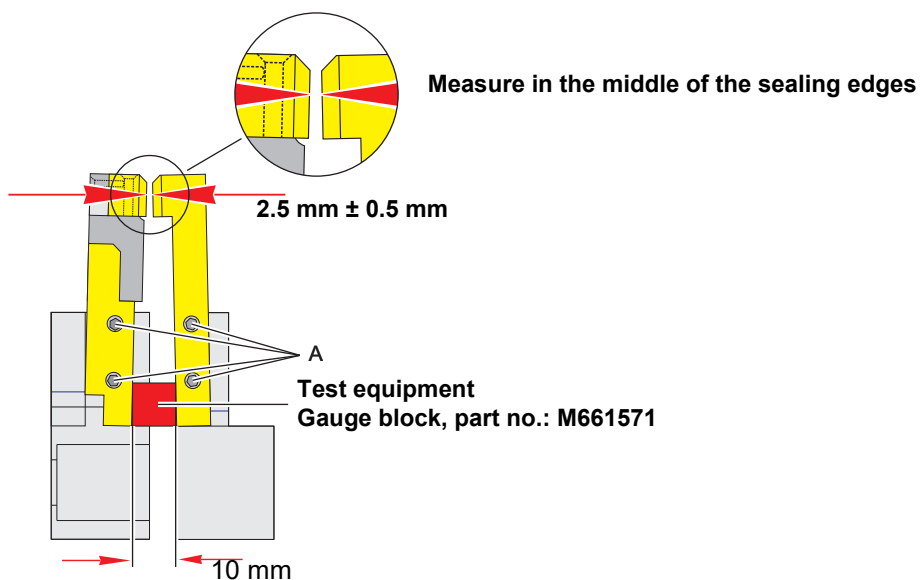
Applicable to devices > E-code 025 (switching plate replacing switching spring):

5.8.1 Checking the HF mechanism

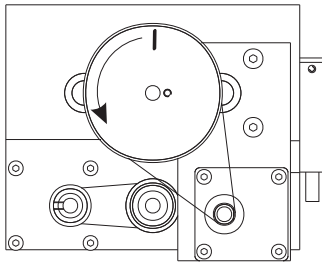
- Select the **HF motor** service menu and open the electrodes
- The HF electrodes must open so far that a 4.2 mm tubing may be easily inserted
- Select the **Y motor** service menu and move the tube clamps to position X1 / Y1.
- Manually close the electrodes
- When being closed, both HF electrode pairs have to touch each other at first on the top (maximum difference: ≤ 0.1 mm)



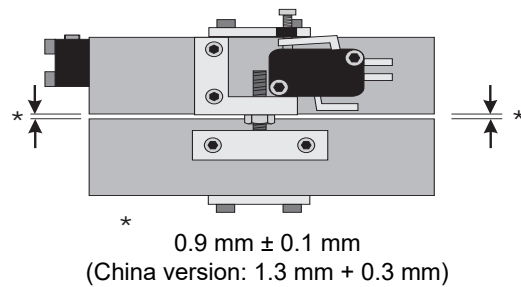
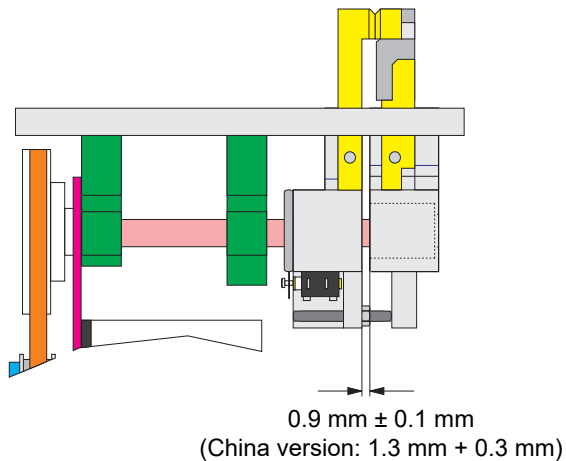
- Insert the gauge blocks (part no. M661571) between both electrode pairs. The labeling on the gauge blocks must be visible.



- Remove the toothed belt.



- Manually move the HF electrode pairs towards each other by turning the HF gear wheel until the two electrode pairs touch the gauge blocks. Mark the HF gear wheel and overtighten it by one quarter of a turn.
- Determine the mean distance between the electrode pairs.
Desired value:
per electrode pair $2.5 \text{ mm} \pm 0.5 \text{ mm}$
(maximum difference between the two electrode pairs: $\leq 0.1 \text{ mm}$)
- Remove the gauge blocks
- Lift the base plate
- Manually move the electrodes to the mechanical end position
Mechanical end position $0.9 \text{ mm} \pm 0.1 \text{ mm}$ (China version: $1.3 \text{ mm} + 0.3 \text{ mm}$)



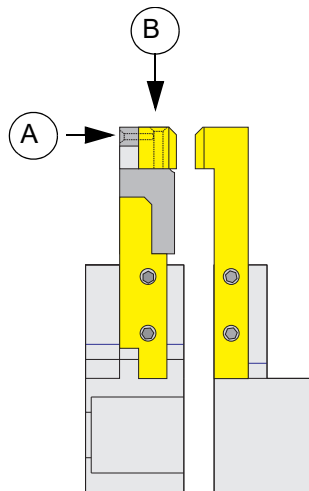
5.8.2 Removal of the electrodes

- **Requirements**

- Tubing cover removed (see chapter 5.6 , page 5-21)
- Central block removed (see chapter 5.7.1 , page 5-25)

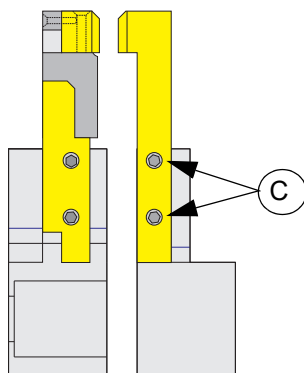
- **Hot electrode:**

- Remove the two fixing screws (A) and (B) of the electrode
- Remove the insert of the hot electrode



- **Cold electrode:**

- Remove the fixing screws (C) of the cold electrode.



5.8.3 Installation of the electrodes

- Follow the procedure in reverse order for installation of the electrodes.
- (see chapter 5.8.4.1 on page 5-30) Adjust electrode distance

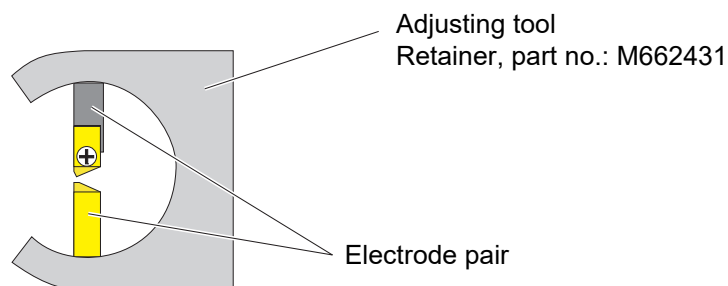
5.8.4 Adjustment of the HF mechanism

● Requirements

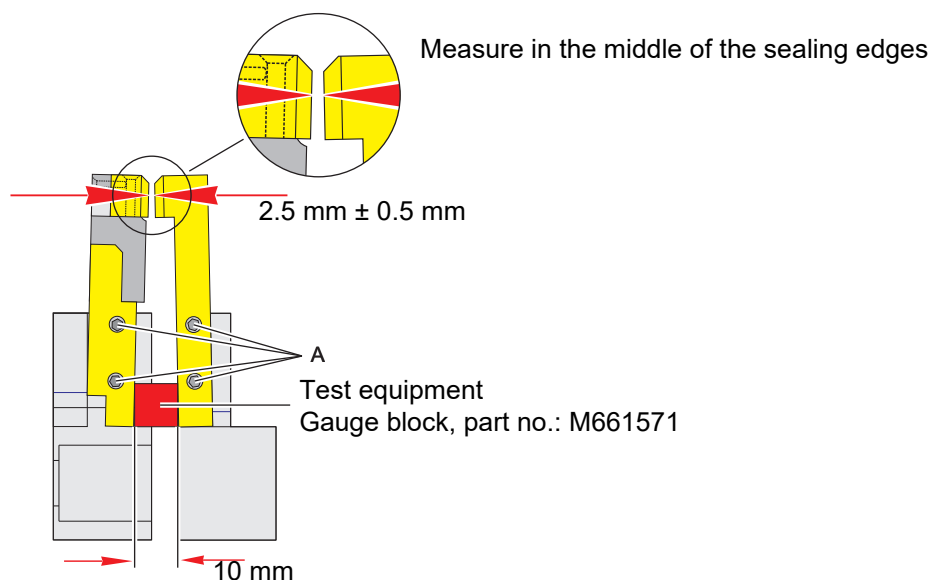
- The central block must not be mounted.

5.8.4.1 Adjusting the electrode distance

- Move the tube clamps into position Y1.
- Loosen the fixing screws (A) of the electrodes.
- Remove the toothed belt.
- Press the top of the electrodes together and place the retainer (part no: M662431) onto the electrode pair
- Manually move the HF electrode pairs away from each other by turning the HF gear wheel until there is a considerably strong resistance.



- Tighten the fixing screws (A).
- Repeat this adjustment on the second electrode pair.



- Insert the gauge block (part no. M661571) between both electrode pairs. The dimension indicated on the gauge block must point up.

- Manually move the HF electrode pairs towards each other by turning the HF gear wheel until the two electrode pairs touch the gauge blocks. Mark the HF gear wheel and overtighten it by one quarter of a turn.
- Determine the mean distance between the electrode pairs (see figure).
Desired value:
per electrode pair $2.5 \text{ mm} \pm 0.5 \text{ mm}$
(maximum difference between the two electrode pairs: $\leq 0.1 \text{ mm}$)

If necessary, adjust the exact value by tilting one electrode.



Note

Please ensure that the electrodes do not change their position while being tightened.

Tighten the fixing screws alternately.

- Slightly open the fixing screws of the electrode and readjust the electrode.
- Tighten the fixing screws (A).
- Open the HF mechanism and remove the gauge block.
- Manually move the HF electrode pairs towards each other by turning the HF gear wheel. When being closed, both HF electrode pairs have to touch each other at first on the top.

If one electrode pair closes earlier, increase the electrode distance of this electrode pair.

- If necessary, readjust the electrode.



Note

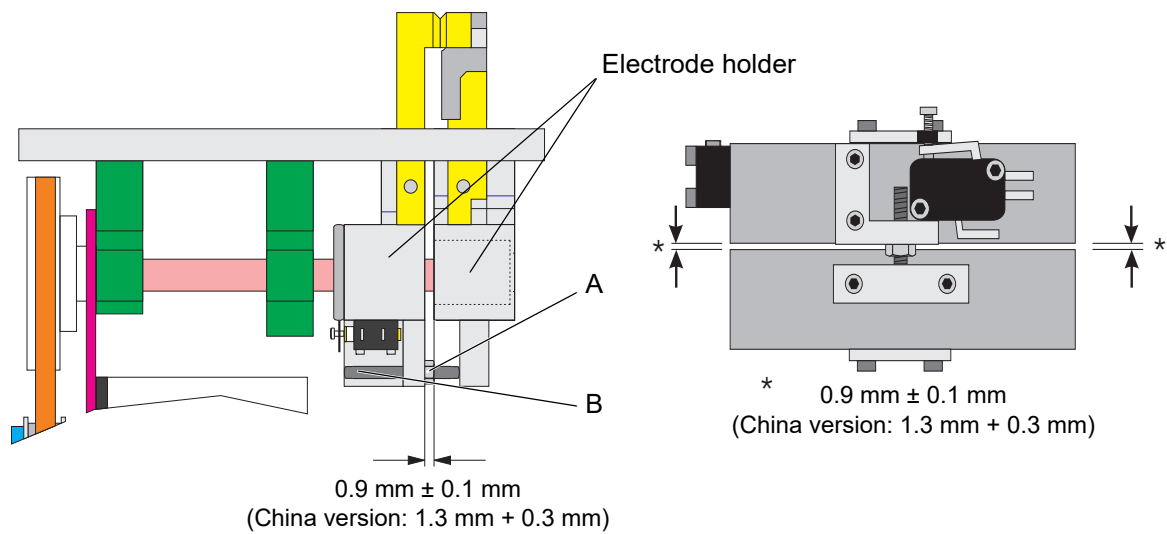
Check the following after working on the HF electrodes unit:

- Tube clamps (see chapter 5.5.1.2 , page 5-18)
- Central block (see chapter 5.5.1.3 , page 5-19)
- HF mechanism (see chapter 5.8.1 , page 5-27)

- Install the toothed belt.

5.8.4.2 Adjusting the mechanical end position of the HF unit

- Unscrew the lock nut (A), size 7, of the center stop of the HF unit.
- Adjust the threaded pin (B) of the center stop of the HF unit so that on both sides the distance between the electrode holders is smaller than $0.9 \text{ mm} \pm 0.1 \text{ mm}$ (China version: $1.3 \text{ mm} + 0.3 \text{ mm}$).
- Tighten the lock nut (A), size 7, of the center stop of the HF unit.
- Check the adjustment



Note

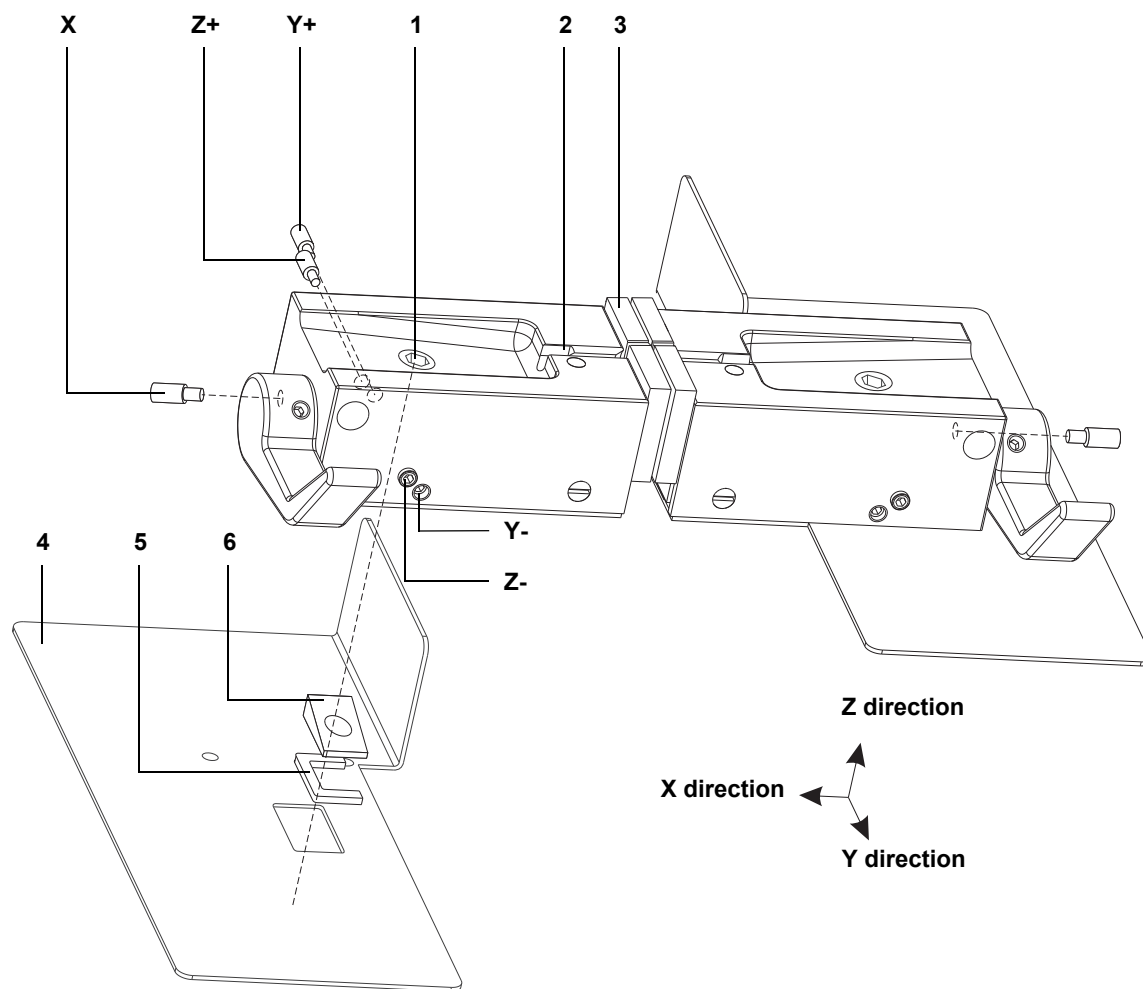
Check the following after working on the mechanical end position:

- Y0 position switch (see chapter 5.11.2 on page 5-48)
- HF-0 position switch (see chapter 5.9.6 on page 5-41)
- HF-1 position switch (see chapter 5.9.7 on page 5-42)
- HF voltage (see chapter 5.13 on page 5-56)

5.9 Tube clamps

● Description

The tube clamps are used to position and to secure the tubings. The tubings to be connected are placed in the line holders of the tube clamps. When the clamping levers are closed, the tubings are held in place by the clamping jaws. During the sealing process the tube clamps move to different sealing positions.



- X** Clamping screw
- Y** Adjusting screws for adjusting the central position of the clamps
- Z** Adjusting screws for adjusting the height of the clamps
- 1** Fixing screw (size 3)
- 2** Line holder
- 3** Clamping jaws
- 4** Cover plate
- 5** U piece
- 6** Wedge

5.9.1 Test

- Check the position of the tube clamps (see chapter 5.5.1 on page 5-18).

5.9.2 Removal of the tube clamps

- Open the device (see chapter 5.1 on page 5-1).
- Select the **Xmotor pos.dock** service menu and move the tube clamps to the docking position (X1/Y1).
- Unscrew the adjusting screws (Y+,Y-) and the clamping screw (X) by 1/4 turn.
- Remove the two fixing screws M4x20 (1) of the tube clamps and then the tube clamps.

5.9.3 Installation of the tube clamps

● Requirements

Electrode distance (see chapter 5.8.4.1 on page 5-30) and mechanical end position (see chapter 5.8.4.2 on page 5-31) are adjusted correctly.



Note

The basic adjustments for the installation of the clamps must be made with the clamps removed.

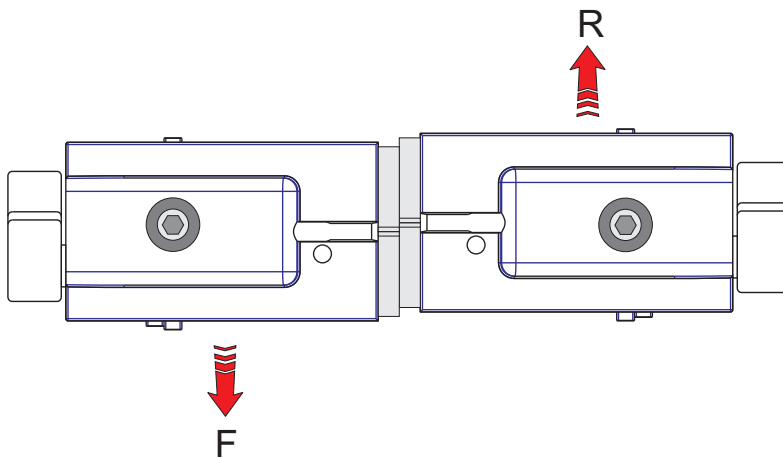
- Open the device
- Lift the base plate
In this position, the wedges of the tube clamps remain securely in the bottom position.
- Loosen all clamp and adjusting screws so that the U piece and the wedge can be adjusted to the maximum.
- Push the wedge down
- Screw the cover plate (4) onto the tube clamps
- Install both tube clamps.
- Tighten the tube clamps with **spring cramps**.
- Hinge the base plate down

5.9.4 Adjustment of the tube clamps (central position)

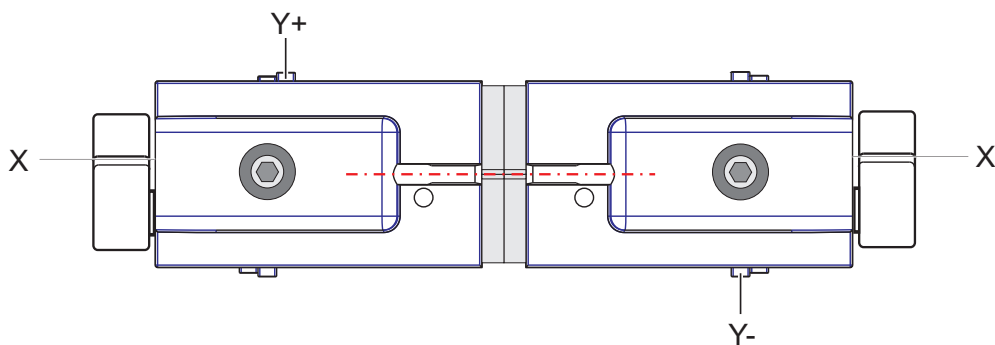
5.9.4.1 Basic adjustment of the tube clamps

● Adjustment for all E-codes

- Move the tube clamps together into the docking position (X3/Y1)
- Close the tube clamps.
- Push the right tube clamp back (direction R)
- Push the left tube clamp forward (direction F)



- Gently tighten the clamping screws (X).
- Move the right tube clamp with the adjusting screw (Y-) to the central position
- Align the left tube clamp with the adjusting screw (Y+) in one line with the right tube clamp (central position) without any offset.
- If the tube clamps are positioned, tighten the screws on the opposite side until a resistance is felt.



5.9.4.2 Adjustment of the central block

- Slightly loosen both Allen screws, size 3, of the central block.
- Select the **Xmotor pos dock** service menu and move the tube clamps to the initial position X0 / Y0.
- Move the tube clamps by one turn at the maximum in X1 direction (to the outside)

- Manually move the tube clamps back in X0 direction, until they just touch the central block
- The central block can be moved easily in Y direction within the tolerance of its fixing holes
- Close the electrodes
- Align the central block centrally to the play of the hole in the Y direction.
- Tighten both Allen screws, size 3, of the central block
- Check and, if necessary, slightly readjust the position of the Y0 position switch

5.9.4.3 Adjustment of the Y0 position switch

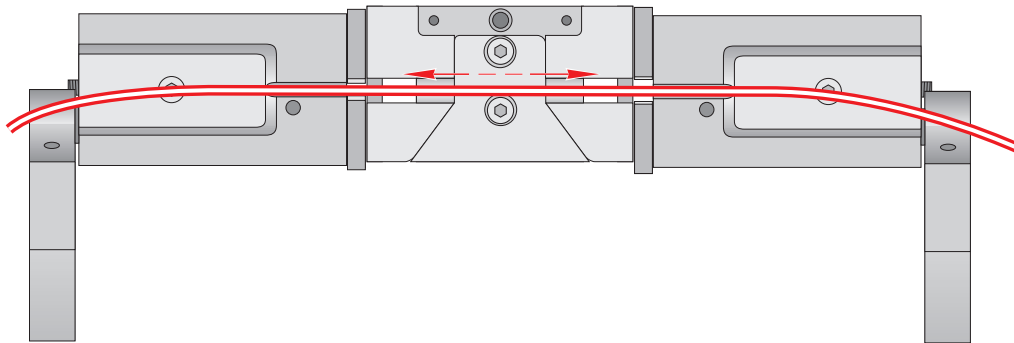
● Requirements

For adjusting the Y0 position switch, the tube clamps must be set correctly. A tubing inserted into both tube clamps and the central block must easily be moved in axial direction.

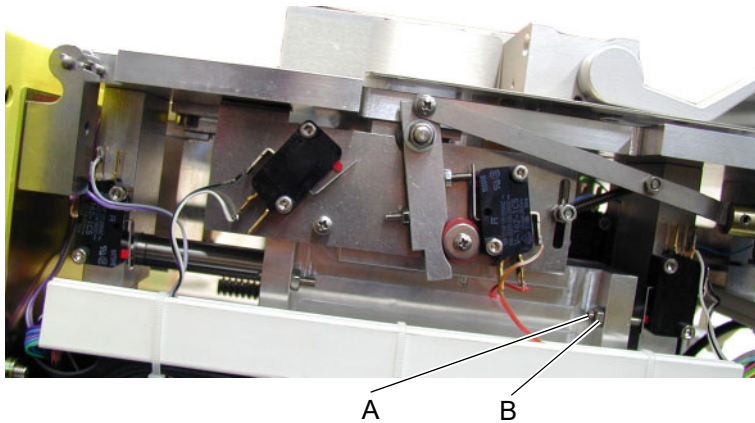
- Basic adjustment of the tube clamps performed
- Adjustment of the central block performed
- Select the **Xmotor pos dock** service menu and move the tube clamps to the X0 / Y0 position.

● Adjustment

- Open the electrode pairs to their maximum
- Open the tube clamps
- Insert a 4.2 mm tubing (part no. M600241) (China version: 4.55 mm tubing (part-no. M600251))
- Move the Y spindle to the left and the right until the tube can be moved in the guide.



- Unscrew the lock nut (B).
- Turn the adjusting pin to the switching point of the Y0 position switch.
- Turn the adjusting pin by one quarter of a turn
- Secure the adjusting pin using the lock nut (B).

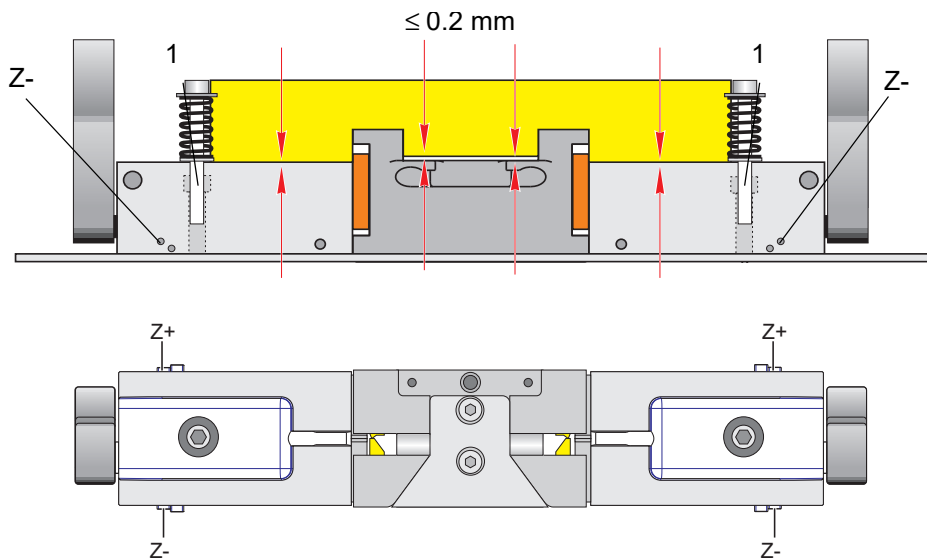


- Repeatedly check the adjustment via the **Xmotor pos.dock** menu.
- If necessary, repeat the adjustment

Applicable to tube clamps <E-code 030:

Using the adjusting gauge (part no. M65 168 1), align the tube clamps with the central block.

- Raising the tube clamps:
Unscrew the front adjusting screw (Z-), screw-in the rear adjusting screw (Z+).
- Lowering the tube clamps:
Unscrew the rear adjusting screw (Z+), screw-in the front adjusting screw (Z-).
- This adjustment has to be repeated until the alignment is correct.
Distance between adjusting gauge and central block: $\leq 0.2 \text{ mm}$



● **Applicable to tube clamps $\geq$ E-code 030:**

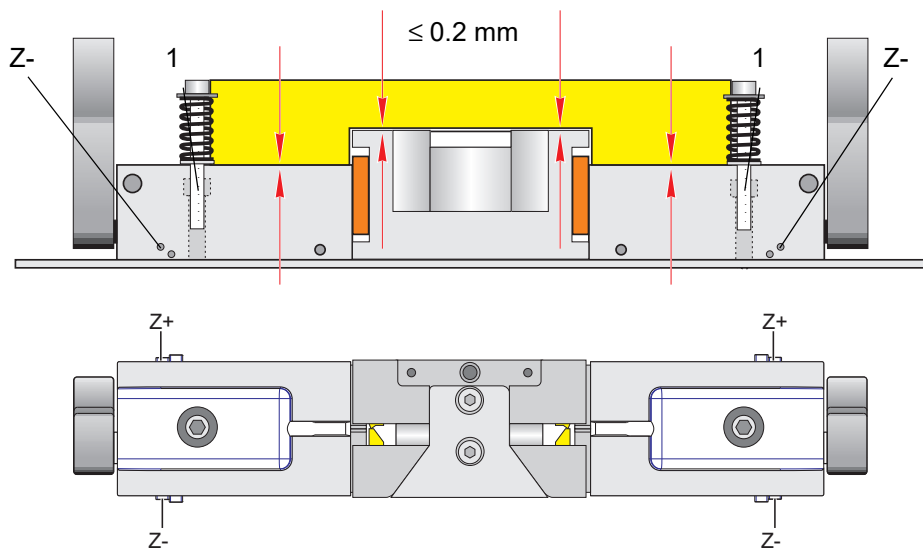


Note

These are the tube clamps with metallic U piece (part no.: 6613141) and the new central block (part no.: 6613131).

Using the adjusting gauge (part no. M659261), align the tube clamps with the central block.

- Raising the tube clamps:
Unscrew the front adjusting screw (Z-), screw-in the rear adjusting screw (Z+).
- Lowering the tube clamps:
Unscrew the rear adjusting screw (Z+), screw-in the front adjusting screw (Z-).
- This adjustment has to be repeated until the alignment is correct.
Distance between adjusting gauge and central block: ≤ 0.2 mm



● **Applicable to all tube clamps**

- Remove the spring cramps.
- Insert and securely tighten the fixing screws M4x20 (1).
- Uniformly tighten all adjusting screws and clamping screws.



Note

After performing the adjustment, proceed as follows:

- Check the Y0 position switch
- Check the Y1 position switch
- (see chapter 5.5.1.2 , page 5-18)
- Check the HF-0 position
- Seal test
- Dock test

5.9.5 Basic adjustment of the HF unit

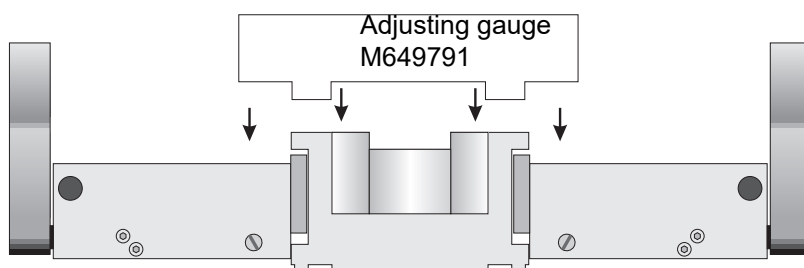
It has only to be performed if the central block was readjusted.

● Requirements

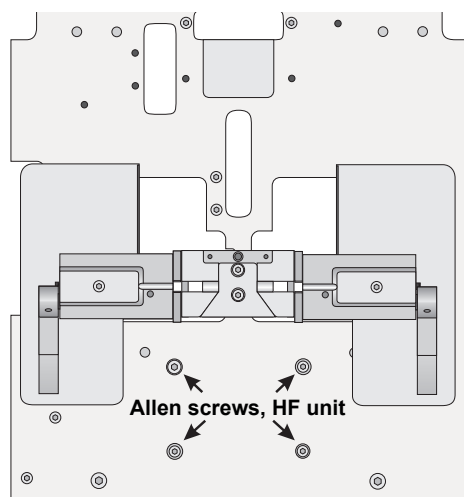
- Electrode distance adjusted (see chapter 5.8.4.1 on page 5-30)
- Tube clamps and central block correctly installed and adjusted (see chapter 5.9.4 on page 5-35)

● Basic setting

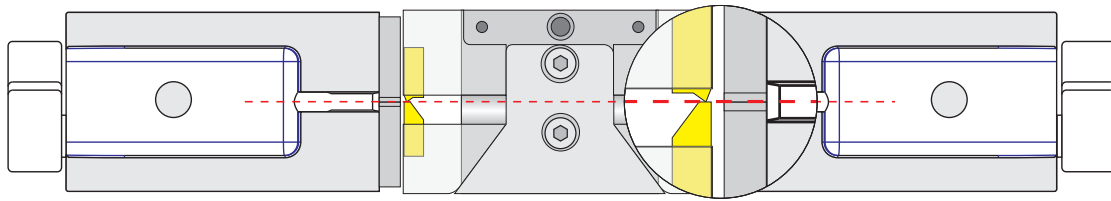
- Select the **Xmotor pos dock** service menu and move the tube clamps to the X0 / Y0 position.
- Insert the adjusting gauge (part no. M649791) into both tube clamps.



- Close the tube clamps.
- Remove the toothed belt of the HF unit.
- Close and carefully overstrain electrodes and hold them
- Unscrew the 4 Allen screws, size 4, of the HF unit by 1/2 turn.



- Slightly vibrate the HF unit in order to stop mechanical friction
- Remove the adjusting gauge.
- Close the electrodes (LED D15 is illuminated)
- Move the HF unit until the closed tube clamps are in a line with the closed electrodes.



- If necessary, repeat the adjustment
- Securely tighten the 4 Allen screws size 4 of the HF unit.
- Install the toothed belt.
- Visually check the adjustment.
Select the **Xmotor pos dock** service menu and move the tube clamps to the X0 / Y0 position.
The lateral distance of the electrodes to the clamping jaws has to be the same on both sides.
The closed tube clamps have to be in a line with the closed electrodes (LED D15 is illuminated).
- Adjust the Y0 position switch (see chapter 5.9.4.3 on page 5-36)



Note

After performing the basic adjustment, proceed as follows:

- Adjust the HF-0 position switch
 - Adjust the HF-1 position switch
-

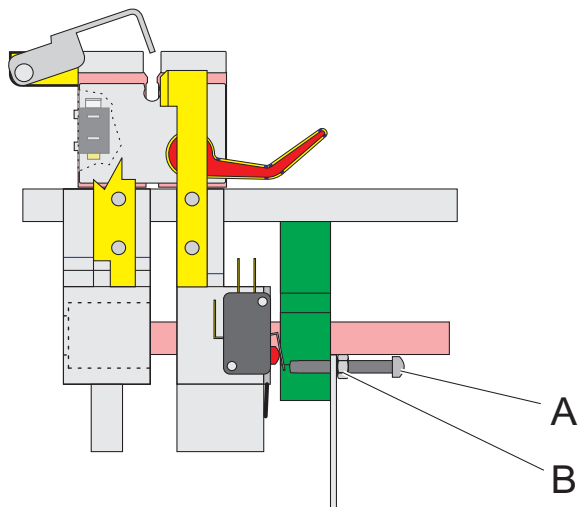
5.9.6 HF-0 position switch

● Test

- The HF electrodes must open so far that a 4.2 mm tubing may be easily inserted

● Adjustment

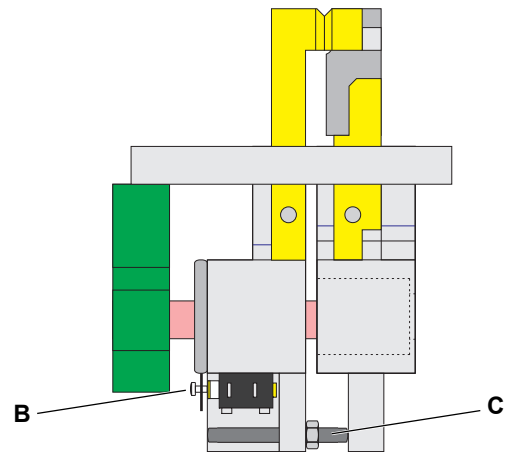
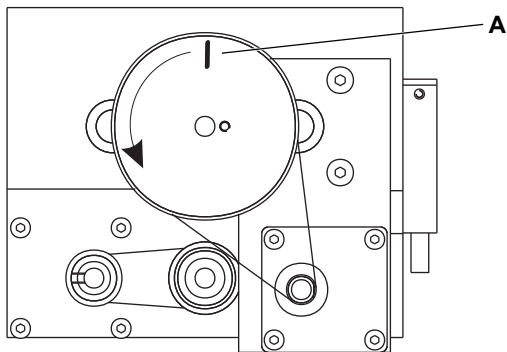
- Select the **Ymotor** service menu and move the tube clamps to the position (X1/Y1).
- Remove the toothed belt of the HF unit.
- Lift the base plate
- Open the electrodes until there is a resistance (mechanical end position).
- Turn the gear wheel 1/4 turn counterclockwise.
- In this position, use the adjusting screw (A) to adjust the switching point of the HF position switch HF-0 (LED D14).
- Secure the adjusting pin using the lock nut (B).
- Install the toothed belt
- Via the **HF motor** service menu, repeatedly open and close the electrodes.
- The HF unit has to be positioned one quarter of a turn before the mechanical end position.



5.9.7 HF-1 position switch

● Check and adjustment of the HF-1 position switch

- Remove the toothed belt.
- Lift the base plate
- Move the HF-1 position together until the mechanical stop (C).
- Mark the gear wheel (A).
- Rotate the gear wheel by $120^\circ \pm 10^\circ$ to the left.
- In this position, use the adjusting screw (B) to adjust the switching point of the HF position switch HF-1.
- Check the adjustment
Via the **HF motor** service menu, repeatedly open and close the electrodes.
- Apply lacquer to the screw after having performed the test.



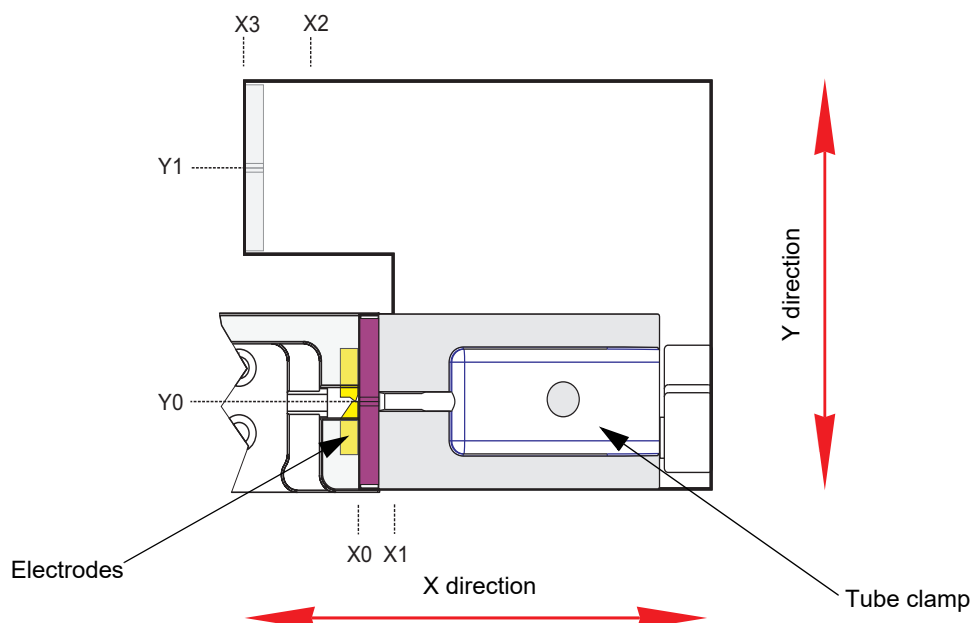
5.10 X system

● Description

The X motor moves the tube clamps in the X (horizontal) direction.

The optical sensor senses the pulses of the rotating perforated disk of the X motor. The number of the detected pulses is used to determine the X movement which is then displayed as Counts.

- X0 / initial position (reference position)
- X1 / outside position
- X2 / heating position
- X3 / docking position



5.10.1 X1 / Y0 position - separation of the tubings

● Test

- Select the **Xmotor Outside pos** service menu and move the tube clamps to the X1 / Y0 position (separation of the tubings).
Desired value: 0.2 s - 1.5 s
- If the value displayed is higher than the desired value, the motor is defective or the X mechanism is blocked.
Remedy:
Correct the jam and remove the foreign object, if necessary.
Check the motor gear and the spindle for smooth running.
- It must be possible to rotate the X-motor at least one quarter of a turn before it reaches the mechanical end position and blocks.
- Check the default values in the **Xmotor Outside pos** menu.
time (= X motor in outer position)
< E-code 030 = 95 counts
≥ E-code 030 or ≥ software 2.2x = 100 counts

● Adjustment

- If the tube clamps move outwards too far when moving to position X1 (separation of the tubings), and if it is no longer possible to perform a 1/4 turn of the motor, reduce the value **Xmotor Outside pos. / Counts**.
- If the tube clamps do not move outwards far enough when moving to position X1 (separation of the tubings), increase the value **Xmotor Outside pos. / Counts**.
- Check the adjustment

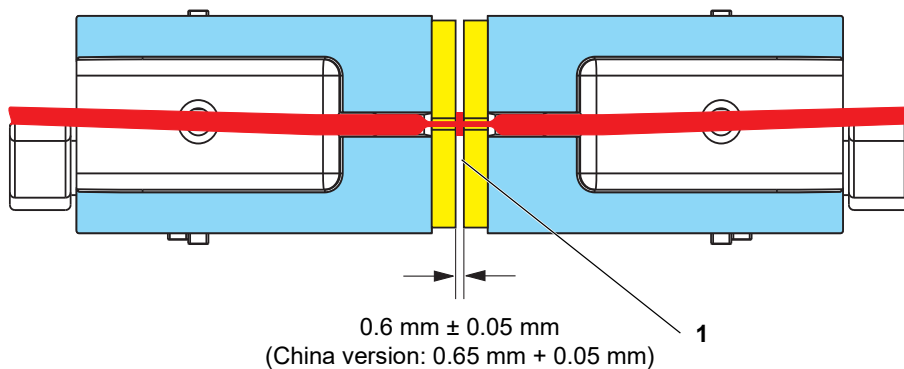
5.10.2 Docking of the tubings - X3 / Y1 position

● Description

The docking position X3 / Y1 is the position of the tube clamps in which the two extreme ends of the tubing are connected to each other.

● Test

- Make a test dock with each tubing.
- Do not open the tube clamps after the docking process.
- Use the feeler gauge to determine at point (1) the distance of the tube clamps in docking position.
Desired value: 0.6 mm + 0.05 mm (China version: 0.65 mm + 0.05 mm)



● Adjustment

- Close the two tube clamps.
- Select the **Xmotor pos.dock** service menu.
- Move the tube clamps into initial position X0 / Y0 and set the desired value 628, using the S1 or S2 key.
Use the key S1 / LP 1058 to increase the Counts value in order to reduce the distance.
Use the key S2 / LP 1058 to decrease the Counts value in order to increase the distance.
- Move the tube clamps into docking position and use the feeler gauge to determine the distance of the tube clamps in docking position. Repeat this adjustment process until the distance between the tube clamps in docking position is 0.6 mm ± 0.05 mm (China version: 0.65 mm + 0.05 mm).
- Check the adjustment
- Perform a dock test (see chapter 5.15 on page 5-63).

5.11 Y system

● Description

The Y motor moves the tube clamps in the Y (vertical) direction. The Y motor then moves the Y system to the Y0 and Y1 position switches.

- Y0 / initial position (reference position) of the tube clamps
- Y1 / docking or heating position

5.11.1 Y0 position switch

● Requirements

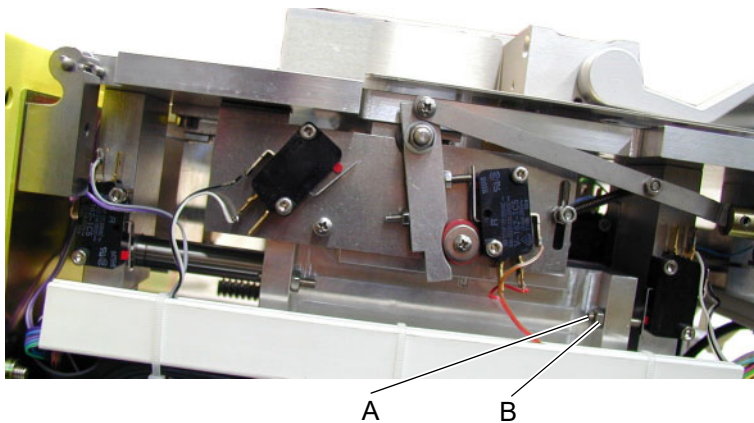
The tube clamps have been adjusted.

● Test

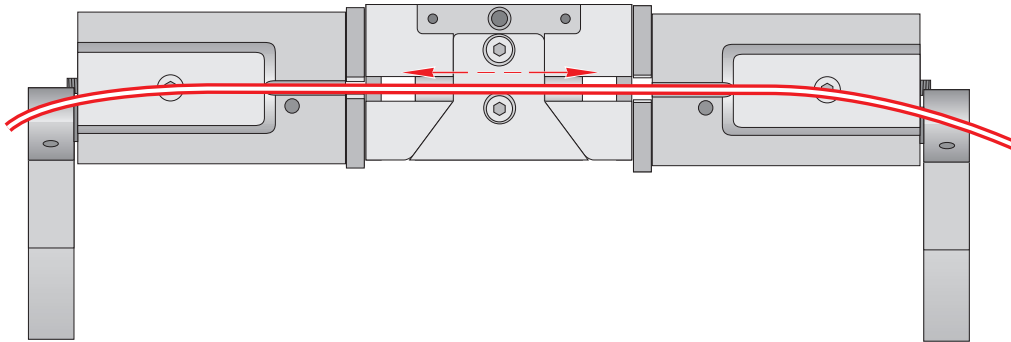
- Select the **Xmotor pos. dock** service menu and move the tube clamps to the initial position X0 / Y0.
- Close the tube clamps.
- Select the **HFmotor** service menu and close the electrodes (LED D15 / P.C.B. LP 1058 is illuminated).
- The two closed tube clamps must be in line with the closed electrode pairs.
- Via the **Xmotor pos.dock** menu, move the tube clamps once more to the docking position and then return them to their initial position X0 / Y0, in order to check for proper functioning.

● Adjustment

- Open the electrode pairs to their maximum.
- Open the tube clamps.
- Insert a 4.2 mm tubing (part no. M600241) (China version: 4.55 mm tubing (part-no. M600251)).
- Move the Y spindle to the left and the right until the tube can be moved in the guide.
- Unscrew the lock nut (B).
- Turn the adjusting pin to the switching point of the Y0 position switch.
- Turn the adjusting pin by one quarter of a turn.
- Secure the adjusting pin using the lock nut (B).



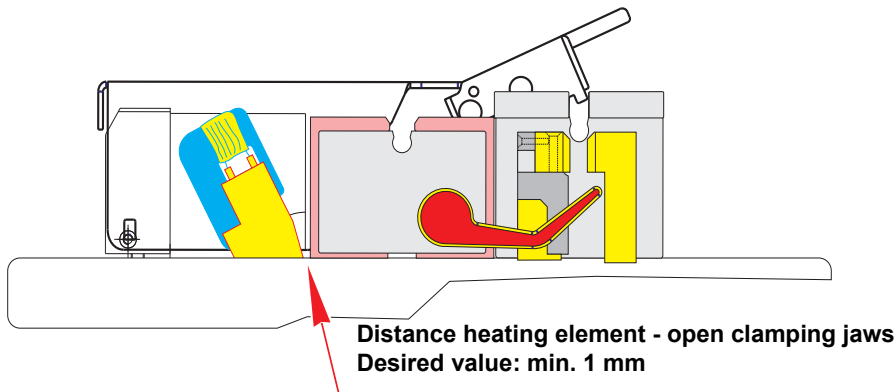
- Repeatedly check the adjustment via the **Xmotor pos.dock** menu.



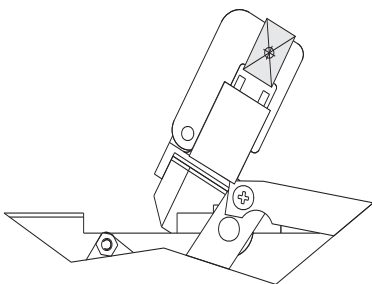
5.11.2 Y1 position switch

● Test

- Open the tube clamps
- Select the **Xmotor pos. dock** service menu and move the tube clamps to the docking position X3 / Y1.
- The lower corner of the open clamp jaws must not touch the heating element.
- Check the play by using the switching lever of the heating element.



- Select the **Xmotor pos. dock** service menu and move the tube clamps to the initial position X0 / Y0.
- Open the tube clamps, insert a tubing, Ø 4.2 mm (part no. M600241) (China version: 4.55 mm tubing (part-no. M600251)) into both tube clamps, and close the tube clamps.
- Select the **Seal test** service menu and seal off the two tubings.
The tube clamps remain closed.
- Select the **Xmotor Heat pos** service menu and move the two tube clamps to the heating position.
- Manually move the heating element out of its enclosure. The two tubing ends must be positioned in the middle of the heating element.



5.11.3 Y1 position switch

● Adjustment

- Place one tube in each of the two tube clamps.
- Select the **Seal test** service menu and seal off the two tubings.
- Select the **Xmotor Heat pos** service menu and move the two tube clamps to the heating position.
- Manually move the heating element out of its enclosure. The two tubing ends must be positioned exactly in the middle of the heating element.

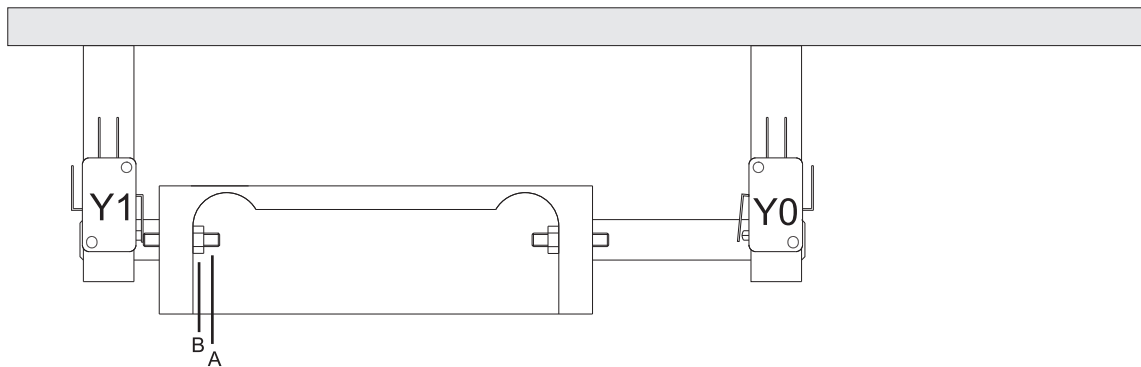


Note

Changes may only be performed in small steps.

If the changes are too extensive, the tube clamps may move into the central block.

- Unscrew the locking nut (B) and adjust the adjusting pin (A) so that the Y1 position switch will be activated when both tubing ends are positioned exactly in the middle of the manually extended heating element.
- Select the **Xmotor pos.dock** service menu and move the tube clamps to the docking position X3 / Y1.
- The lower corner of the open clamp jaws must not touch the heating element.
- Check the play by using the switching lever of the heating element.
- Secure the adjusting pin using the lock nut (B).



- Check the adjustment
- Perform a dock test (see chapter 5.15 on page 5-63).

5.12 Heating element

● Description

After turning the CompoDock power on using the main power switch, the heating element is heated to a temperature of approx. 465 °C (China version: 470 °C ± 5 °C). The heating element will not cool down before the CompoDock has been turned off.

Before starting the docking process, the sealed tubing ends must be heated. The heating element exits its protective enclosure and heats the sealed off tubing ends for approx. 8 seconds.

After heating the tubings, the heating element will return to its original position.

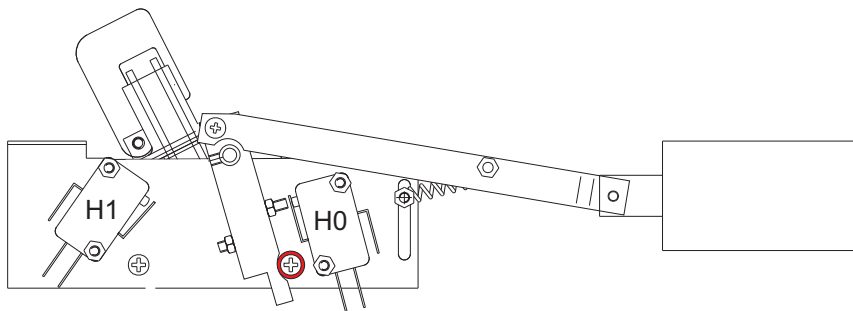
5.12.1 Test of the heating element

● Heating temperature

- In the service menu, select **Heater dock temp.**
- Use the infrared thermometer to measure the temperature of the heating element.
Desired value: 465 °C ± 5 °C (China version: 470 °C ± 5 °C)

● H0 position switch

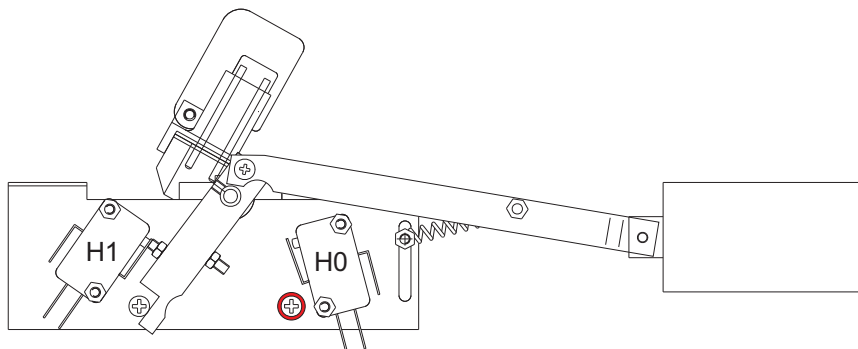
- Set the switching lever mechanism of the heating element to the H0 position (idle position).
- The pressed down switching lug of the H0 position switch must have approx. 1 mm play.



● H1 position switch

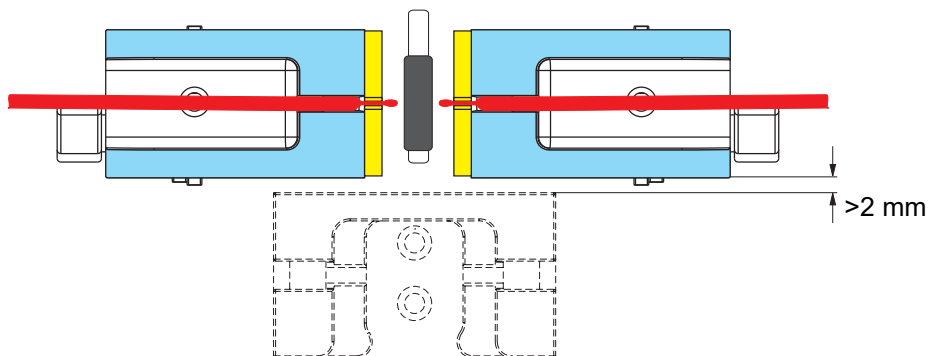
- Manually move the switching lever mechanism of the heating element towards the end stop of the H1 position. (heating element in heating position).

- The pressed down switching lug of the H1 position switch must have approx. 1 mm play.

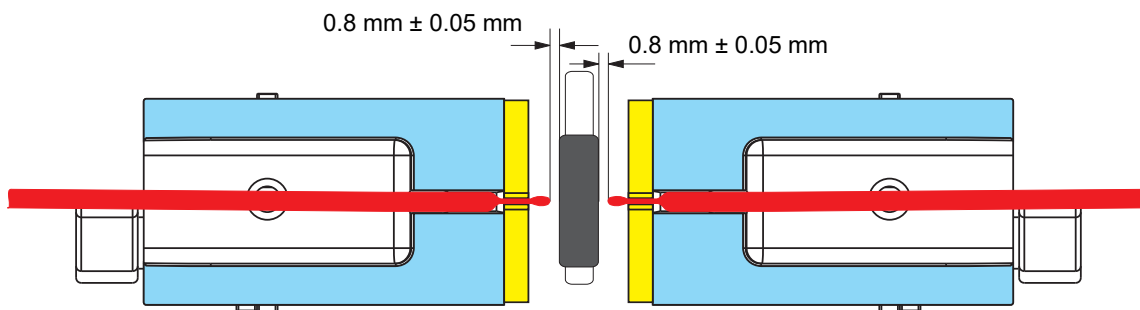


● Heating position (axial adjustment of the heating element)

- Select the **Seal test** service menu and seal off the two tubings.
- Keep the tube clamps closed.
- Select the **Xmotor Heat pos** service menu and move the two tube clamps to the heating position.
- Manually move the heating element out of its enclosure. The two tubing ends must be positioned exactly in the middle of the heating element.
- Measure the distance between the two tube clamps and the central block.
Desired value: min. 2 mm



- The heating distance between the tubing ends and the heating element must be $0.8 \text{ mm} \pm 0.05 \text{ mm}$.
($0.8 \text{ mm} \pm 0.05 \text{ mm}$)



5.12.2 Adjustment of the heating temperature

- In the service menu, select **Heater dock temp.**
- Set a heating temperature of $465\text{ °C} \pm 5\text{ °C}$ (China version: $470\text{ °C} \pm 5\text{ °C}$).
- Perform a sealing test (see chapter 5.14 on page 5-62).

5.12.3 Removal of the heating element



Caution

Risk of burning!

Turn the CompoDock off and allow the heating element and adjacent components to cool down.

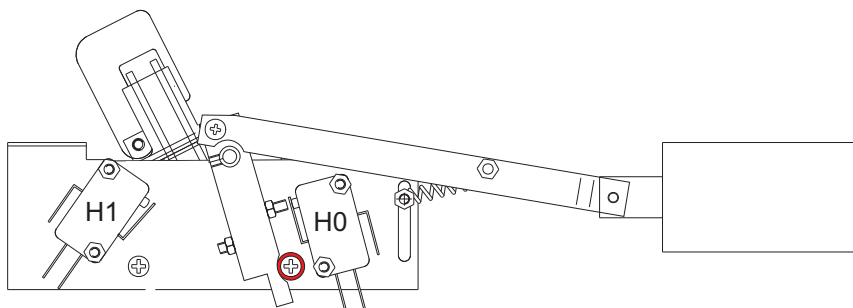
- Remove the heater cover and the protective enclosure.
- Unscrew the two Allen screws, size 3, on the upper side and rear side of the heating element.
- Remove the heating element.

5.12.4 Installation of the heating element

- Carefully insert a new heating element and ensure not to push the springs down.
- The heating element must be in a right angle to the aluminum mounting plate using the two Allen screws, size 3.
- Check the heating element (see chapter 5.12.1 on page 5-50).
- Perform a dock test (see chapter 5.15 on page 5-63).

5.12.5 Adjustment of the H0 position switch

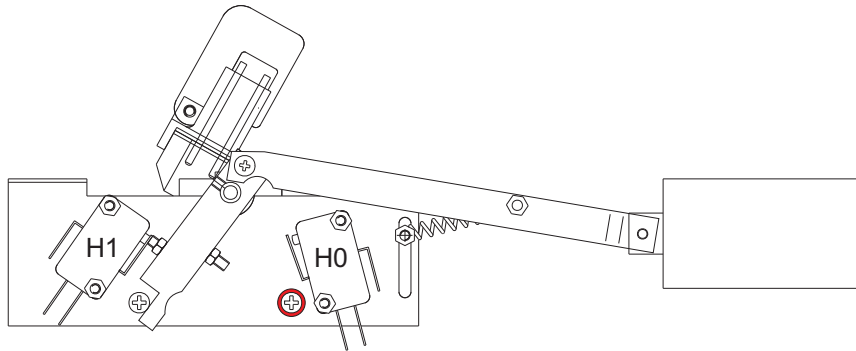
- Set the switching lever mechanism of the heating element to the H0 position (idle position).
- Adjust the adjusting pin of the H0 position switch so that the pressed down switching lug has approx. 1 mm play at the end stop.



5.12.6 Adjustment of the H1 position switch

● H1 position switch

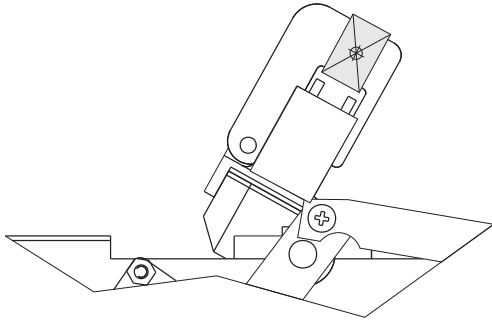
- Manually move the switching lever mechanism of the heating element towards the end stop of the H1 position. (heating element in heating position).
- Adjust the adjusting pin of the H1 position switch so that the pressed down switching lug has approx. 1 mm play at the end stop.



5.12.7 Adjustment of the heating position

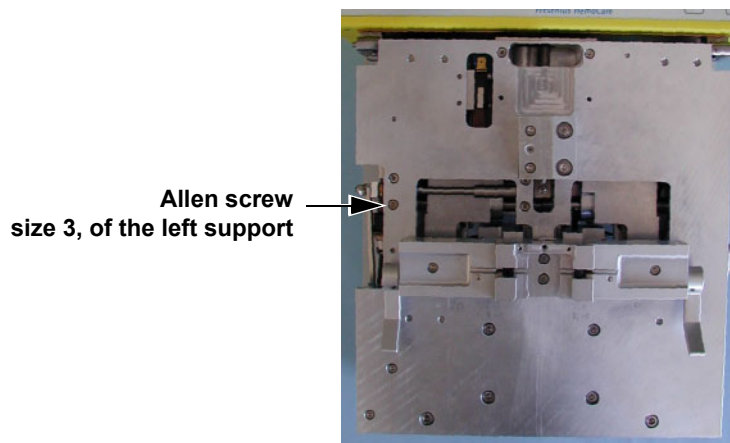
● Position Y1

- Adjust the Y1 position switch so that the tubing ends are positioned in the middle of the heating element (see chapter 5.11.2 on page 5-48).

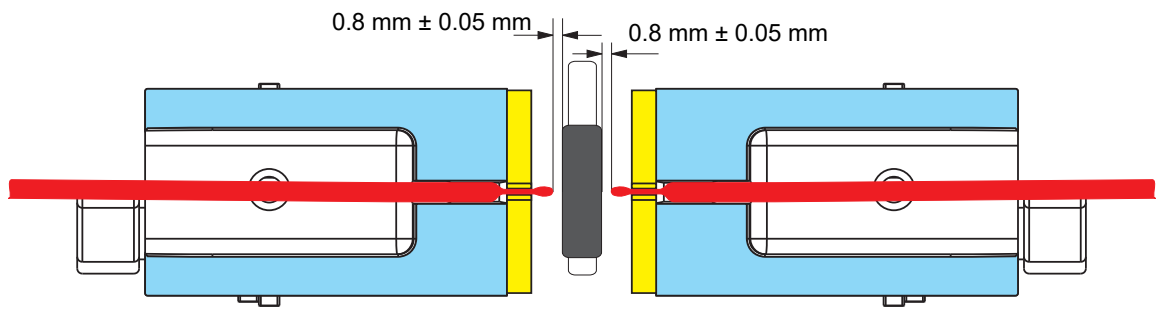


● Axial position of the heating element

- Select the **Seal test** service menu and seal off the two tubings.
- Select the **Xmotor Heat pos** service menu and move the two tube clamps to the heating position.
- Manually move the heating element out of its enclosure. The heating element must be positioned in the middle of the two tubing ends. If the heating distance of the two tubing ends is correctly adjusted, the distance between the heating element and each of the two tubing ends must be $0.8 \text{ mm} \pm 0.05 \text{ mm}$.
- Unscrew the Allen screw, size 3, of the left support of the heating element shaft and adjust the position.



- Retighten the Allen screw, size 3, and check the shaft for smooth running.
- Check the adjustment (see chapter 5.12.1 on page 5-50).
- The heating distance between the tubing ends and the heating element must be $0.8 \text{ mm} \pm 0.05 \text{ mm}$. ($0.8 \text{ mm} \pm 0.05 \text{ mm}$)



- Measure the distance between the heating element and the clamping jaws.
Reducing the distance: Increase the Counts value
Increasing the distance: Reduce the Counts value

5.13 HF voltage

● Description

HF generator

The CompoDock is provided with a maintenance-free HF generator, with an output of $84 \text{ W} \pm 5 \text{ W}$ at a frequency of 40.68 MHz.

Repair or adjustment of the HF generator may be performed by Fresenius Kabi only, as these tasks require special equipment and expert knowledge.

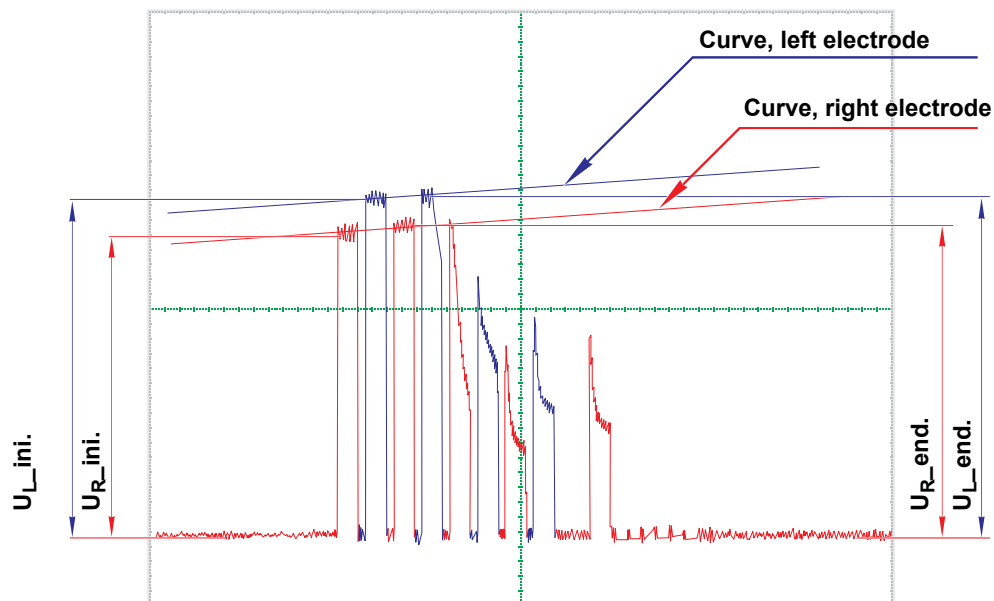
Seal head box

Controlled by a micro processor, the seal head box alternately switches the HF signal over to the left and right electrode.

The curves of the HF signals for the two electrode pairs can be detected via measurement points on the control board. After replacing the HF generator or the seal head box, the curves must be re-adjusted (set) by means of the seal head box.

● Description of the oscilloscope screen

The oscillogram shows two sealing curves for the left and for the right electrode pair. The two electrode pairs are clocked alternately. The right electrode pair is clocked first.



$U_{L_ini.}$: Initial value, left electrode pair

$U_{R_ini.}$: Initial value, right electrode pair

$U_{L_end.}$: Final value, left electrode pair

$U_{R_end.}$: Final value, right electrode pair

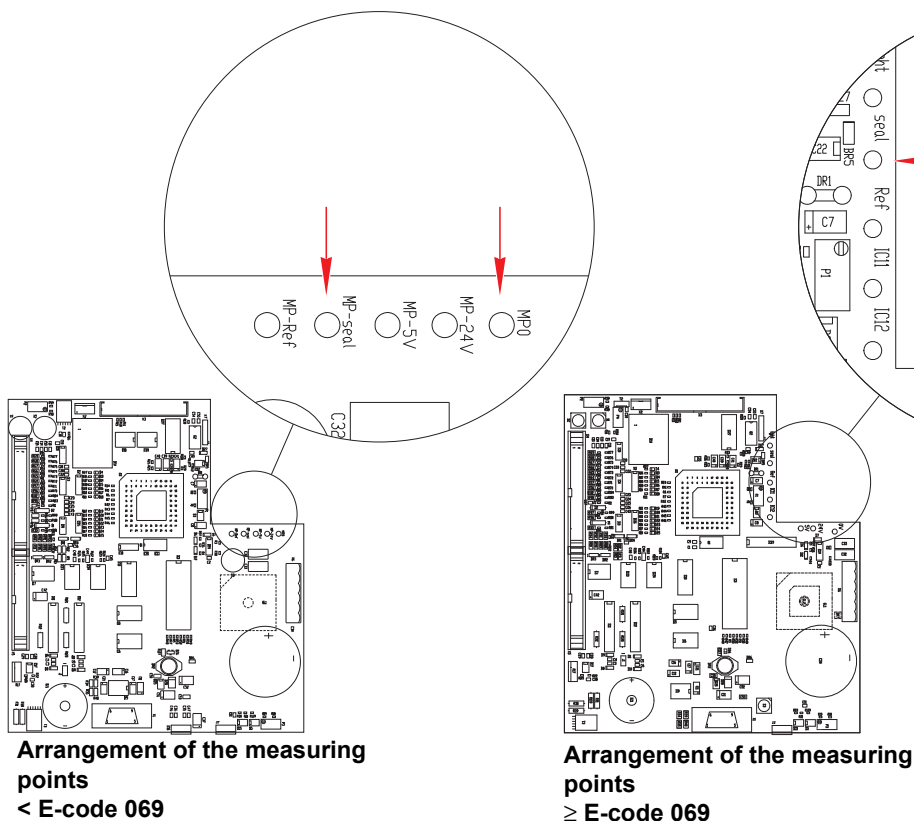
5.13.1 Checking the HF voltage

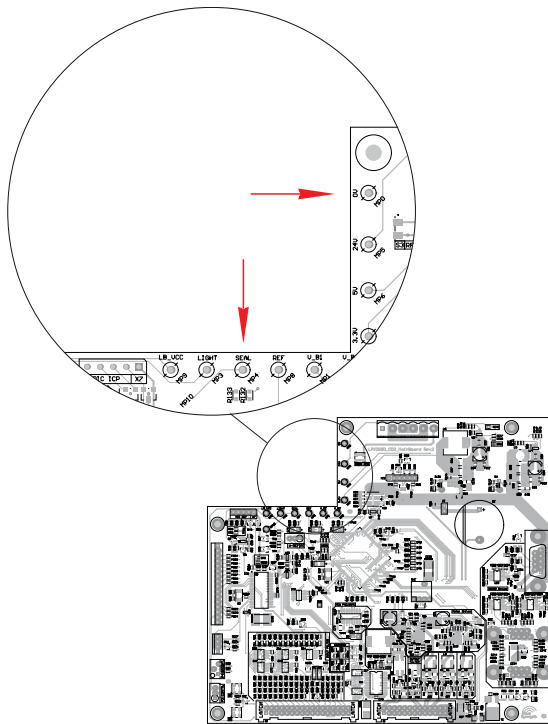
The following changes require the HF voltage to be checked:

- Generator replacement
- Electrodes replacement
- Electrodes adjustment
- Seal head box replacement
- Mainboard replacement

● Oscilloscope settings

- Pretrigger: 50%
Timebase: 0.5 s/Div / DC
channel A: 1 V/Div / DC
storage: on
- Perform three seals with an empty tubing Ø 4.2 mm (part no. M600241) (China version: Ø 4.55 mm, part.no. M600251).
- Connect the oscilloscope to **MP 0 / P.C.B. LP 1058** and **MP-seal / P.C.B. LP 1058**.
- Connect oscilloscope to **MP 0 / P.C.B. LP 1058D** and **MP-seal / P.C.B. LP1058D** ($\geq$ E-Code 140)





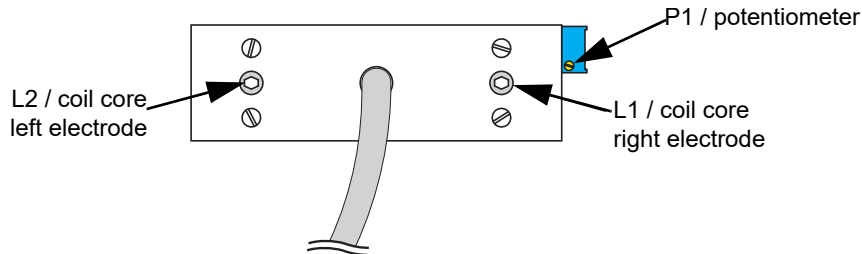
Arrangement of the measuring points
≥ E-code 140

Select the **Seal test** service menu.

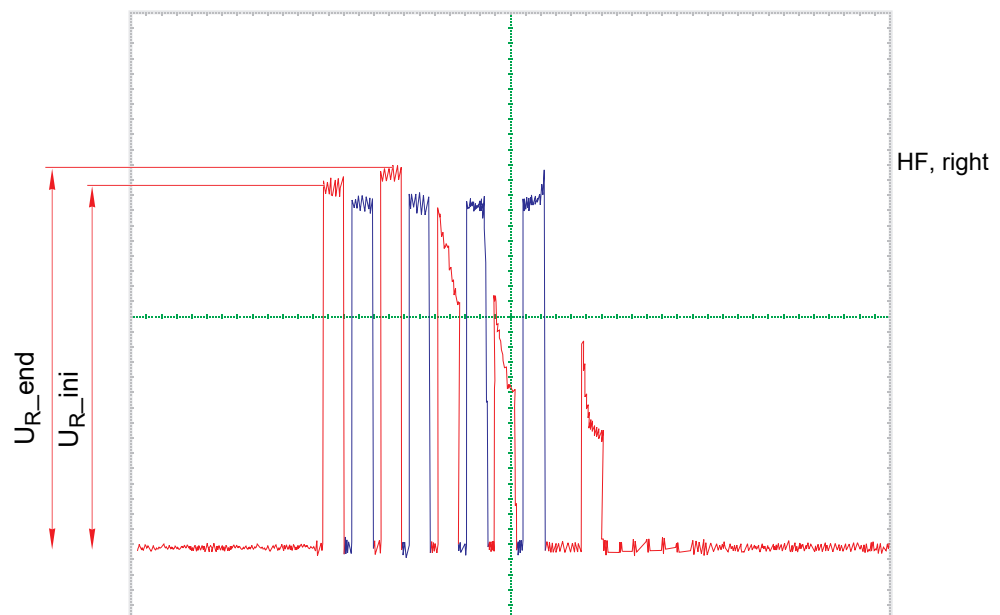
- Perform one seal with an empty tubing Ø 4.2 mm (part no. M600241) (China version: 4.55 mm (part-no. M600251)) and record the sealing curve.
- The gradients of the sealing curves should be slightly ascending.
- The smaller of the two sealing curves must show an initial voltage value of $4.1 \text{ V} \pm 0.1 \text{ V}$.
- Record the initial voltage values of the left and right electrode.
 $U_{R_ini.} (\text{empty}) = ______ \text{ V}$
 $U_{L_ini.} (\text{empty}) = ______ \text{ V}$
- Perform one seal with a tubing filled with NaCl Ø 4.2 mm (part no. M600241) (China version: 4.55 mm (part-no. M600251)) and record the sealing curve. The gradients of the sealing curves are flattening (become horizontal).
- Record the initial voltage values of the left and right electrode voltage.
 Right electrode: $U_{R_ini.} (\text{filled}) = ______ \text{ V}$
 Left electrode: $U_{L_ini.} (\text{filled}) = ______ \text{ V}$
- The difference between the initial values for the filled tubing and the empty tubing must range between -0.1 V and $+0.3 \text{ V}$.
 $U_{R_ini.} (\text{filled}) - U_{R_ini.} (\text{empty}) = ______ \text{ V}$
 $U_{L_ini.} (\text{filled}) - U_{L_ini.} (\text{empty}) = ______ \text{ V}$

5.13.2 Adjustment of the HF voltage

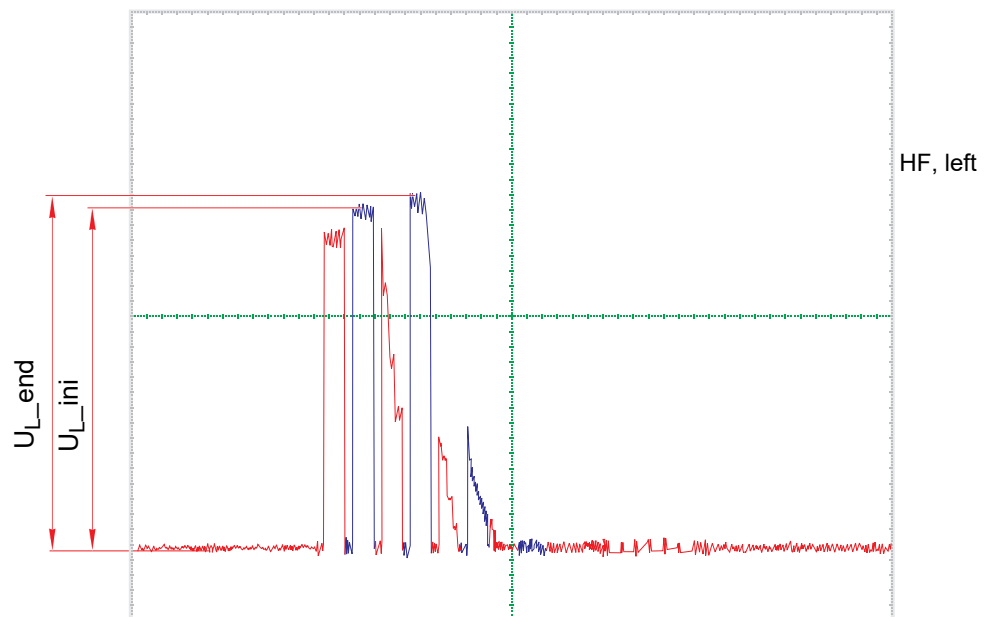
- Turn the potentiometer P1 / seal head box towards the left limit stop.
- Adjust the coil cores L1 and L2 so that they are in line with the housing of the seal head box.



- Connect the oscilloscope to **MP 0 / P.C.B. LP 1058** and **MP-seal / P.C.B. LP 1058**.
Oscilloscope settings:
 Pretrigger: 50 %
 Timebase: 0.5 s/Div / DC
 channel A: 1 V/Div / DC
 storage: on
- Select the **Seal test** service menu.
- Perform three seals with an empty tubing Ø 4.2 mm (part no. M600241) (China version: 4.55 mm (part.-no. M600251)).
- The gradients of both sealing curves must be ascending.
- Turn in the right coil core L1 until the relation between final value and initial value of the right electrode voltage has achieved the value 1.04 ± 0.02 .
 Desired value (U_{R_end}/U_{R_ini}) right electrode pair: 1.04 ± 0.02

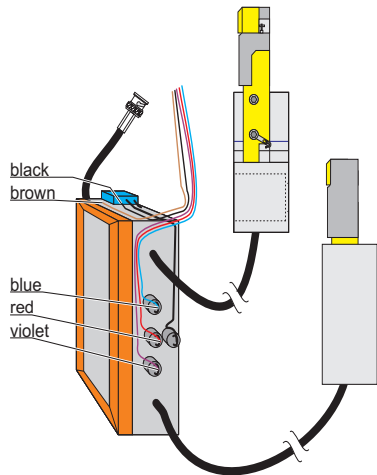


- Turn in the left coil core L2 until the relation between final value and initial value of the left electrode voltage has achieved the value 1.04 ± 0.02 .
 Desired value (U_{L_end}/U_{L_ini}) left electrode pair: 1.04 ± 0.02



- Perform the seals with an empty tubing Ø 4.2 mm (part no. M600241) (China version: 4.55 mm (part-no. M600251)) and record the sealing curve.
- The gradients of the sealing curves should be slightly ascending.
- The smaller of the two sealing curves must show a minimum peak voltage of $4.1 \text{ V} \pm 0.1 \text{ V}$.
- Use the potentiometer P1 to adjust the initial value of the smaller of the two sealing curves to a nominal voltage of $U_{\text{L_ini.}} = 4.1 \text{ V} \pm 0.1 \text{ V}$.
- Record the initial voltage values of the left and right electrode.
 $U_{\text{L_ini.}} = \text{___} \text{ V}$
 $U_{\text{R_ini.}} = \text{___} \text{ V}$
- Perform the seals with a tubing filled with NaCl Ø 4.2 mm (part no. M600241) (China version: 4.55 mm (part-no. M600251)) and record the sealing curve.
- The gradients of the sealing curves are flattening (become horizontal).
- Record the initial voltage values of the left and right electrode.
 $U_{\text{L_ini.}} = \text{___} \text{ V}$
 $U_{\text{R_ini.}} = \text{___} \text{ V}$
- The difference between the initial values for the filled tubing and the empty tubing must range between -0.1 V and +0.3 V.
 $U_{\text{R_ini.}} (\text{filled}) - U_{\text{R_ini.}} (\text{empty}) = \text{___} \text{ V}$
 $U_{\text{L_ini.}} (\text{filled}) - U_{\text{L_ini.}} (\text{empty}) = \text{___} \text{ V}$
- Insert an empty tubing
- Check the adjustment and apply lacquer to the coil cores.

5.13.3 Removal of the seal head box



Note

In the event of a defective seal head box, the complete modification kit (part no.: M612421) must be installed.

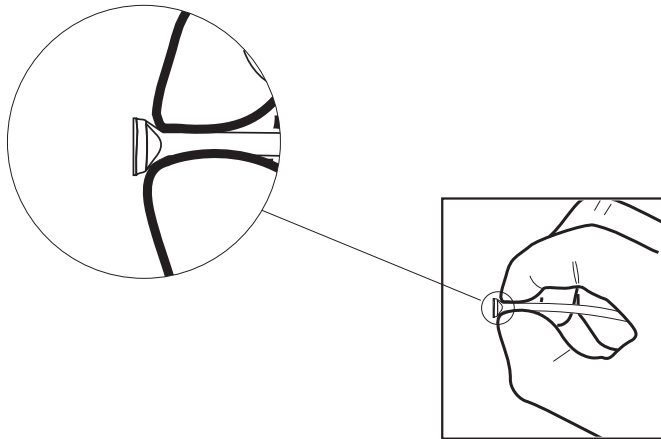
- Unsolder the leads (violet, red, blue, brown and black) of the defective seal head box.
- Unsolder the coaxial cable at the hot electrodes.
- Separate the coaxial connector from the HF generator and open the cable ties.
- Unscrew the angle bracket of the seal head box.

5.13.4 Installation of the seal head box

- Install the new seal head box. Solder the cable harness.
- Solder the inner and outer conductors of the coaxial cable to the hot electrodes.
- Close the cover of the seal head box and seal it using the copper band.
- Measure the HF voltage (see chapter 5.13.1 on page 5-57).
- Perform a sealing test (see chapter 5.14 on page 5-62).

5.14 Sealing test

- Select the **Seal test** menu item in the service menu and seal a filled tubing.
- Insert the line into both clamps.
- Close clamps.
- Seal the tubings.
- Remove the tubing ends and check that the sealing will not open under slight pressure.



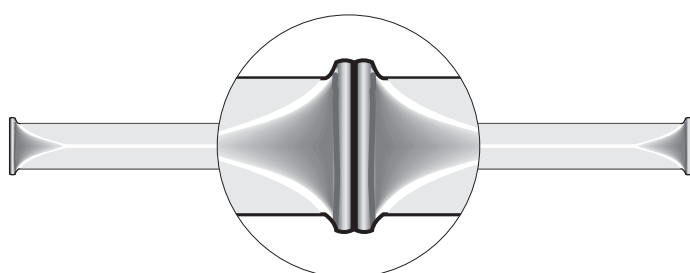
5.15 Docking test

Measuring equipment and accessories:

- Saline solution
- 2-kg weight
- Stopwatch
- Two strippers, one tube clamp

5.15.1 Visual check

- Make a test dock with two empty tubings and check that the test dock is symmetrical.



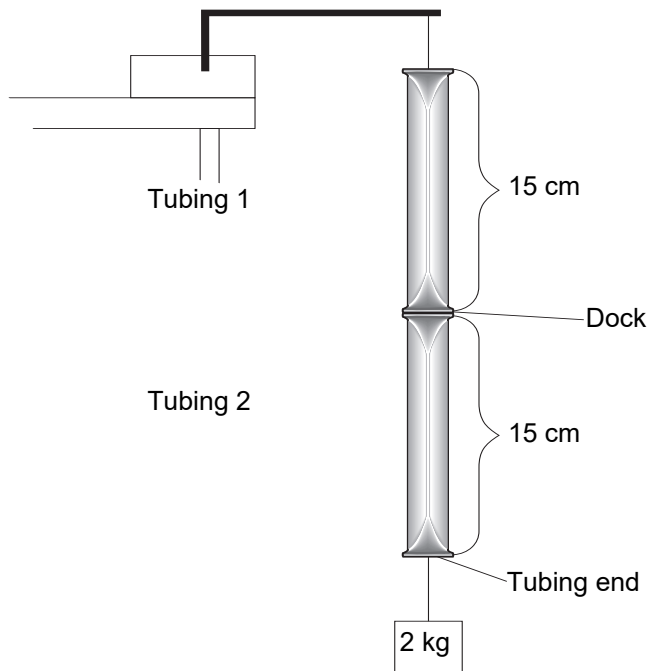
5.15.2 Tensile strength

- Dock combinations:

Europe / USA / Japan versions 9028651 / 9028721 / 9028851		China version 9028201	
Tubing 1	Tubing 2	Tubing 1	Tubing 2
Fresenius 4.2 empty	Fresenius 4.2 empty	Fresenius 4.55 empty	Fresenius 4.55 empty
Fresenius 4.2 filled	Fresenius 4.2 filled	Fresenius 4.55 filled	Fresenius 4.55 filled
Fresenius 4.55 filled	Fresenius 4.55 filled	Fresenius 5.2 empty	Fresenius 4.55 empty
Fresenius 4.55 empty	Fresenius 4.55 empty	Fresenius 5.2 filled	Fresenius 5.2 filled
		Fresenius 4.55 empty	Fresenius 5.2 empty
		Fresenius 5.2 empty	Fresenius 4.55 empty
		Fresenius 4.55 empty	Fresenius 5.2 filled
		Fresenius 4.55 filled	Fresenius 5.2 empty
		Fresenius 5.2 empty	Fresenius 4.55 filled

- Attach tubing 1, letting it hang down freely. Attach a 2-kg weight to the end of tubing 2 and allow the weight to hang for 15 s.

- After the loading test, the dock must not be damaged.



5.15.3 Tightness

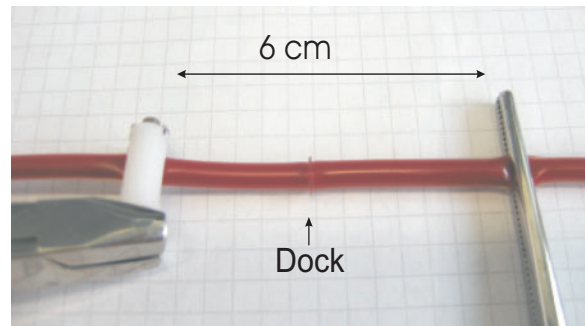
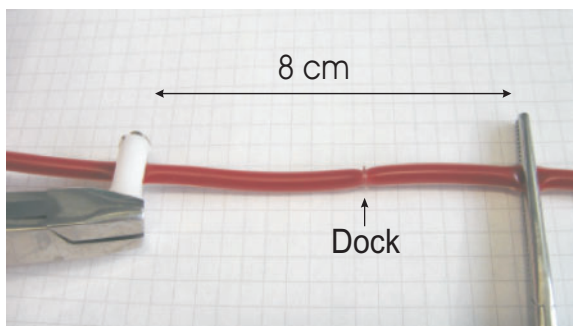
- Perform a test dock of two tubings filled with saline solution.
- Use a tube clamp or a stripper to clamp the docked tubing at a distance of 4 cm to the dock.



Tip

Preferably use a ruler to measure the tubing lengths and mark them with a waterproof pen.

- Use the stripper to strip down the included tubing length to 6 cm. This will generate an internal pressure of > 3 bar.
- After one minute's time, the dock must not be damaged, and there must not occur any leakage.



5.16 Electrical test

- **Protective conductor check**

Measurement according to DIN VDE 0701 - 0702

- **Protective conductor contact resistance**

Using the probe of the measuring instrument, measure the following points one after the other:

- Housing (front screws 2x, rear screws 6x)
- Central block (Allen screws 2x)

Desired value: $< 0.2 \, \Omega$

5.17 Initial start-up



Caution

Never allow any fluid to directly enter into the device. The fluid may come in contact with electronic components and lead to a contamination of the electronics and subsequent failure of the device.



Caution

When the CompoDock is installed, the air opening on the rear must always stay uncovered, so that the ventilated air can flow out undisturbed.



Note

Repeatedly switching the CompoDock on and off may shorten the life of the heating element substantially. It is recommended to switch on the CompoDock only once a day.

Unpacking

Take the CompoDock out of its protective transport container. Keep the container for a possible future transport.

Place of installation

Place the CompoDock on a solid base.

Variations in temperature

An appropriate acclimatization time must be ensured in case of great variations in temperature.

Scope of delivery

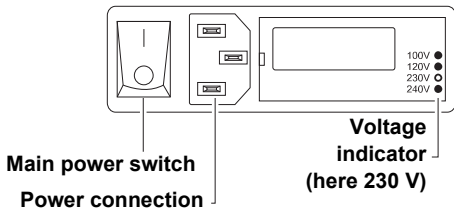
Before starting the initial start-up, check that all parts have been accounted for.

- Power cable 1x
- CompoDock Counter 1x
- Side tables 2x
- Operating Instructions

General external checks

Check the CompoDock on the outside for possible damage in transit or other damage.

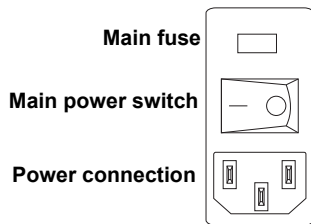
Power connection
< E-code 069



The rear panel of the CompoDock is provided with a connection for a grounded power cable. Please ensure that the voltage label indicates the voltage required. If not, please immediately contact Fresenius Kabi or your local distributor.

Connect the power cable to the power connection on the rear of the device and to a grounded receptacle outlet.

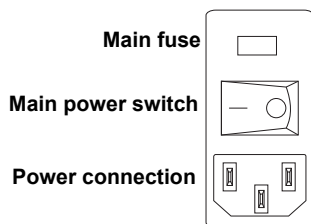
Power connection ≥ E-code 069



The rear panel of the CompoDock is provided with a connection for a grounded power cable.

Connect the power cable to the power connection on the rear of the device and to a grounded receptacle outlet.

Power connection ≥ E-code 140



The rear panel of the CompoDock is provided with a connection for a grounded power cable.

Connect the power cable to the power connection on the rear of the device and to a grounded receptacle outlet.

CompoDock Counter

Insert the CompoDock Counter into the counter compartment.

Turning the device power on

- Turn on the device using the main power switch on the rear panel of the system.

The green Power LED on the operating panel is illuminated.

The system check is performed.

The motors are reset.

If the DockMaster is available, a connection to the **DockMaster** will be established. (Only with >E-code 069)

```
* COMPODOCK V2.431 *
Fresenius Kabi
System check    | / |
```

The display shows the following messages:

(The display messages may differentiate from the illustration depending on the system software used).

This is followed by an audible alarm to check the audible alarm generator.

```
*   xxxx docks left
Insert tubes and
press tube cover
```

The CompoDock is ready for operation.

5.17.1 Seal test

(see chapter 5.14 on page 5-62)

5.17.2 Dock test

(see chapter 5.15 on page 5-63)

5.18 Alarm processing

5.18.1 Fatal (system) error

● Cause

The counter was removed while the device was switched on or off

Power failure during the docking process

● Remedy

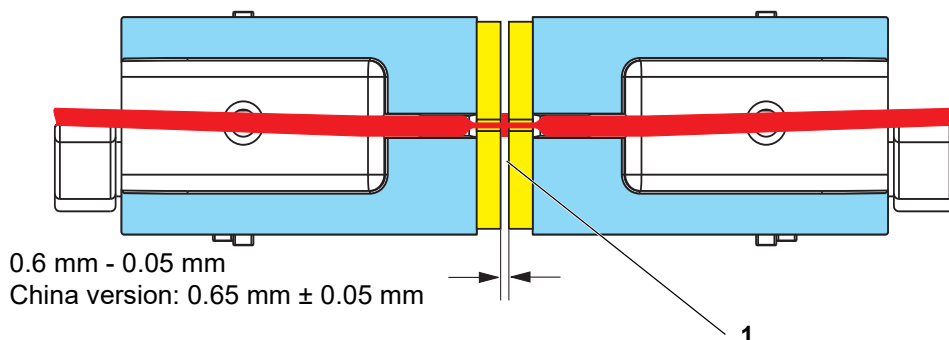
- Insert the counter
- Simultaneously press and hold the service keys S1 and S2 / LP 1058 and turn on the device with the master switch.
- Note down the following values:
Xmotor Outside pos
Xmotor pos.dock
Heater dock temp.
Xmotor heat pos
heating Parameter/x-MotorSpeed Heat
- Select Set Default Values and follow the instructions displayed
- Set the values noted down and perform the following tests

Xmotor Outside pos

- Select the **Xmotor Outside pos** service menu and move the tube clamps to the X1 / Y0 position (separation of the tubings).
 Desired value: 0.2 s - 1.5 s

Xmotor pos.dock

- Make a test dock with each tubing.
- Do not open the tube clamps after the docking process.
- Use the feeler gauge to determine at point (1) the distance of the tube clamps in docking position.
 Desired value: 0.6 mm + 0.05 mm (China version: 0.65 mm + 0.05 mm)

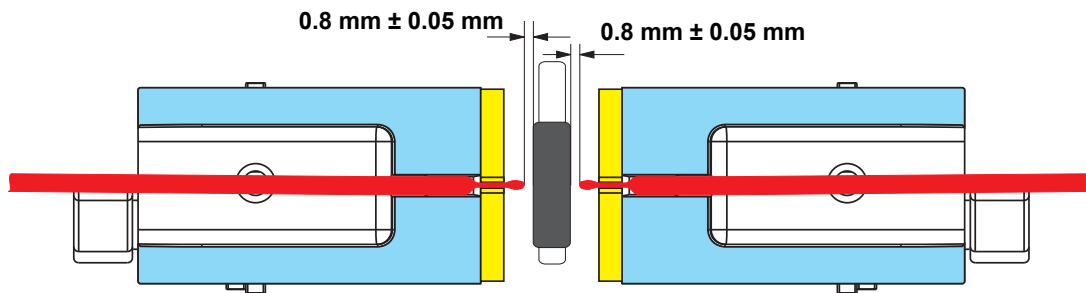


Heater dock temp.

- In the service menu, select **Heater dock temp.**
- Use the infrared thermometer to measure the temperature of the heating element.
Desired value: $465\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$ (China version: $470\text{ }^{\circ}\text{C} \pm 5\text{ }^{\circ}\text{C}$)

Xmotor heat pos

- Select the **Seal test** service menu and seal off the two tubings.
- Keep the tube clamps closed
- Select the **Xmotor Heat pos** service menu and move the two tube clamps to the heating position.
- Manually move the heating element out of its enclosure. The two tubing ends must be positioned exactly in the middle of the heating element.
- Measure the distance between the two tube clamps and the central block.
Desired value: min. 2 mm
- The heating distance between the tubing ends and the heating element must be $0.8\text{ mm} \pm 0.05\text{ mm}$.
($0.8\text{ mm} + 0.05\text{ mm}$)



- Turn the device off and back on again, using the main switch

5.19 Software update

5.19.1 Software versions 2.20.R11 and 2.20.R12

- **Measure**

External influences (temperature, contamination or dirt etc.) can lead to a blocking of the X-axis (movement from preheating to heating position) and thus cause the error message **Maximum time moving to heating position**. The adjustment range of the „X-MotorSpeed Heat.“ was changed from 5-15 counts to 10-20 counts in the new software in order to correct this error.

The new software version only has to be used if the error described above occurs.

- **Compatibility**

- Applicable to devices with software version 2.1.x and higher and equipment code (E-code) ≥ 030 .

- **Logistics**

- EPROM software V 2.20.R11 (EN/IT/DE), part no.: M659211
- EPROM software V 2.20.R12 (EN/NL/FR), part no.: M659241

- **Replacing the software**

- Read out and note down the parameters (see chapter 5.5.4 , page 5-20)
- Open the device (see chapter 5.1 , page 5-1)
- Lubricate threaded pins with silicone spray (part no. M642401)
- Select menu option **Heating Parameter**
- Select submenu **X-MotorSpeed Heat**.
- Increase the Speed (count) displayed to 20
- Close the device
- Verify the parameters
- Functional test
 - Make 3 test docks and check the result

5.19.2 Software versions 2.21.R0, 2.21.R1 and 2.21.R2

● Measure

The CompoDock software was revised The following changes were made:

- The Fresenius HemoCare logo was replaced by the Kabi logo.
- A failure of the HF generator is detected.
- In the service mode, the X motor moves to the docking position at a constant speed.
- The languages Spanish and Portuguese were implemented.
For this purpose, a separate EPROM was created (V 2.21.R2). This EPROM also contains the English software.

● Compatibility

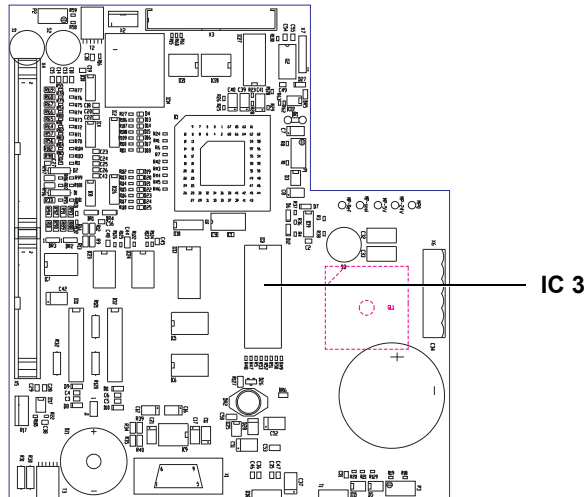
- Applicable to devices with software version 2.1.x and higher and equipment code (E-code) ≥ 030 .

● Logistics

- EPROM software V 2.21.R0 (EN/IT/DE), part no.: M659211
- EPROM software V 2.21.R1 (EN/NL/FR), part no.: M659241
- EPROM software V 2.21.R2 (EN/ES/PT), part no.: M665051

● Replacing the software

- Read out and note down the parameters (see chapter 5.5.4 , page 5-20)
- Open the device (see chapter 5.1 , page 5-1)
- Replace EPROM IC 3 on P.C.B. LP 1058 (observe ESD precautions)



- Access the service menu and set the language (see chapter 4.4.15 , page 4-11)
- Verify the parameters
- Close the device
- Functional test (see chapter 5.15 , page 5-63)
Make 3 test docks and check the result

5.20 Modification to CompoDock with interface (E-code 059 or higher)

● Preparation

- Open the rear panel. (see chapter 5.1 , page 5-1)



Caution

Electric shock!

Do not touch live parts when working on the open system.

- Connect the power plug.
- Start the Service menu (see chapter 4.3.1 , page 4-2)
- Read the system parameters from the Service menu and note them down.



Tip

The table below can be used as a master copy for the parameters to be read.

- Measure the temperature of the heating element (see chapter 5.12.1 , page 5-50).
- Measure the distance between the tube clamps in the dock position (see chapter 5.10.2 , page 5-45).

Master copy for system parameters

System number:	E-code:	Language:
Function	Possible value or desired value	Actual value
Autostart	on / off	
Voltage supply 5V:	4.60 V - 5.40 V	
Voltage supply 24V:	23.65 V - 24.31 V	
Blower current:	8.6 mA - 20.9 mA	
Xmotor Outside pos:	< E-code 030: 95 / > E-code 030: 100	
Ymotor Outside pos: Front	0.8 sec - 2.5 sec	
Ymotor Outside pos: Back	0.8 sec - 2.5 sec	
Dock cool down time	004	
Xmotor pos. Dock (Count):	Variable	

System number:	E-code:	Language:
Function	Possible value or desired value	Actual value
Xmotor pos. Dock (gap):	0.65 mm $\pm$ 0.5 mm	
Heater dock temp.	Standard device: 465°C / China: 470°C	
Tube heating time.	12 sec	
Xmotor Heat pos:	Variable	
Heating Parameter Offset Heating Pos:	030	
Heating Parameter Pre heating Time	4 sec.	
Heating Parameter X-MotorSpeed Heat	015 - 029	
Seal stretch Tubes:	005	
Seal Times: Pulse time	026	
Seal Times: HF-Motor Move In	999	
Seal Times: HF-Motor Move time	000	
Seal Times: Pausetime	008	

- Switch off the CompoDock and pull out the power cable.
- Pull out all plug connections from P.C.B. LP 1058
- Clamp the white and blue cable from the toroidal transformer to the screw plug X6.
- Remove the PE connection to the housing.

● Fit a new rear panel.

- Make the PE connection to the housing.
- Connection up power supply unit as shown below.
As can be seen in the photo, the blue cable crosses over with the red cable.



Caution

The unit can be damaged.

Pay attention to the correct polarity of the cables!

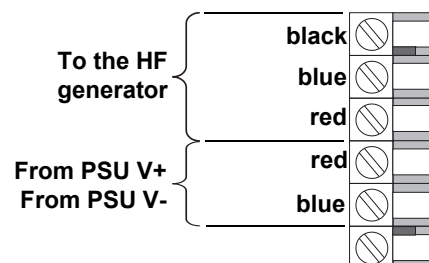
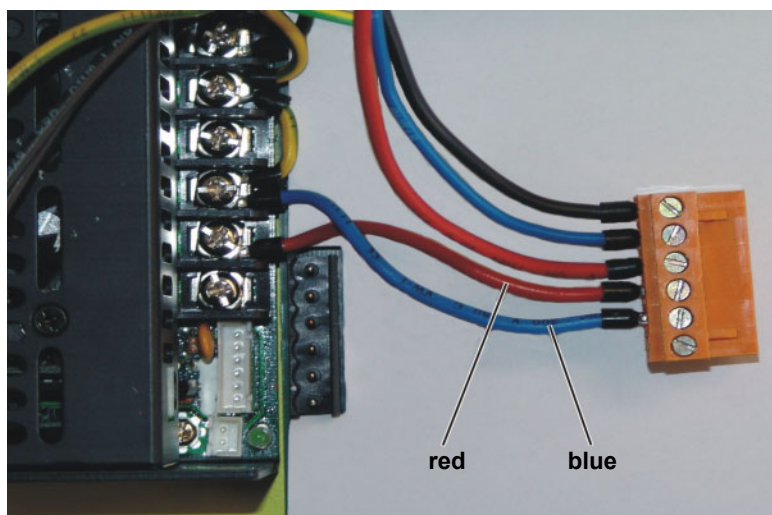


Caution

The unit can be damaged.

The wiring configuration of the plug is different to the previous version!

Connect up power supply unit part as shown below!



- Reconnect all plug connections.
- Turn the device on.
- Set/monitor all voltages (Chapter 5.3.3 on page 5-8))
- Start the Service menu (see chapter 4.3.1 , page 4-2)
- Monitor the system parameters in the service menu and carry over the variable values as noted.



Note

After the new PCB has been fitted, the temperature of the heating element can change owing to component tolerances.

The temperature of the heating element has to be monitored with an immobile temperature measuring device and possibly corrected.

- The signal duration at the end of the docking procedure can be set with the new parameter **Dock End Beep Time**. Recommendation: 0.4 sec.

5.21 Software upgrade (E-code 069 - 140)



Caution

The CompoDock and/or the **PC** may be damaged.

Both units have to be switched off before the connection is made!

- **Connect the CompoDock to the PC/laptop using the zero modem cable:**

- Connect the CompoDock at the **Network** connection.
- Connection the PC/laptop to a free **COM interface**.



Tip

Should there not be a free COM-interface on the PC/laptop, the PC/laptop can be connection via to **USB to COM**adapter. (available as an accessory see Appendix C).

- **Switch on the PC and install the *CompoDock-Service-Tool* (M726331) (PC software):**

- In the master directory of **C:**, create the folder **CDockServiceTool**.
- Copy the file **CDockServiceTool.exe** into the above folder.
- Copy all language files (CD Flash_xx.bin) into the above folder.

- **Start CompoDock in update mode:**

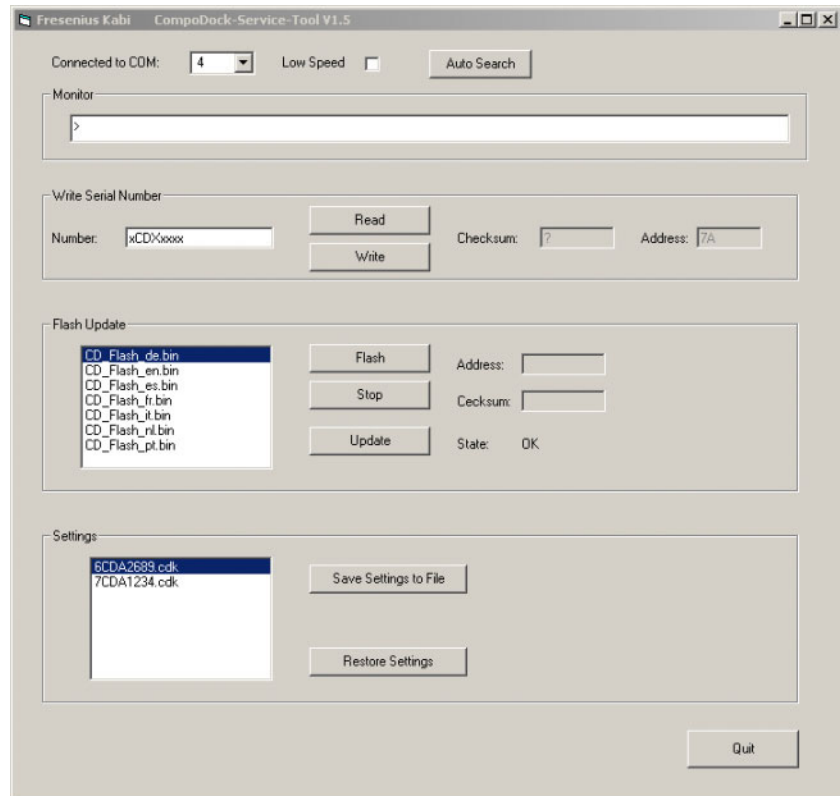
- Hold down keys S1 and S2 and switch on the CompoDock.

The CompoDock starts with **bootload V1!**

- **Start the CompoDock-Service Tool:**

- In the directory **C:\CDockServiceTool**, double click on the file **CDockServiceTool.exe** .

The window below will open.



- Click on **Auto Search**.

The assigned COM interface will be sought.

The connection to the CompoDock is created.

- **Save the serial number**

- Enter the serial number of the CompoDock into the **Number** field.
- Click **Write**.

The serial number is transferred to the CompoDock.

- **Checking the saved serial number**

- Click on **Read**.

The saved serial number is read from the CompoDock.

- **The language files are transferred**

- Select the desired language file using the mouse.
- Click on **Flash**.

The language file is transferred into the CompoDock. (visible by the timer symbol)

Transfer takes around 5 minutes.

Transfer is complete as soon as the cursor reappears.

5.22 Software upgrade (E-code 140 and higher)

5.22.1 Installation of the Service-Tool V2

Depending on the operating system (.NET is already installed or not), there are 2 ways to install the PC program.

The installer shows a popup after the start, if .NET is not installed. If the .NET Framework is already installed, the step "install .NET Framework" will/can be skipped.

- **Start the installer**

- Start "CD2_Service_Tool_V2_Setup.msi".

If there is a Popup "**This setup requires the .NET Framework version v4.7.2. Please install....**", follow the instructions in the next chapter. Otherwise skip the next chapter.

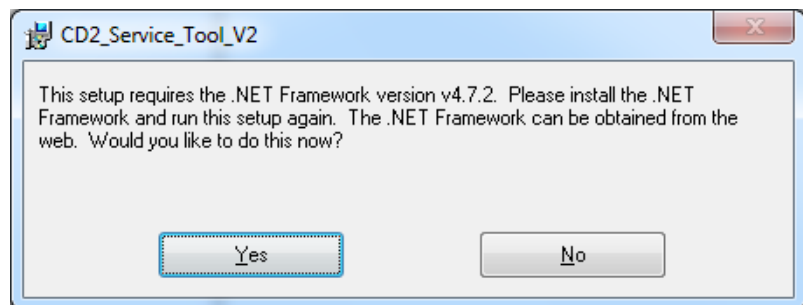
- **Install .NET Framework**

If there is a Popup "**This setup requires the .NET Framework version v4.7.2. Please install...**" after starting the installer, follow these instructions and install the .NET Framework (internet access is required).

If there is no internet access on your PC, you can use the ".NET-offline installer":

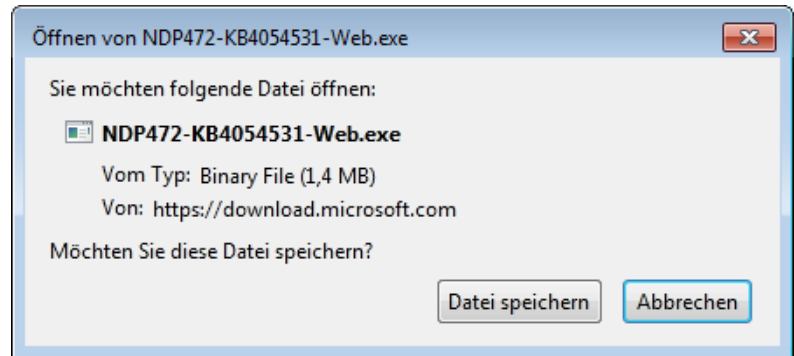
NDP472-KB4054530-x86-x64-AllOS-ENU.exe

"Link: <http://go.microsoft.com/fwlink/?linkid=863265>



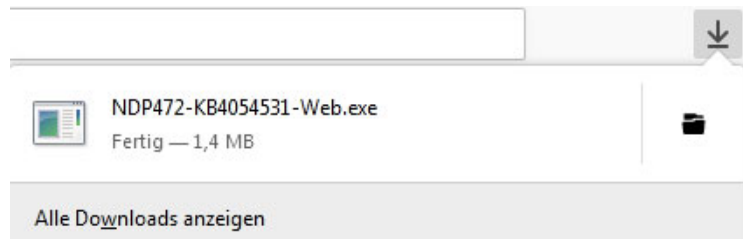
- Click **Yes**.

Download the .NET installer.



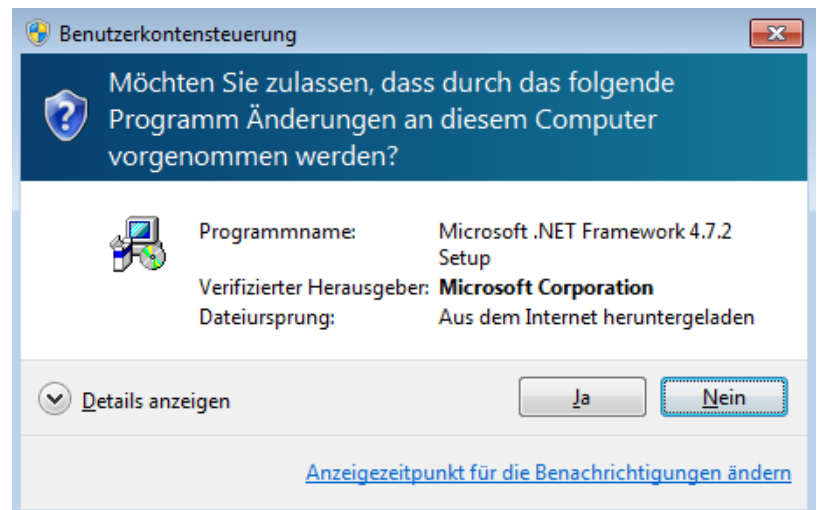
- Click **Datei speichern (Save file)**.

Start the .NET installer



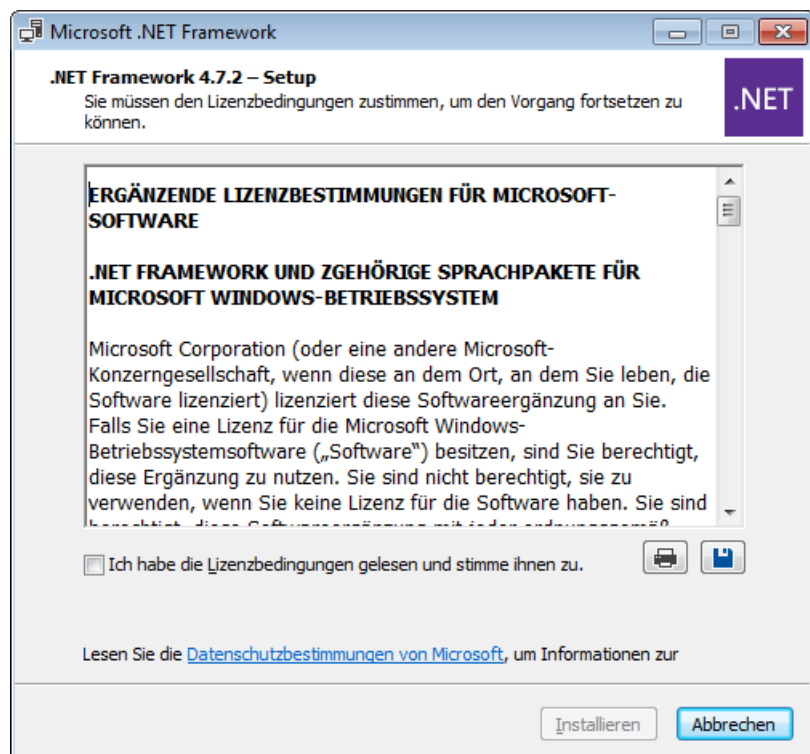
- Click to start.

Allow/permit, that installer can do changes...



- Click **Ja (Yes)**.

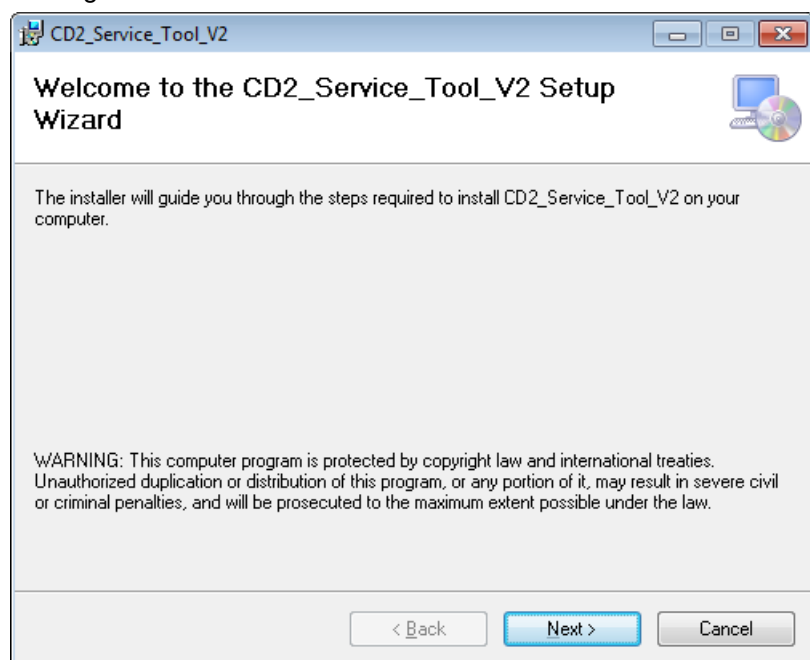
Accept the License.



– Click **Installieren (Install)**.

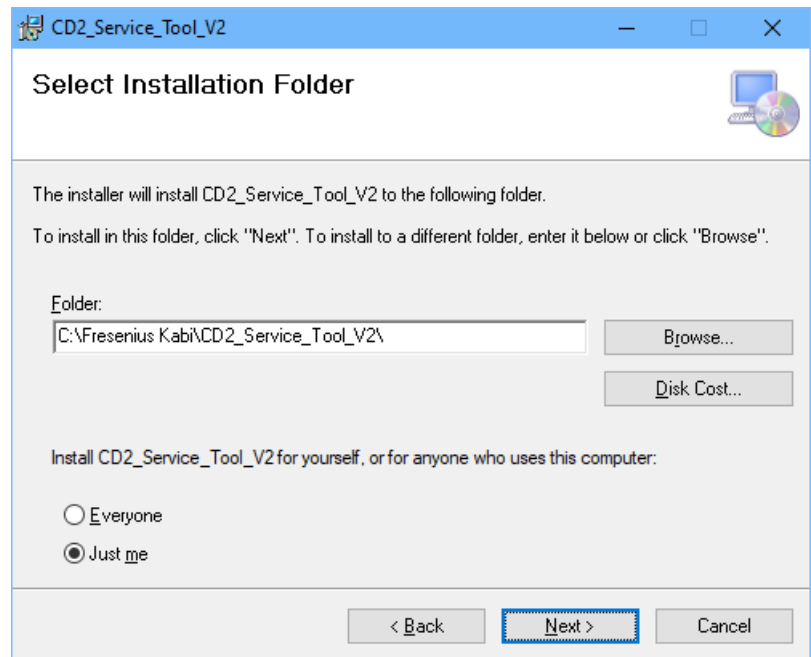
● **Service-Tool-V2 (if .NET is already installed)**

The following dialog appears after starting the installer.

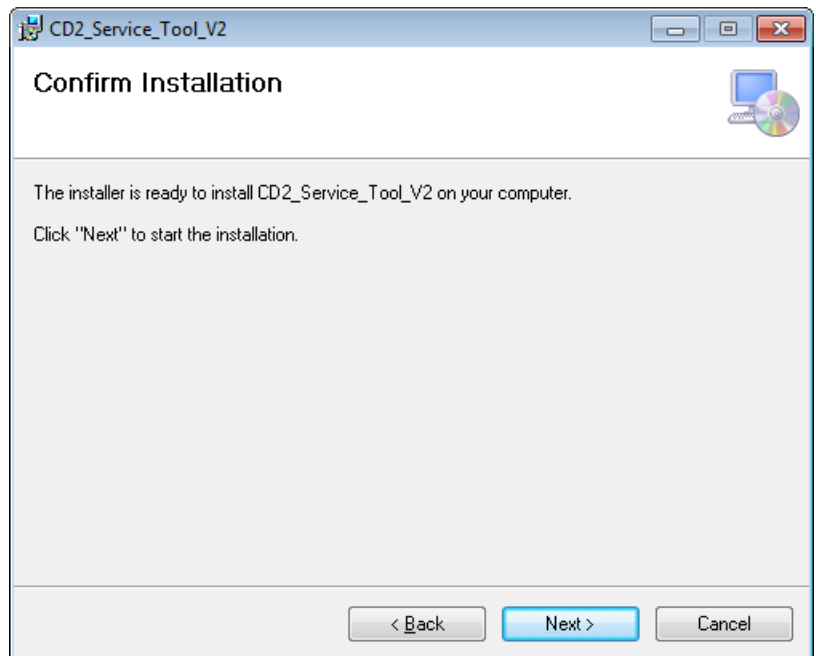


– Click **Next**.

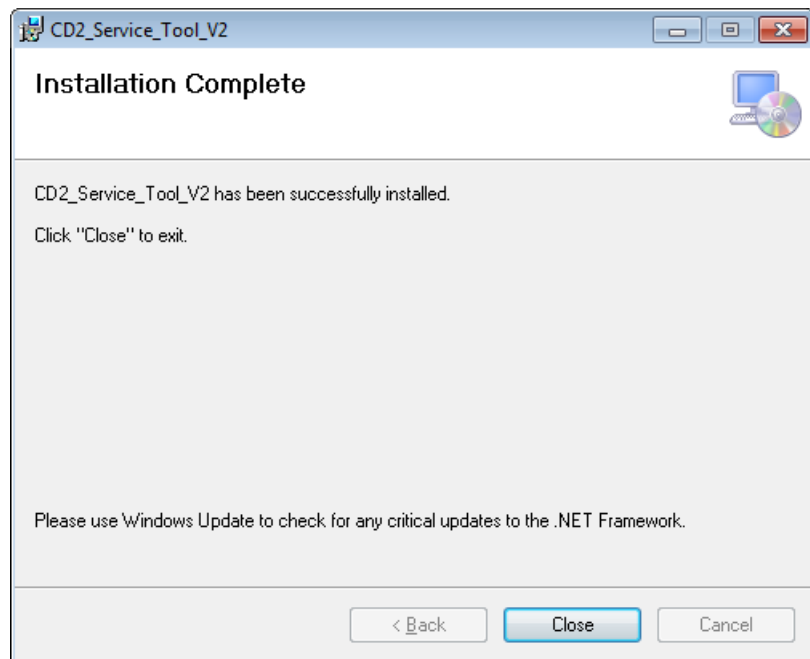
- Click **Browse**.



- Select Folder: Default folder (C:\Fresenius Kabi\CD2_Service_Tool_V2\)
- Choose **Everyone** or **Just me**.
- Click **Next**.



- Click **Next**.



- Click **Close**.

● How to start Service-Tool-V2

You can start the Service-Tool-V2 via STARTMENU or manually under "C:\Fresenius Kabi\CD2_Service_Tool_V2\CD2_ServiceTool_V2.exe".

5.22.2 GUI Elements

- Use a null modem cable to connect the CompoDock with the personal computer.
- Start the software "CompoDock-Service-Tool V2".
- Switch on the CompoDock.

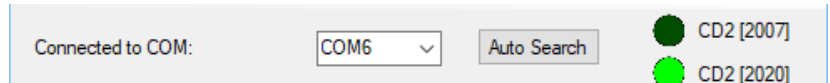
CompoDock2 [2007]:

- If no program was transmitted yet, the CompoDock runs in the Bootload-Mode.
- If there still exists a functional program on the CompoDock you can run the Bootload-Mode by pressing the pushbuttons **S1** and **S2** at the same time during the switch on. The pushbuttons are located on the PCB ore they are plugged to the Scanner-Interface.
- You can also run the Bootload-Mode by switching on the CompoDock three times during "system check".
- If CD2 is in Bootload-Mode. **Bootload V1!** is shown in the display.

CompoDock2 [2020]:

- If there still exists a functional program on the CompoDock you can run the Bootload-Mode by pressing the pushbuttons **S1** and **S2** at the same time during the switch on. The pushbuttons are located on the PCB ore they are plugged to the Scanner-Interface.
- If CD2 is in Bootload-Mode. **Bootload V2!** is shown in the display.

● Connection



- Click the button **Auto Search** to search the COM port of the connected CompoDock2.

You can change manually the chosen COM port via DropDown-Box.

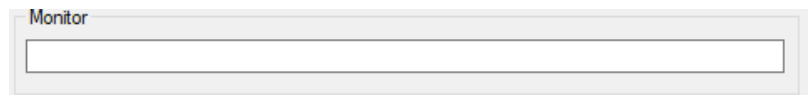
The Service-Tool-V2 shows the type of the connected CompoDock2 device:

CD2 [2007], produced until summer 2019

CD2 [2020], produced since April 2020

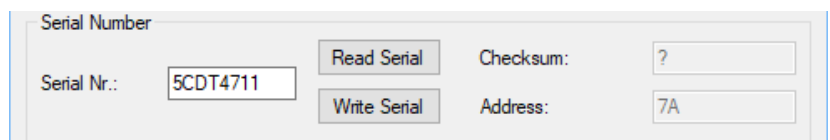
If no CompoDock was found, there will be a popup error message.

● Monitor



In the field "Monitor" the current data progress is displayed.

● Serial number



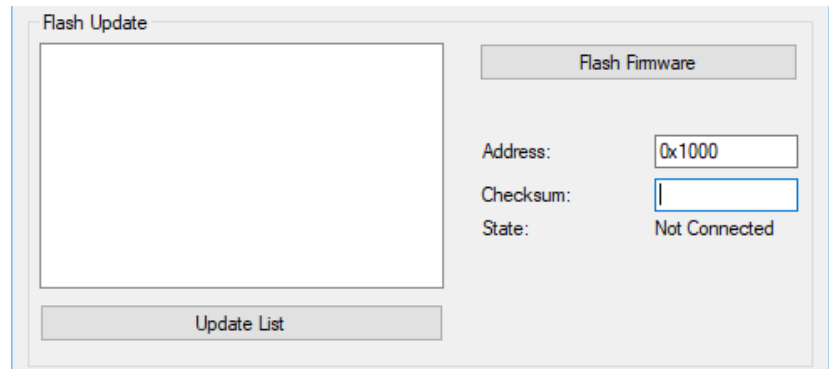
- Read out the serial number from the CompoDock by clicking the button **Read Serial** in the **Serial Number** section.
- Type in the serial number in the field **Serial Nr.**

With the button **Write Serial**, the serial number can be written to the EEPROM (Flash) of the CompoDock.

The checksum of the EEPROM is displayed in the field **Checksum**.

The address, where the checksum of the EEPROM is stored is displayed in the field **Address**.

● **Flash update**



The section **Flash Update** shows a list of firmware/language files (*.bin) which are stored in the same folder as the Service Tool V2.

With the button **Flash Firmware** the selected firmware/language file is transferred to the CompoDock.

The transfer can be stopped with the button **Stop**.

After using the **Stop** button, the CD2 device must be switched OFF in order to establish further communication.

By clicking the button **Update List** the list of the firmware/language files is updated.

Address: shows the current address during programming process
Checksum: shows the checksum of the last transmitted firmware block
State: shows the current state of process

5.23 Backup (E-code 069 - 140)

- **Reading out parameters**

- Click on **Save Settings to File** .

The system-specific parameters are stored in folder **C:\CDockServiceTool** . The file name consists of the **serial number** and the suffix ***.cdk** .



Tip

If the system-specific parameters are available as a file, the parameters do not have to be re-identified after necessary replacement of the PCB. The parameters can be simply transferred using the **CDServiceTools** .
*)

*) The temperature of the heating element has to be monitored/corrected.

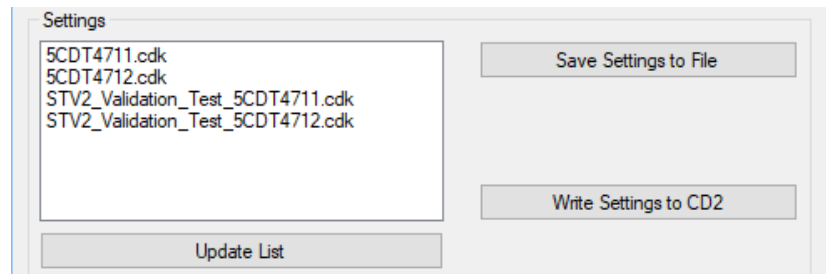
- **Write the parameters into the unit**

- Select the file from the **Settings** list.
- Click on **Restore Settings** .

The system-specific parameters are written into the CompoDock.

5.24 Backup (E-code 140 and higher)

- **Settings**



The section "Settings" shows a list of the setting files (.cdk) which are stored in the **same folder** as the Service Tool V2.

- Click the button **Save Settings to File** to store the settings of the CompoDock in a file named like the serial number of the connected CompoDock.
If this file already exists, there will be a popup **overwrite?**
- Click the button **Write Settings to CD2** to write the settings from the file back to the CompoDock.

6 Initial Start-up Test report

– See Appendix A.

7 Maintenance Test report

– See Appendix B.

8 Spare Parts Catalog

– See Appendix C.

